

Labor Market Flexibility and Gender Heterogeneity in Labor Allocation

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Abstract

Does greater flexibility create job opportunities and improve labor allocation? This paper examines how firms and workers respond to a labor market flexibility shock in a setting characterized by high regulation costs. Focusing on the Brazilian labor market, I leverage linked formal employer-employee data to analyze the impacts of the 2017 Labor Reform, a policy that removed constraints on flexible work schedules. I show that this policy change led to an increase in job opportunities and employment, especially in part-time positions. Notably, women experienced greater employment gains. At the state level, I observe that while the reform did not significantly impact unemployment rates, it contributed to reduce informality, which accounted for 40% of the workforce in 2016. Gender-specific analysis indicates that this effect is primarily driven by women transitioning from informal to formal employment within the private sector in the short run, followed by both women and men in the state-level economy in the medium run. These findings underscore the interaction between labor market flexibility and gender disparities, underscoring the potential of such reforms to reconfigure employment allocation. The case of the Brazilian reform offers a more nuanced perspective when informality is considered.

Keywords: labor market flexibility, gender differences, informality, part-time employment

JEL: J16, J21, J38, O17

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1. Introduction

As economies transition towards more flexible work schedules, with the prevalence of part-time employment increasing in most countries over the past decades,¹ labor regulations may constrain economic agents from freely choosing their preferred or optimal working hours (see Lachowska et al., 2025; Carry, 2024). By incorporating constraints into the worker's and firm's decision problem, labor regulations introduce rigidity into labor markets and might lead to an inefficient allocation of labor. Importantly, gender-neutral policies can have different implications for men and women when gender differences in the propensity to enter a job are stark. For instance, restrictions on part-time employment or changes in weekly hour limits have been proven to disproportionately affect the female labor supply, as women are more likely to sort into jobs with fewer hours and with more time flexibility (see for example Wasserman, 2023; Carry, 2024). Yet, as labor regulations become more flexible in terms of relaxing legal constraints on the use of part-time or temporal contracts, there is little evidence on whether policies aimed at enhancing flexibility can improve overall market efficiency and promote greater job opportunities.²

In this paper, I leverage unique features of the 2017 Brazilian labor reform, a policy that unexpectedly and dramatically modified labor legislation, to investigate whether and how flexibility in the labor market affects job creation, employment, and labor allocation across genders and economic sectors. Within the scope of the changes, the reform streamlined the establishment of flexible work contracts by providing employers and employees with new avenues for flexible work arrangements, such as freelance and temporary contracts, as well as additional legal adjustments relevant to part-time jobs. Additionally, it facilitated labor recruitment and termination by halving severance payments and weakening the bargaining power of labor unions, thereby allowing firms to adjust their workforce as needed.

In the first part of this paper, I use matched employer-employee data together with an event study design to empirically evaluate the impact of the 2017 Brazilian reform at the firm level. The administrative data allow me to study in detail the reform's effects on job creation, workforce composition, and employment, particularly, the unequal effects across different groups (e.g., part-time vs. full-time workers, women vs. men). To assess the reform's effects, my empirical strategy leverages variation in work-hour constraints across firms. Constraints on hours capture the idea that workers might be restricted to schedules that do not align with their preferred hours due to factors such as a firm's specific technology. In the Brazilian context, firms may have also imposed constraints due to the high regulation costs³, which restricted or prohibited freelance work, part-time jobs, and home office arrangements before 2017. The analysis suggests that this was the case for small firms, which were more bidden by the constraints dictated by the regulation and the resulting labor costs. Specifically, the presence of substantial fixed costs in hiring and dismissal prior to the reform led small firms to op-

¹For instance, on average in the OECD, 16% of workers worked less than 30 hours per week in 2022.

²See, for instance, Daruich et al. (2023) that studies an Italian reform that relaxes several legal constraints on hiring temporary contract workers and shows that new temporary jobs substituted existing permanent ones without generating an increase in employment.

³Botero et al. (2004) demonstrate that Brazil's labor regulation costs are approximately 20% higher than the mean and median of the 85 countries surveyed.

timally employ a limited number of full-time workers instead of maintaining a workforce of rotating part-time employees.

To gauge firm-level constraints on working hours, I construct a measure based on the standard deviation of average hours worked across skill groups within the firm in the pre-reform period, following Labanca and Pozzoli (2022). This metric captures within-firm variation in mean hours across workers with different skills and heterogeneous labor supply preferences. Lower values indicate that workers in different skill groups work similar hours, consistent with tight firm-level constraints on working-time arrangements. Conversely, higher values reflect greater dispersion in average hours across skill groups, indicating looser constraints and the presence of heterogeneous work-hour contracts within the firm. I exploit this firm-specific measure of hours constraints as a source of cross-sectional variation. Specifically, I compare firms with a pre-reform standard deviation of hours equal to zero to firms with a strictly positive standard deviation, which define the treated and control groups, respectively.

The key identifying assumptions in this paper are twofold. First, the flexibility shock induced by the reform primarily impacted firms that were highly constrained by rigid work schedules, as they faced higher costs in implementing flexible work arrangements compared to firms with fewer pre-existing constraints (i.e., no spillover effects to already flexible firms). Second, other concurrent labor market reform shifts—such as severance-pay provisions—influenced both treated and control firms similarly, as this was a nationwide policy. Therefore, the empirical strategy hinges on the assumption that the reform’s effects were mediated through firms’ varying propensities to adopt new flexible schedules, specifically driven by their pre-reform hour constraints.

Consistent with the reform’s goal of easing hiring restrictions and promoting flexible work, the firm-level results indicate that firms with high pre-reform constraints experience larger gains in job creation and are more likely to change their worker composition. I find that the reform fostered job creation and increased the share of new part-time jobs in firms with ex-ante strict constraints relative to firms with looser constraints. The observed increase in the share of part-time jobs is explained by the combination of an increase in part-time jobs and a decline in full-time jobs. This pattern, therefore, suggests that the reform led to a substitution of full-time workers with part-time workers. The positive effect on employment provides further evidence of job creation, indicating that the reform increased employment within firms that were highly constrained before the reform. The substitution between job arrangements is additionally supported by a negative effect in the average weekly hours per job and in the total number of hours worked within firms. These findings demonstrate how firms adjust to flexibility in the labor market, suggesting that relaxing legal labor constraints cause a reallocation effect toward jobs with relatively fewer hours.

The firm-level analysis also reveals heterogeneity in the effects across different firms. I find that the reform had more pronounced job creation and composition effects on firms (i) in service industries, (ii) with a relatively small pre-reform number of workers (5-19) and a single-establishment structure, and (iii) with a high pre-reform fraction of female employees. In contrast, I find that the reform led to a significant decrease in size within large firms (above 50 workers or multi-establishments), suggesting that such firms with strict hours constraints were negatively affected relative to less con-

strained firms due to a significant decline in full-time jobs. Overall, the observed job effect appears to be driven by the extent to which firms substituted full-time employees with part-time ones.

Who is driving the main effects of the reform? I explore whether the reform had heterogeneous effects between men and women. The increase in job opportunities at the firm level seems to be mainly explained by a rise in female jobs. I find that the reform particularly fostered part-time job creation among women. While both men and women experienced positive job and composition effects, the point estimates are significantly larger for women. These findings speak to the literature assessing heterogeneous preferences towards work schedules between genders (e.g., Wiswall and Zafar, 2017; Goldin, 2015; Maestas et al., 2023), highlighting that flexibility in the formal market might boost female employment as women tend to place a higher value on flexible work hours compared to men.

I provide further evidence in support of the gender-based heterogeneity in labor allocation and evaluate whether the effect varies across occupations. I find that the heterogeneous effects between men and women are mostly explained by gender differences in pink-collar occupations (e.g., retail clerks, salespersons), where the reform's impact on part-time job creation and employment is particularly more pronounced among women. The observed effects show no significant gender differences within managerial positions, high-skill professionals, low-skill white-collar roles, or blue-collar job categories. These results indicate that after the reform, job positions requiring flexible work schedules were more likely to be filled by women. Moreover, the within-industry analysis reveals that this gender difference corresponds to higher pre-reform levels of part-time jobs held by women compared to men. This suggests that "egalitarian" industries exhibit similar employment and composition effects for both men and women.

Does the creation of these new part-time jobs result in overall higher employment? I find that this is not necessarily the case in developing economies. This is partly due to the labor regulations and institutions that push firms and workers towards partial compliance or informal, non-written contracts.⁴ Informal employment provides a flexible alternative that can enhance the efficiency of hiring, firing, and production processes (Ulyssea, 2018; Dix-Carneiro et al., 2025). In Brazil, where 40% of the labor force lacks formal contracts, it is uncertain whether firms are recruiting informal workers or unemployed workers after the reform's implementation. The relaxation of legal constraints driven by the reform would impact overall employment if firms are indeed recruiting unemployed workers. However, if informal workers are being hired instead, we should expect unemployment rates to remain unchanged.

The second part of this paper, hence, investigates the effects of the 2017 reform on unemployment and informality at the state level. To do so, I leverage variation across states on the share of jobs offered by firms with strict hours constraints (i.e., those with a standard deviation of hours across skill groups equal to zero). Using this continuous exposure measure, I provide evidence that unemployment in more (relative to less) exposed states was essentially unaffected after the reform, suggesting that this policy did not materialize in a reduction of unemployment as anticipated.

⁴This trend is particularly pronounced in Latin America, where informality ranges from 35% in Chile to as high as 80% in Peru (Perry, 2007).

Conversely, I find that the probability of working informally in the private sector and state-level informality rates decline after the reform's implementation. A gender-specific analysis further reveals that the reduction in informality is primarily driven by women, particularly mothers, transitioning from informal to formal employment within the private sector in the short run. This is followed by both women and men moving to formal employment in the state-level economy in the medium run. This effect could be attributed to the fact that men were already more likely to have job-hour arrangements matching their preferences before the reform, which reduces the scope for reallocation within the private sector immediately after the reform compared to their female counterparts. Taken together, these results indicate that the reform led firms to draw new hires mainly from the informal sector rather than from the pool of unemployed workers, which explains the limited response of aggregate unemployment to the policy.

This paper contributes to several strands of literature. First, this article advances the growing empirical literature that explores gender disparities in labor allocation and work hour preferences (e.g., Wasserman, 2023; Carry, 2024; Wiswall and Zafar, 2017; Flabbi and Moro, 2012; Bustelo et al., 2023; Mas and Pallais, 2017; Goldin, 2015; Maestas et al., 2023), by leveraging a first-time reform that boosted flexibility into the labor market. The sharp bite of the reform, combined with high-quality administrative data on the universe of formal workers and firms, allows me delve into the gender-specific labor allocation and employment responses to an enhanced market flexibility. My findings confirm that women tend to put more value onto flexible work schedules compared to men, while also shedding light on how flexibility can ameliorate the misallocation of female workers.

Second, this study adds to the existing literature on labor supply and the role of hour constraints (e.g., Chetty et al., 2011; Kahn and Lang, 1991; Altonji and Paxson, 1986; Labanca and Pozzoli, 2022; Lachowska et al., 2025; Wasserman, 2023), wherein hour constraints play a pivotal role in shaping the decision of working hours. By leveraging the quasi-experimental variation induced by the Brazilian labor reform, I provide additional insights into both the origin and the extent of the effects by which hour constraints distort labor allocation. More specifically, I shed light on how labor regulations and the associated costs are a source of hours constraints, thereby influencing the composition of the workforce within a firm.

Third, my results speak to the wide literature on working time regulations and labor policies that relax legal constraints towards flexible employment contracts (e.g., Daruich et al., 2023; Hunt, 1999; Chemin and Wasmer, 2009; Bentolila and Saint-Paul, 1992; Blanchard and Landier, 2002), which has been mainly focused on developed countries. By bringing the analysis to a developing country, this paper sheds light on some aspects of labor demand-supply adjustments that are not common in developed countries. In particular, the state-level analysis allows me to understand how policies intended to introduce labor market flexibility create limited opportunities for formal employment and are ineffective at declining unemployment due to the presence of informality.

Last, this paper contributes to the literature highlighting informality as a consequence of excessively rigid formal labor markets subject to burdensome and costly regulatory frameworks (Meghir et al., 2015; Ulyssea, 2018). Importantly, without the

presence of informal jobs, the participation of women in the labor force would be substantially lower. Empirical findings, such as those put forth by Berniell et al. (2021) in the context of Chile, indicate that this phenomenon ensues as women strategically opt for informal employment to mitigate labor misallocation in the face of a rigid formal market. Leveraging the 2017 Brazilian labor reform as a case study, this paper provides novel causal evidence on the interplay between flexibility in the formal labor market, informality, and gender.

The rest of the paper is organized as follows. Section 2 provides institutional background on the Brazilian labor market and the 2017 Labor Reform. Section 3 describes the data sources and samples used in the empirical analysis. Section 4 discusses the empirical framework for examining the effect of the reform on firm-level outcomes, presents the main results, and explores the heterogeneity in the effects by gender. Section 5 characterizes the implications of the reform for unemployment and informality at the state-level. Section 6 concludes.

2. Institutional Background: The Brazilian Labor Regulation and the 2017 Labor Reform

The Brazilian labor market has long been characterized by its rigidities and extensive regulations (Almeida and Carneiro, 2012). At the core of this system lies the *Consolidação das Leis Trabalhistas* (CLT), a labor code based on the civil law system, which was established in 1943. The CLT governs all aspects of the labor market, delineating the permissible types of labor contracts, their associated conditions, and termination terms.

Over the years, the CLT has shaped a notably rigid labor market, offering limited scope for bilateral negotiations between employers and employees (Ponczek and Ulyssea, 2022). Its stringent provisions have contributed to the inflexibility and high labor costs prevalent in the Brazilian labor market. To compound matters, the amendments introduced by the Federal Constitution of 1988 strengthened labor regulations and resulted in a substantial increase in workers' protection, significantly elevating Brazil's labor regulation expenses well above international averages. For instance, the employment index in Botero et al. (2004) indicates that Brazil's labor regulation costs are approximately 20% higher than the mean and median of 85 countries surveyed (Ponczek and Ulyssea, 2022). These costs encompass hiring and firing expenses, social security contributions, payroll taxes, as well as contributions to the severance fund. When compared to the United States, Brazil's labor regulation costs are more than 2.5 times larger.⁵

The cumbersome and costly nature of the labor regulation in Brazil incentivized both firms and workers to resort to partial compliance or opt for informal labor contracts to evade its high costs.⁶ In 2015, informal employment accounted for 45% of

⁵Prior to the reform, firing a worker in Brazil was not significantly more difficult compared to other Latin American countries, although it appears costlier on paper (Almeida and Carneiro, 2012). For example, dismissed workers were entitled to compensation paid by the employer and the firing firm had to pay the worker a net of 40% of the balance in the worker's social insurance account.

⁶Throughout this paper, informal workers are defined as those employees who do not hold formal labor contracts and are hired "off the books" by firms.

total employment in Brazil, nearly half of the employed population, while the unemployment rate averaged 8.6%, up from 6.8% the previous year. In response to the high degree of informality and unemployment, the Brazilian government enacted the most extensive legislative package in the Brazilian Labor Law, the 2017 Labor Reform (Law 13.467). The reform aimed to increase formal labor employment by reducing labor costs and introducing flexible employment practices, alongside other substantial changes to the existing employment legislation.

The 2017 Labor Reform brought forth a range of new legal instruments. Amendments were made to allow the engagement of freelance workers and remote work, thereby introducing the concepts of "intermittent work" and "home office" into the CLT.⁷ Additionally, part-time work underwent changes in legislative provisions⁸, granting further flexibility in this domain and providing employers and employees with new avenues for flexible work arrangements. The 2017 reform also granted greater autonomy to employees in negotiations with employers. The terms of negotiations extend to holidays, labor relations at the firm level, and the process of dismissal.⁹ For instance, employers do not need the Union's approval for mass layoffs, and collective agreements on banked hours can be directly negotiated between employer and employees. In line with these changes, the reform offered new venues for bilateral negotiations between employers and employees to terminate their labor contracts and substantially reduced the severance payment from 40% to 20% of the balance in the worker's social insurance account.¹⁰

In conclusion, the 2017 Labor Reform in Brazil marked a crucial milestone in shaping the dynamics of labor relations in the country. While upholding essential worker rights guaranteed by the CLT, this reform established a new regime with greater flexibility and less labor rigidities in the recruitment of employees. In this paper, I assess whether the labor market flexibility induced by the reform led to an increase in formal employment and the extent to which this new regulation had heterogeneous effects among different workers.

3. Data and Sample

In order to examine the effects of the 2017 labor reform, I use two datasets in my empirical analysis. First, I rely on administrative linked employer-employee data to study firm-level outcomes. Second, I use the National Household Survey which contains information on individuals' labor market outcomes and informality status to conduct an aggregate state-level analysis.

⁷Freelance or intermittent work was forbidden before the reform. Employers may now hire workers on demand and by hour, as long as the worker is given a three-day notice (Articles 442–443 of Law 13.467). Prior to the reform, there was no legal basis for working from home. After November 2017, work from home is allowed and the employer is required to pay for the related costs such as equipment (Article 75-B of Law 13.467).

⁸Employers may hire part-time workers for up to 30 hours a week, without overtime; or for no more than 26 weekly hours, with overtime limited to 6 hours per week (Article 58-A of Law 13.467). Before the reform, part-time employment was allowed up to 25 hours per week. Overtime was forbidden for these workers.

⁹Articles 134, 444, and 477 of Law 13.467.

¹⁰Article 484-A of Law 13.467.

3.1. Matched Employer-Employee Data

The empirical analysis presented in this study relies on a comprehensive dataset derived from the *Registro Anual de Informações Sociais* (RAIS). This dataset comprises Brazilian administrative records on employment spells, providing the full employment history of individuals who at some point in their career were employed by a firm in the formal sector. *Formal* firms are defined as those registered with the tax authorities, which means that they possess the tax identification number required for Brazilian firms (*Cadastro Nacional de Pessoa Juridica* (CNPJ)).

Each observation in RAIS is an employer-employee match, or spell, in a given year. For each spell, the dataset includes worker-level characteristics (e.g., age, gender, race, educational attainment, and a unique worker identifier), firm-related variables (e.g., sector of economic activity, establishment size, geographical location, and unique establishment and firm identifiers), and job-related attributes (e.g., monthly earnings, contractual weekly hours, tenure, occupation, date of hiring and separation).

Sample Selection. My empirical analysis focuses on the years 2014 to 2019. Using the job spell data, I restrict the sample to private sector firms, excluding public servants, public firms, and NGOs. The sample includes working-age population between 18 and 64. I omit job spells that registered zero contractual working hours. Finally, the sample is composed of firms active between 2014 and 2019 that had at least 5 registered workers before the year of implementation of the 2017 reform.

For each year and employer, I construct a yearly measure of employment by taking the mean of the number of employees reported across months. I also compute the share of part-time jobs (and new part-time jobs) as the fraction of jobs¹¹ with workweek less than or equal to 25h/wk in the firm in a given year. Similarly, I compute the share of jobs with workweek between 26 and 39h/wk, and above 39h/wk at the firm level. I aggregate job-level data to recover the number of total hours worked within the firm for a given year, using the number of contractual weekly hours, along with the start and end dates of each job spell. Additionally, I aggregate job-level data to recover employment, share of part-time jobs, and total hours worked separately for men and women.

3.2. The Continuous National Household Sample Survey (PNAD-Continua)

To study the effects of the reform on unemployment and informality, I use the Continuous National Household Sample Survey (PNAD-Continua)¹² from 2012 to 2019. Each quarter, about 211,000 households are interviewed, covering approximately 16,000 census sectors of 3,500 municipalities.

The PNAD contains information on individuals' labor market outcomes (e.g., work hours, wages and employment), including formality status and detailed sociodemo-

¹¹Jobs are defined by the intersection of a unique worker identifier and the most detailed occupational classification, which is represented by a 6-digit occupational code, within a firm for a given contractual period.

¹²PNAD is a national household survey, similar to the Current Population Survey in the US.

graphic characteristics.¹³ I define as *informal* workers those employees who do not hold a formal contract, which in Brazil is characterized by having a “signed work booklet”, *carteira de trabalho*. Workers who report being employees are directly asked whether they have a formal contract. The measure of informality excludes the self-employed.

Individual-level Panel. The PNAD-continua follows and surveys a designated set of households, visiting them every three months, thus generating a panel dataset. Leveraging this dataset, I construct a rotating individual-level panel including individual attributes related to labor status (informality and unemployment status), demographics (age, sex, years of schooling), and household characteristics (indicator of head of household, household members) across five consecutive quarters. I restrict the sample to individuals aged between 18 and 64 years old.

State-level Dataset. For each of the 27 states in Brazil, I separately compute unemployment and informality rates. To do so, I collapse the share of unemployed workers and the share of informal workers aged between 18 and 64 in a given state. Specifically, the share of informal workers refers to the proportion of employed individuals lacking formal employment documentation, as indicated by the absence of a signed work booklet. Conversely, individuals are classified as unemployed if they are not engaged in any form of employment or self-employment.

4. The Effect of Labor Market Flexibility: Firm-Level Analysis

In this section, I analyze the effects of the 2017 labor reform on firm-level outcomes. I begin by describing the measure of exposure leveraged by my research design. I then analyze the reform’s effect using an identification strategy that exploits two dimensions: the timing of implementation and firm-level heterogeneity in exposure.

4.1. Exposure to the 2017 Reform: Hours Constraints within the Firm

To gauge the extent to which relaxing labor restrictions affect firm-level outcomes, I construct a measure of exposure based on hours constraints at the firm level. Constraints on hours capture the idea that workers have limited discretion over hours and may be constrained to work undesired hours. Workers may want to work longer or shorter hours, but the formal market or firms do not offer the option. Moreover, employers may prefer certain work hours or work schedules due to firm-specific technology, worker setup costs, fixed employment costs per worker, team synergies, and other considerations of scheduling and coordination (Lachowska et al., 2025). For instance, before the reform, the fixed costs associated with hiring and firing workers in the formal sector may have incentivized employers to hire a small number of full-time workers rather than a larger contingent of part-time employees. Consequently, this structural rigidity might have prevented workers seeking more flexible or part-time employment opportunities from securing positions that aligned with their preferences.

¹³The main advantage of this data source over RAIS is that it contains information on workers employed in the informal sector and unemployed workers.

Building on the measure proposed by Labanca and Pozzoli (2022), my measure of hours constraints is the standard deviation of the mean hours worked across skill groups within the firm:

$$\sigma_{ft} = \left[\frac{1}{S_{ft}} \sum_{s=1}^{S_{ft}} (\tilde{h}_{sft} - \mu_{ft})^2 \right]^{\frac{1}{2}}, \quad \tilde{h}_{sft} = \frac{1}{N_{sft}} \sum_{i=1}^{N_{sft}} h_{isft}$$

where h_{isft} is the number of weekly hours worked by employee i in skill group s in firm f at year t , $\tilde{h}_{sft}$ is the average of h_{isft} across workers in sft , and μ_{ft} is the average of $\tilde{h}_{sft}$ across skill groups in firm-year ft . N_{sft} and S_{ft} are the number of workers in sft and the number of skill groups in ft , respectively. For consistency with the frequency of the data on hours used in the analysis, I measure σ_{ft} at a weekly basis. I construct a measure of skill given by the intersection of the education and occupation categories. The education categories are: (1) less than primary education, (2) complete primary education, (3) complete secondary education, and (4) post-secondary education. The occupation categories are: (1) management occupation, (2) professionals, (3) technicians and associate professionals, (4) clerical support workers, (5) service and sales workers, (6) skilled agricultural, forestry and fishery workers, (7) craft related trade workers, (8) plant and machine operators, and assemblers, and (9) elementary occupations. This variable includes a total of thirty-six skill groups.

I average the yearly standard deviation of hours over three pre-reform years (2014-2016) to obtain a time-constant measure for each firm:

$$(1) \quad \overline{\sigma}_f = \frac{1}{3} \cdot \sum_{t=2014}^{2016} \sigma_{ft}$$

The measure $\overline{\sigma}_f$ captures the degree to which workers with different skills at a firm are constrained to work undesired hours. I focus on constraints on hours among workers with different skills since workers in different skill groups generally differ in their desired hours.¹⁴ Under the assumption that observed work hours reflect underlying workers' preferences for hours, equation (1) allows me to capture differences in desired work hours among coworkers.¹⁵ Using the National Household Survey that contains detailed information on work hours reported by employees laboring both formal and informally, Table 1 provides a first indication that work hours vary across skills. For instance, the least skilled service and sales workers work approximately 3.5% more hours (1.5 weekly hours) compared to their more skilled counterparts in the same occupation. This trend extends beyond specific occupations, suggesting a broader pattern where the least skilled workers in Brazil often work longer hours than their more skilled counterparts, even within the same occupation. In Section A of the Appendix, I show that the average work hours of informal workers also varies across skill groups. In contrast to formal workers, in the informal sector where the labor legal restrictions

¹⁴For instance, Labanca and Pozzoli (2022) show that the least skilled workers in Denmark desire to work approximately 15% fewer hours than the most skilled workers.

¹⁵Ideally, I would use survey data that contain information on desired hours (see Carry (2024) and Labanca and Pozzoli (2022)), but there are not such data in Brazil.

are not binding, workers are more likely to work part-time (less than 40h/wk)¹⁶, suggesting that formal workers may be constrained from below in choosing their work hours.

In equation (1), a low value of $\bar{\sigma}_f$ indicates that coworkers with different desired hours tend to work a similar number of hours, suggesting that they face strong constraints. Conversely, a high value of $\bar{\sigma}_f$ implies that workers with different preferences for hours are likely to work their desired hours, suggesting that they face loose constraints on hours. Figure 1 shows the distribution of this measure. Approximately 60.3% of the firms have a standard deviation of hours across skill groups equal to zero, which, in turn, implies that workers in those firms work the same number of hours regardless of their skill.¹⁷ The limited variation observed in this measure suggests that in Brazil the vast majority of employers either face strong restrictions in offering diverse work schedules or exhibit strong preferences for full-time workers. The relaxation of hiring constraints driven by the reform would have an impact on the demand-supply behavior if the former scenario holds, whereas if the latter is true, we should not anticipate significant changes.

In my empirical analysis, I dichotomize the exposure measure $\bar{\sigma}_f$ and leverage variation from the fact that some firms have loose constraints on hours. Because the 2017 reform relaxed hiring restrictions and allowed for flexible work arrangements, I would typically expect firms with strong constraints on hours to experience more changes in their work schedules than less constrained firms. My empirical strategy, thus, exploits both firm-specific hours constraints and the timing of the reform by comparing firms with pre-reform constraints on hours $\bar{\sigma}_f$ equal and above zero, which represent treatment and control firms, respectively.

4.2. Descriptive Statistics

In order to document the bite of the 2017 reform, I separately consider firms with strict hours constraints ($\bar{\sigma}_f = 0$) and firms with loose constraints ($\bar{\sigma}_f > 0$). First, in Panel (a) of Figure 2, I plot the evolution of the share of new hires for jobs with a workweek less than or equal to 25h/wk. From 2014 to 2016, there is an upward trend in the share of new part-time hires in firms with loose constraints, whereas the trend is flat near zero for firms with stricter constraints. Starting in 2017, the year of the reform, the share for these firms gradually increased, reaching 1% by 2018. In contrast, the evolution of the share of part-time hires in firms characterized by loose constraints followed a similar trajectory as the pre-reform period. Panel (b) and Panel (c) show that the increase in the share of part-time hires in firms with strict hours constraints is driven by a sharp increase in the number of part-time hires rather than a decline in the number of full-time hires¹⁸.

Figure 2 provides suggestive evidence that firms with strict constraints experienced larger increases in part-time hires and were more likely to change their worker compo-

¹⁶In the informal sector, employers do not comply with wage and employment regulation or pay taxes. In addition, part-time workers are overrepresented in the informal sector: in 2015, nearly 40% of informal workers were working less than 40 hours a week compared to 11% in the formal sector (Derenoncourt et al., 2025).

¹⁷In the sample, more than 80% of the jobs have a workweek with 40h/wk or above.

¹⁸Jobs with workweek with at least 40h/wk.

sition after the reform than less constrained firms. These heterogeneous changes in the aggregate share and number of part-time hires suggest that firms with strict constraints were more exposed to the 2017 reform than firms with looser constraints. Frictions, as in Chetty et al. (2011), preclude workers with heterogeneous preferences from working their desired hours. In my setting, the rigid labor institutional structure and high costs of hiring workers with different arrangements explain workers' inability to work their desired hours in firms with strict constraints.

Table 2 presents average characteristics of firms with loose and strict constraints one year prior to the reform. The average firm size in the sample is 83.1 workers in firms with loose constraints and 16.3 in firms with strict hours constraints. Relative to firms with loose constraints, firms with strict constraints are less likely to have female employees, white employees, and workers with tertiary education. In addition, these firms have a higher percentage of employees whose hourly wage is less than $1.2 \times$ the minimum wage (24.8%), compared to firms with looser constraints (18.7%). These differences on firm size and wage suggest that small firms are less likely to bear labor costs associated with hiring and regulation, indicating that pre-reform constraints were more binding for these firms.

4.3. Empirical Strategy

To estimate the effect of the 2017 reform on firm-level job creation, workforce composition, and employment, I use the following event study specification that considers the timing of the reform and the firm-level variation plotted in Figure 1:

$$(2) \quad y_{ft} = \alpha_f + \lambda_{j(f),t} + \sum_{\substack{k=-3 \\ k \neq -1}}^{k=2} \beta_k \times \mathbb{1}\{\bar{\sigma}_f = 0\} \times \mathbb{1}\{t = 2017 + k\} + \varepsilon_{ft}$$

where y_{ft} is a *detrended* firm-level outcome—such as the share of part-time jobs observed in firm f and year t . $\mathbb{1}\{\bar{\sigma}_f = 0\}$ indicates whether the firm has a pre-reform standard deviation of hours across skill groups equal to zero and $\mathbb{1}\{t = 2017 + k\}$ is an event study indicator. The terms α_f and $\lambda_{j(f),t}$ denote firm and state-by-year fixed effects, where $j(f)$ is the state associated with firm f . The inclusion of firm and state-by-year fixed effects implies that the difference-in-differences coefficients $\{\beta_k\}$ in equation (2) are identified by comparing within-employer changes in outcome y_{ft} . Specifically, these comparisons are made by observing changes in y_{ft} for firms with strict hours constraints, both before and after the reform's implementation, relative to firms with looser hours constraints within the same state.

To account for the possibility that more and less constrained firms evolved differently prior to the implementation of the reform, I detrend outcome y_{ft} .¹⁹ To do so, I first estimate a linear trend using only observations prior to the reform. Then, I subtract out the estimated linear time trend from the observations updated for the post-reform years.²⁰ This allows me to visualize any trend breaks in outcomes at the time of the

¹⁹For instance, Panel (a) of Figure 2 depicts differential pre-reform linear time trends between firms with loose constraints and those with strict constraints, specifically in terms of the creation of new part-time job positions.

²⁰See Bhuller et al. (2013), Goodman-Bacon (2018), Gruber et al. (2021), and Dustmann et al. (2022) for examples.

reform's implementation.

In equation (2), the coefficients $\{\beta_k\}$ when $k \geq 0$ identify the causal effect of the reform on firm-level outcomes, k years after the reform relative to the year before its implementation.²¹ The identification of $\{\beta_k\}$ hinges on the assumption that firms with high exposure to the reform (strict constraints on hours) would have had a similar evolution of outcomes to firms with weak exposure to the reform (loose constraints on hours), had the reform not been implemented. Estimates of the coefficients over the pre-reform period allow me to test whether this parallel trend assumption holds. This corresponds to the coefficient estimates β_k being statistically and economically indistinguishable from zero for the pre-reform years (i.e., for $k < 0$).

Although I report estimates on *linearly* detrended outcomes, the findings in Appendix E.1 highlight that my results continue to hold when I allow for non-linear violations of parallel trends; that is, when I allow the slope of the differential trend to change by no more than M across consecutive periods, as in Rambachan and Roth (2023).

4.4. The Effect of the 2017 Reform on Firm-Level Outcomes

This section delves into the 2017 reform's effect on a range of labor market outcomes, including the number of jobs, the share of part-time jobs, the creation of part-time jobs, employment, average weekly hours per job, and the total hours worked in the firm. Within the Brazilian context, where the option to hire workers off the books exists—as underscored by Ulyssea (2018)—the effect of the reform on formal job creation and labor allocation efficiency remains puzzling. While hiring informally may seem less costly due to the absence of costs associated with pension, health, and severance benefits, the threat of detection by labor authorities and consequent hefty fines are not trivial (Almeida and Carneiro, 2012). This setting offers a unique vantage point to explore the impacts of a flexibility shock in the formal labor market, especially in a context where informality is a prominent structural feature, underscoring a key novelty of the findings discussed in this section.

Job Creation. Figure 3, Panel (a) plots the event study coefficients $\{\beta_k\}$ from equation (2) when the dependent variable represents the log total number of jobs (spells) in the firm for a given year. The estimates over the pre-reform period (2014-2016) are very close to zero and economically insignificant²², providing suggestive evidence in favor of the common-trend assumption. Following the implementation of the reform in 2017, estimates show a consistent and almost linear increase in the log number of jobs within firms with strict hours constraints relative to firms with loose constraints. Table 3, column (1), shows that there is a substantial jump of 1.2% in the number of jobs at the year of the reform's implementation, and this figure increases to approximately 3.0% two years thereafter.

²¹I normalize β_{-1} to 0.

²²Due to the very large sample of the administrative data, the standard errors are particularly small. As a robustness check, I compute in Section 4.9 alternative confidence intervals accounting for potential differential pre-trends using the method proposed by Rambachan and Roth (2023) and find similar results, less precisely estimated.

To shed light on whether the reform's effect on the total number of jobs stems from an increase in all jobs, part-time (flexible) jobs, or full-time jobs, I estimate equation (2) on the number of part-time and full-time jobs as proportions of the total number of jobs in 2014.²³ Panels (b) and (c) of Figure 3 show the estimated coefficients $\{\beta_k\}$ corresponding to the proportion of part-time jobs with weekly hours up to 25 and those between 26 and 39 hours, respectively. The estimates reveal a significant increase in the number of part-time jobs. Specifically, two years post-reform, firms with strict hour constraints experienced a 2.4 percentage point increase in jobs below 25 hours per week and in jobs ranging from 26 to 39 hours per week, relative to firms with loose constraints on hours. Conversely, the analysis of full-time jobs, depicted in Figure 3, Panel (d), suggests that the reform did not lead to an increase in full-time jobs; if anything, there appears to be a decline in jobs with 40 hours or more per week.

Job Composition Effect. Consistent with the previous set of results, I find that the reform shifted the composition of jobs towards part-time work schedules. Panels (a), (b) and (c) of Figure 4 plot the event study coefficients $\{\beta_k\}$ from equation (2) when the dependent variable represents an indicator of at least one part-time job in the firm, the share of part-time jobs, and the share of new part-time jobs, respectively. These event study coefficients are relatively flat and close to zero in the years before the reform, providing evidence in favor of the common-trend assumption. Figure 4, Panel (b), shows that the share of part-time jobs increases at a linear rate in more (relative to less) constrained firms. At the year of the reform, there is a significant increase of about 0.6 percentage points in the share of part-time jobs. The effect increases to around 1.9 percentage points two years after the reform. Consistent with these findings, Panel (c) depicts that the reform significantly increased the creation of part-time jobs. The share of new part-time jobs increased by 2.1 percentage points in 2017, and this trend persists, increasing even further post-reform by up to 4.0 percentage points in 2019.

Employment Effect. To what extent are employers substituting full-time workers with part-time workers? I investigate whether the increase in part-time jobs translates into an increase in the monthly average number of employees within firms or if, conversely, firms are reallocating their workforce from full-time to part-time jobs. Panel (d) of Figure 4 illustrates that, up to one year post-reform, the effect on the log monthly average number of employees remains statistically insignificant, indicating no discernible effect.²⁴ However, two years post-reform, there is an uptick of 1.0% in firm employment. While the reform initially drives an increase in part-time jobs within firms with strict hour constraints, this increase does not immediately translate into an increase in the monthly average number of employees until two years post-reform. This result potentially reflects a substitution effect, wherein full-time workers are substituted by part-time workers, coupled with an increase in rotating contracts, which may explain the increase in yearly spells within firms but not in the count of

²³I compute the number of part-time jobs with weekly hours under 25, part-time jobs with hours ranging from 26 to 39 per week, and full-time jobs with at least 40 hours per week for each firm-year. Then, I normalize these counts by the total number of jobs in 2014 within each firm, thereby circumventing data loss due to zero counts in the part-time job variable.

²⁴Exhibits positive values but lacks statistical significance.

employees.

To summarize, the evidence shows that constrained firms (relative to less constrained ones) appear to have shifted their composition towards more flexible work arrangements after the implementation of the reform. Notably, the increase in the share of part-time jobs (and job creation) achieved through the reform appears to translate into a substitution of full-time workers with part-time workers and a delayed increase in firm-level size within firms with strict pre-reform constraints on hours, providing evidence that relaxing hiring constraints in the formal labor market can generate higher employment in the medium run.

4.5. Adjustment in the Average Weekly Hours per Job and Total Hours Worked

Having analyzed the reform's main effects on the number of jobs, workforce composition, and employment, I now turn to assess its effect on the job's intensive margin, specifically the average weekly hours per job, and the aggregate hours worked within firms.

Panel (a) of Figure 5 shows estimates from equation (2) using the log average weekly hours per job within the firm as an outcome. The event study coefficients exhibit a flat profile centered at zero in the pre-reform period. Upon the reform's implementation, the average weekly hours per job drop significantly by 0.6%. The effect linearly decreases by up to 1.7% two years after the reform. This result follows the effect on the composition of workers within the firm, wherein, after the reform, firms with strict constraints on hours are more likely to hire part-time workers than firms with looser constraints.

Bringing together the evidence of the reform's positive effect on the number of jobs (extensive margin) and its negative effect on the weekly hours per job (intensive margin), the analysis now shifts towards assessing its implications for the aggregate number of hours worked to determine whether these effects translate into a net change in overall firm employment. Figure 5, Panel (b), shows a significant decline in total hours worked within firms post-reform, indicating that the observed change is predominantly driven by the intensive margin, namely the substitution of full-time jobs with part-time jobs. This contraction in total work hours implies that hours of work and workers can be flexibly substituted within firms in a more flexible labor market, suggesting that in a rigid labor market, firms and workers facing strict hour constraints experience a misallocation of work hours, attributable to various factors, including the high costs associated with recruiting part-time labor.

4.6. Firm Heterogeneity

To shed light on the mechanisms underlying the reform effects on the number of jobs, workforce composition, part-time job creation, and employment, I estimate equation (2) separately across various firm subgroups. These subgroups are delineated based on pre-reform characteristics expected to impact the likelihood of hiring part-time workers: industry, firm size, number of establishments, and the fraction of female employees. Appendix B presents estimates of β_2 , two years post-reform, for a range of firm-level outcomes. The estimates in column (1) of Tables B.2 and B.3 reveal that the reform

had larger positive job effects on firms (i) in service industries (retail and accommodation and food services), (ii) with a relatively small pre-reform number of workers (5-19) and single-establishment, and (iv) with a high pre-reform fraction of female employees. Conversely, the estimates indicate that the reform led to a significant decline in the number of jobs in large firms (those with over 50 workers or multiple establishments), suggesting that large firms with strict hours constraints were negatively affected relative to less constrained firms.

What accounts for the observed heterogeneous treatment effects on job creation? Estimates provided in columns (2)-(4) of Tables B.2 and B.3 shed some light on the heterogeneous effects on the number of jobs, notably the negative job effect in larger firms. The negative job effect seems to be primarily driven by a reduction in full-time positions, despite a rise in part-time jobs following the reform. This contrasts with the increase in jobs experienced by smaller firms. Consistent with these results, the composition effect shown in columns (5) and (6) further underscore that the job effect varies according to the extent to which firms substitute full-time jobs with part-time job positions. While the positive and significant effect on the share of part-time and new part-time jobs is consistent across all samples, its magnitude is particularly small in large firms with at least 50 workers or with multiple establishments. The effect on the monthly average number of employees reflects a similar trend to that on the annual number of jobs, as column (7) shows. The substitution between job arrangements is additionally supported by estimates in column (9), which suggest a reduction or no change in the total number of hours worked within firms, particularly in service industries, where firms appear to have entirely shifted from full-time to part-time jobs without an overall change in total hours worked.

4.7. Market Labor Flexibility and Gender Differences

Several studies in the U.S. have analyzed gender differences in preferences for work arrangements, revealing that women tend to place higher value on flexible work hours compared to men (e.g., Wasserman, 2023; Wiswall and Zafar, 2017; Mas and Pallais, 2017; Goldin, 2015; Goldin and Katz, 2011; Maestas et al., 2023). Similar to gender patterns in the U.S., women in Brazil also exhibit greater inclination towards part-time jobs. In 2016, women represented 63% of the workforce engaged in jobs with weekly hours below 40, as opposed to 39% in jobs requiring more than 40 hours per week. Hence, the labor reform of 2017 is likely to have affected men and women differently, potentially through these existing employment preferences on labor supply.

To test this hypothesis, I estimate equation (2), where the dependent variable is now gender-specific, for instance, the number of jobs executed by women (or men) in the firm. Exposure to the reform is, as before, indicated by $\mathbb{1}\{\bar{\sigma}_f = 0\}$ which denotes whether the pre-reform standard deviation of hours across skill groups equals zero, i.e., the firm has strict hours constraints. For each outcome of interest, I estimate two separate regressions, by gender.

I summarize the key findings on gender heterogeneity in Table 4, where I report estimates of β_0 and β_2 from equation (2) across a range of job-related outcomes, including total number of jobs, part-time and full-time jobs, the share of part-time jobs, the share of new part-time jobs, the monthly average number of employees, log of av-

erage weekly hours per job, and total hours worked within the firm.²⁵ Figures 6 and 7 show that, in the years preceding the reform, estimates are close to zero and economically insignificant for the outlined outcomes. After the reform's implementation, Panel (a) of Figure 6 shows that the number of jobs for both, men and women, significantly increase at a nearly linear rate in firms with strict pre-reform hours constraints, relative to less constrained firms. The slope, however, is steeper for women throughout the period 2017-2019. At the year of the implementation, the number of female jobs increased by 0.9 percentage points relative to the year prior to the reform, whereas the number of men merely increased by 0.5 percentage points. Two years after the reform, the effect of the reform reached 2.5 percentage points for women, while the effect for men increased to 1.8 percentage points. Notably, Panels (b) and (c) of Figure 6 show that the increase in the number of jobs is stemming from an increase in the number of part-time jobs. I find no significant effect on the number of full-time jobs for both men and women (see panel (d) of Figure 6).

Additional insights can be gleaned from Figure 7. Specifically, Panels (b) and (c) provide further evidence that the differential increase in the number of female and male jobs subsequent to the reform can be attributed to the shift in the workforce composition towards part-time contracts. In the years following the reform's implementation, estimates are larger in magnitude for women compared to men. At the reform's implementation, the share of part-time jobs experienced a 1.0-percentage point increase for women and 0.6-percentage point increase for men. Moreover, the share of new part-time jobs for women and men increased by 2.6 and 1.5 percentage points, respectively. This gap between male and female jobs widened over time. Two years after the reform, the effect on the share of part-time jobs reached 2.7 percentage points for women and 1.6 percentage points for men, whereas the rate of part-time job creation experienced an increase of 5.1 percentage points for women and 2.6 percentage points for men.

While the reform has a clear, positive effect on number of jobs and workforce composition within the firm for both, men and women, Panel (d) of Figure 7 shows that the reform only led to a significant effect on female employment, measured as the average number of female employees with respect to the average number of workers in 2014. Two years post-reform, firms with strict hours constraints experienced a 2.7 percentage points increase in female employment, relative to firms with loose constraints on hours. In contrast, the estimates on male employment depict a null effect, suggesting that the reform did not lead to a change in the monthly average number of male employees.

Figure 8, Panel (a), illustrates that the shift towards more flexible work schedules, particularly part-time jobs, results in a significant, nearly linear decrease in the average weekly hours per job for both genders. However, this change is marked by gender disparities, with women experiencing a notably sharper reduction, at 2.3%, compared to 1.6% for men two years after the reform. In line with the employment evidence, Panel (b) of Figure 8 presents contrasting estimates for total hours worked within firms by

²⁵Since not all of the firms in the sample hire men and women, I normalize the number of jobs, part-time jobs with workweek below 25h/week, part-time jobs with workweek ranging from 26 to 39h/week, and full-time jobs above 40h/week in terms of the total number of jobs in the firm in 2014. Similarly, I normalize the monthly average number of female and male employees by the average number of employees in the firm in 2014, and the total hours worked in the firm by men and women by the total hours worked in the firm in 2014.

gender. Women in firms subject to pre-reform strict hours constraints experienced an increase of 2.2 percentage points in their total hours worked, relative to those in firms with looser constraints. Conversely, men in more constrained firms (compared to less constrained ones) witnessed a reduction of 2.4 percentage points in their total hours worked two years post-reform. Given that the average number of women within the average firm represented 62% of the male workforce in 2016, the estimates on the average number of employees and the total hours worked within a firm imply that the reform's introduction of flexible work arrangements might have progressively narrowed the employment gender gap over time.

Firm Heterogeneity. Demand-side factors play a pivotal role in shaping working hours and work schedules (Labanca and Pozzoli, 2022; Lachowska et al., 2025). Specifically, the industry factor can wield influence over the supply of work hours, given that industries often operate with varying technologies. For instance, in 2016 the share of part-time jobs with weekly hours below 25 was 34% in the educational services industry, in contrast, the construction and manufacturing industries had a part-time job participation of approximately 1% (see Table D.1 in the Appendix). Motivated by this, I test whether there is heterogeneity in the effect of the reform across industries²⁶, firm sizes, and the pre-reform proportion of women within firms.

Figure 9 illustrates estimates of β_2 from equation (2), obtained separately for various firm subgroups, when the outcome variable is the proportion of jobs held by men and women, the share of new part-time jobs (for both genders), the proportion of female and male employees, and the proportion of total hours worked within the firm by men and women. Post-reform, the empirical findings indicate that the overall positive impact of the reform on the number of jobs for both genders primarily stems from firms in the service industries, small and medium-sized firms, or those with a single establishment (see Panel (a) of Figure 9).

Of particular note are the differential effects of the reform by gender on part-time job creation, proportion of female and male employees, and proportion of total hours worked within the firm by men and women (see Panel (b), (c), and (d) of Figure 9). Significant gender disparities are noticeable, particularly in the retail and manufacturing sectors, as well as among small firms. These disparities correspond to higher pre-reform levels of part-time jobs held by women compared to men. For instance, one year prior to the reform, the share of women working part-time in the retail and manufacturing industries was 3.0 and 5.0 percentage points higher, respectively, than the corresponding share for men in firms with loose hours constraints. This difference, however, is not observed in the accommodation and food services sector; if anything, the share of part-time jobs was slightly higher for men by 0.9 percentage points. The absence of such a gender gap in this sector indicates that the reform had a similar effect on part-time work (and employment) for men and women, possibly due to the nature of the jobs in this industry.

²⁶In this paper, I focus on three industries: retail, accommodation and food, and manufacturing, which are the most prevalent industries in the firm-level data. In the balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform, the retail, food and accommodation, and manufacturing sectors respectively represent the 32%, 16% and 10% of the sample.

Surprisingly, the reform's effect is notably positive and larger for women in terms of employment (both in terms of average number of employees and total hours worked within the firm), while for men, the reform appears to have a negative or null effect in constrained firms compared to those with looser hour constraints. Particularly striking are the near-perfect substitution of men's total hours worked with women's total hours worked within the firm. Specifically, the coefficients β_2 from equation (2), obtained separately for men and women on the proportion of total hours worked, exhibit similar magnitude but opposite signs in firms (i) in the retail and manufacturing industries and (ii) with relatively small pre-reform employee numbers (5-19) and single establishment. To put these estimates into perspective, the reform led to an increase of 1.3 percentage points in women's total hours worked within firms in the retail sector, whereas the corresponding effect on men's total hours worked was -2.1 percentage points. As this outcome variable represents a proportion of the hours worked within the firm, it appears that constrained firms are reallocating hours among their workers, assigning more hours to women and fewer hours to men. These contrasting effects seem to stem from the negative impact on full-time job positions and a substantially higher positive effect on female part-time jobs relative to male part-time jobs (refer to Figure C.1 in the Appendix for estimates on the effect on retail jobs).

These findings align with evidence suggesting that the reform's effect tends to be positive and more pronounced for women, who generally exhibit a greater inclination towards part-time jobs and flexible work arrangements. However, in industries characterized by greater gender equality in part-time job hiring, such differences in terms of job creation and employment are not observed. For a detailed examination of the reform's heterogeneous effect on jobs, part-time job creation, and monthly average number of employees in retail, accommodation and food, and manufacturing industries, refer to Appendix C.

Gender Differences within Occupation. To better understand for which subgroups of women the composition and employment effects are particularly pronounced compared to men, I investigate whether the effects vary across occupations. To address this question, I estimate equation (2) with outcome variables that are occupation and gender specific. The estimates of β_2 are shown in Figure 10, Panel (a), when considering the proportion of jobs in terms of the 2014 count as the outcome. Two years after the reform's implementation, the estimates on both female and male jobs are particularly large and similar for (i) managerial positions (e.g., manager and director); (ii) high-skill professionals (e.g., researchers, teachers, doctors, nurse, engineers, technicians, architects, mathematicians, and statisticians); and (iii) low-skill white collar jobs (e.g., cashier, receptionist, secretary, and library assistant). Remarkably, for pink-collar occupations (e.g., retail clerks, salespersons), the reform's impact on the number of jobs is more pronounced for women in comparison to men. The reform led to a 1.1 percentage points increase in female jobs within service and sales positions, whereas the corresponding effect for men was 0.6 percentage points. I observe small effects on the number of jobs within blue-collar job categories.

Estimates regarding part-time job creation, depicted in Panel (b) of Figure 10, suggest that the impact on job creation partly reflects compositional changes in the workforce. Specifically, the reform's effect on the proportion of new part-time jobs is larger

and statistically significant within white-collar non-managerial positions for both men and women. In high-skilled professional and pink-collar occupations, the effect exhibits a gender-based heterogeneity: two years post-reform, the share of new part-time jobs increased by 7.2 and 2.9 percentage points, respectively, for women, while men experienced an increase of 4.7 and 1.9 percentage points, respectively.

The findings presented in Figure 10, Panel (c), further highlight that post-reform, firms subject to strict hours constraints experienced an increase in employment within white-collar and pink-collar job occupations compared to firms with looser constraints. However, significant gender differences emerge within pink-collar occupations: two years post-reform, the proportion of female employees increased by 3.1 percentage points, whereas the increase for male employees was only 1.1 percentage points. In terms of total hours worked within the firm, Figure 10, Panel (d), reveals that the reform led to a modest yet statistically significant increase in white-collar non-managerial positions. Male employees in craft related trades, in contrast, experienced a decline in total hours worked in response to the reform. For all other occupations, the observed effects appear to be relatively negligible.

These findings underscore the nuanced impacts of the reform, providing evidence on its differential effects across gender and occupations. A possible explanation for the more pronounced increase in women’s employment and part-time jobs, specifically, in service and sales positions (and retail industry in general), is that the reform improved the match between firms’ labor demand and female workers’ labor supply. In the aftermath of the reform’s implementation, offering part-time positions within the formal labor market incurs lower costs.²⁷ In fact, after the reform, firms characterized by strict pre-reform constraints exhibited an increase in their demand for low-hour jobs, compared to less constrained firms (as shown in Panels (b) and (c) of Figure 3). Moreover, men and women differ in their observed work hours. For instance, in 2016, women comprised 63% of the workforce engaged in roles requiring weekly hours below 40, in contrast to 39% in positions requiring more than 40 hours per week. Additionally, the share of part-time jobs held by women (9.5%) in firms with loose constraints ($\overline{\sigma}_f > 0$) surpasses the corresponding share held by men (7.2%) one year prior to the reform. Given the disparities in the preferences for work schedules, the reform notably increased the likelihood that women, who exhibit stronger preferences for part-time work, find a job that aligns with their preferences. Therefore, the combination of reform-induced changes in labor demand and the initial differences in labor supply behavior across genders appears to have been instrumental in shaping the observed gender-specific effects of the reform.

4.8. Transition to the Formal Labor Market

How did the reform impact employment of individuals outside the formal labor market? Empirical findings thus far indicate a discernible rise in job opportunities and overall employment within firms with strict pre-reform hours constraints, mainly explained by a shift in job composition towards part-time jobs. To elucidate the potential mechanisms underlying this observed increase in job creation, I investigate whether

²⁷Furthermore, firms encounter less risks to legal actions and lawsuits (Corbi et al., 2022).

the reform affected employment of individuals newly entering the formal labor market.

Figure 11 presents estimates from equation (2), with the outcome variable denoting the proportion of employees newly entering the formal labor market since 2013.²⁸ After the implementation of the reform, the proportion of new formal workers increased linearly for both male and female workers. However, consistent with the differential gender impacts on job creation, employment, and workforce composition, a notably larger effect is observed among female workers. Specifically, the effect at enactment was 0.9 percentage points, escalating to around 2.0 percentage points two years post-implementation for women aged 18 to 64. Meanwhile, the proportion of newly formalized male employees increased by 0.5 percentage points in the year of enactment, experiencing a subsequent increase to 1.1 percentage points in the medium run. Even upon restricting the sample to employees aged above 35, albeit with smaller point estimates, the differential gender effect persists.

These findings suggest that the positive effect on job creation and employment partly arises from recruiting individuals who were not engaged with the formal labor market prior to the reform. However, it remains unclear whether firms are predominantly absorbing informal workers or those previously unemployed, a question further addressed in Section 5.

4.9. Robustness analysis

In this section, I delve into the validity of the parallel trend identification assumption and examine the robustness of the firm-level results concerning changes in the sample, mean reversion, and the definition of exposure to the reform. The key findings on the reform's effects on the number of jobs, employment, and workforce composition are robust to a series of potential concerns.

Parallel Trend Assumption. Identification of the impact of the 2017 labor reform hinges on the assumption that, had the reform not been implemented, outcomes of firms with strict hours constraints would have evolved similarly as firms with loose hours constraints. First, I assess this assumption by looking at the event study coefficients β_k for $k < 0$ from the two-way fixed-effects model in equation (2), where I adjust for the extrapolation of a linear trend from the pre-reform periods. For all outcomes, the estimates are relatively flat and very small in magnitude, relative to the post-reform estimates. Second, I apply the procedure proposed by Rambachan and Roth (2023) to check the robustness of the results to less restrictive assumptions about the class of possible differences in pre-existing trends. This allows me to compute alternative standard errors that account for: (i) the fact that firms with different exposure to the reform might experience differential trends and (ii) the fact that these trends are unknown and estimated. I compute alternative confidence intervals for the 2018 estimates (one year after the reform was implemented). The new 95% confidence intervals are shown in Figure E.1, Section E of the Appendix. Regarding the number of jobs at

²⁸Identification of individuals newly entering the formal labor market relies on indicator variables derived from their presence in the RAIS dataset for the years 2013-2019. For example, individuals appearing solely in 2019 are classified as having entered the formal market in 2019.

the firm level, as well as the proportion of part-time jobs and total hours worked, the estimated effects of the reform persist in both magnitude and statistical significance.

Firm sample. The main findings are derived from an analysis conducted across a balanced panel of firms that had at least 5 workers during the pre-reform period. However, no constraints were imposed on the number of skill groups, which, according to the reform exposure measure in Section 4.1, delineate whether firms were subject to strict or loose hour constraints. It is possible that firms predominantly hired a limited number (or even a single) skill group, which would imply a negligible shift in their workforce composition after the reform’s implementation if these workers exhibit consistent preferences for full-time employment. To address this concern, an alternative and more restrictive sample selection criterion is applied, considering only those firms employing at least 3 skill groups in the pre-reform period. Appendix E.2 shows that for all firm-level outcomes, the estimates pertaining to $\{\beta_k\}$ remain close and economically similar to those previously estimated.

Mean Reversion. Mean reversion is unlikely to be driving the main results. First, I compute the measure of hours constraints across skill groups over a three-year pre-reform period (2014-2016), anchoring the identification of effects on a source of variation less prone to temporary shocks. Additionally, I exclude small firms from the sample (those with fewer than 5 workers). Second, to provide additional evidence on the robustness of the findings, a difference-in-difference regression is conducted over two-year rolling periods, leveraging the exposure measure outlined in Section 4.1, and defining the second year as the treated year. Figure E.5 presents the estimates on the log number of jobs, the proportion of part-time jobs, and the log total hours worked within the firm. Each estimate is computed in a separate regression. For instance, the estimate in 2016 stems from a difference-in-difference regression estimated on years 2015 and 2016, comparing the outcomes of firms subject to strict hours constraints ($\overline{\sigma}_f = 0$) with those facing looser constraints ($\overline{\sigma}_f > 0$). While estimates for the number of jobs and the share of part-time jobs are statistically significant, they remain small in magnitude relative to those observed in the post-reform implementation. Conversely, pre-reform estimates for total hours worked are statistically indistinguishable from zero, with post-reform negative and significant estimates. Accordingly, any potential bias attributable to mean reversion is expected to be negligible compared to the estimated impact of the reform.

Definition of Exposure to the Reform. In my main analysis, I gauge constraints on working hours by using the standard deviation of hours across skill groups within the firm. As a robustness check, I replicate the analysis using two alternative proxies for a firm’s exposure to the labor reform. First, I compute the standard deviation of hours across all workers within a firm, regardless of their skills, and then I generate an indicator to distinguish between firms with a standard deviation of zero and those with a value greater than zero. The estimates of $\{\beta_k\}$ obtained from equation (2) using this alternative metric are documented in Appendix E.4. This alternative measure exhibits a correlation with my baseline metric, thereby yielding similar and robust results concerning the impact of the reform on the number of jobs, share of part-time jobs, and

total hours worked within the firm.

Second, I use an industry-state level measure of exposure to the reform. Given that labor laws primarily apply to entities within the formal sector²⁹, industries characterized by a high degree of formalization are more likely to (i) comply with regulations outlined in the CLT or labor laws, and (ii) adhere to changes introduced by the 2017 reform. Thus, the second alternative exposure measure represents the formality rate (i.e., the fraction of formal workers) computed for each 1-digit industry and state in Brazil, using data from the PNAD³⁰ spanning the pre-reform period. Each firm is subsequently assigned to an industry-state pair with its corresponding pre-reform formality rate³¹ to estimate the following event study regression:

$$(3) \quad y_{ft} = \alpha_f + \lambda_{j(f),t} + \eta_{j(f),c(f)} + \sum_{\substack{k=-3 \\ k \neq -1}}^{k=2} \delta_k \times \text{FormalityRate}_{c(f),j(f)} \times \mathbb{1}\{t = 2017 + k\} + \varepsilon_{ft}$$

where $\text{FormalityRate}_{c(f),j(f)}$ denotes the average fraction of formal workers in industry c and state j of firm f over the period 2012-2016.³² The terms $\eta_{j(f),c(f)}$ represent industry-by-state fixed effects. δ_k denotes the effect of FormalityRate transitioning from 0% to 100% k years after the reform's implementation, or placebo parameters in the pre-reform years. Figure E.9 shows estimates $\{\delta_k\}$ on the log number of jobs and the share of part-time jobs. The results demonstrate a positive and statistically significant effect of the reform, corroborating a discernible increase in the number of job opportunities and a shift in the workforce composition of highly exposed firms towards more flexible, part-time jobs.

5. The Effect of Labor Market Flexibility: State-Level Analysis

The findings in the last section show that the 2017 labor reform yielded a significant increase in the proportion of part-time jobs and the generation of job opportunities, particularly notable among women and individuals newly entering the formal labor market. The labor reform thereby played a pivotal role in boosting employment in firms with strict hours constraints relative to those with loose constraints, by allowing the former firms to hire workers with flexible work arrangements. Nonetheless, the overall effect of the reform remains uncertain. The reform's impact on firm-level employment could be explained by: (i) the absorption of workers from the informal labor market; or (ii) the hiring of unemployed workers.

To shed light on the aggregate effects of the reform, I provide complementary evidence by leveraging variation in exposure to the labor reform across states. An advantage of this state-level analysis, in comparison to the firm-level approach, is that it

²⁹Comprising firms registered with the tax authority and workers documented by these firms in administrative data, such as RAIS.

³⁰See Section 3.2 for details.

³¹In the PNAD, there is no distinct category specifically designated for the Retail Sector. Instead, Wholesale and Retail are grouped together in the survey. Consequently, I assign the Wholesale-Retail formality rate at the state level to these respective industries in RAIS.

³²The distribution of FormalityRate is depicted in Figure E.8.

provides broader insights on the reform's employment effects. In particular, this analysis measures the effects in terms of aggregate formal employment and allows me to test whether the effect is driven by the recruitment of informal or unemployed workers. For instance, if firms primarily respond to the labor reform by hiring informal workers, without benefiting those who were unemployed at the time of the reform, the state-level approach would uncover no effect on unemployment rates, which the firm-level approach fails to detect.

5.1. State-level Exposure to the 2017 Reform

For each of the 27 Brazilian states, I compute a continuous measure for its exposure to the reform, based on the share of jobs offered by firms with strict hours constraints at the state level:

$$\text{Share}_{jt} = \frac{\sum_{f=1}^{F_{jt}} N_{ft} \cdot \mathbb{1}\{\overline{\sigma}_f = 0\}}{\sum_{f=1}^{F_{jt}} N_{ft}}$$

where F_{jt} and N_{ft} are the number of firms in state j and the number of jobs held within firm f at year t . As previously defined, $\overline{\sigma}_f$ denotes the average standard deviation of hours across skill groups over three pre-reform years (2014-2016) of firm f . Thus, the measure Share_{jt} reflects the fraction of jobs offered by firms with strict hours constraints in state j at year t .

I average the yearly share of jobs offered by firms with strict hours constraints. This yields a time-invariant measure for each state:

$$(4) \quad \overline{\text{Share}}_j = \frac{1}{3} \cdot \sum_{t=2014}^{2016} \text{Share}_{jt}$$

The distribution of the exposure measure varies across states, with an average value of 25.6%. Paraíba is the state with the highest exposure (29.7%), whereas Rio Grande do Sul presents the lowest exposure (12.1%). Figure 12 shows a map of the 27 states, wherein darker colors indicate a stronger exposure to the 2017 reform based on the exposure measure in equation (4). The figure highlights that states in Northeast Brazil were more heavily exposed to the reform than states in South Brazil.

5.2. The Effect of the 2017 Reform on Unemployment and Informality

To examine the impacts of the reform on unemployment and informality, I estimate two distinct sets of specifications employing (i) an individual-level panel and (ii) a state-level panel.

5.2.1. Individual-level Panel

Using the individual panel data from PNAD and focusing on individuals surveyed both before and after the implementation of the reform, I investigate whether residents in states highly exposed to the reform exhibit an increased likelihood of employment and reduced informal employment compared to those in less exposed states. To do

so, I estimate an event-study regression of the following type over a sample of 103,723 individuals:

$$(5) \quad y_{it} = \alpha_i + \phi' X_{it} + \lambda_t + \sum_{\substack{q=-3 \\ k \neq -1}}^{q=2} \gamma_q \times \overline{\text{Share}_{j(i)}} \times \mathbb{1}\{t = 2017\text{Q}3 + q\} + v_{it}$$

where y_{it} is an indicator variable of the employment status or informality status of individual i , residing in state $j(i)$ in quarter t .³³ The vector X_{it} contains indicators for individual i 's high school completion, age, number of household members, and head of household status. The vector α_i denotes individual fixed effects, and λ_t accounts for quarter fixed effects. The coefficients $\{\gamma_q\}$ trace out how individual outcomes in states more affected by the labor reform evolve in comparison to less affected states, relative to the pre-reform quarter 2017Q3. Specifically, γ_q denotes the effect of $\text{Share}_{j(i)}$ transitioning from 0% to 100% q quarters after the reform's implementation, or placebo parameters in the pre-reform quarters. This analysis hinges on the assumption that if the reform had not been implemented, individual outcomes in both more and less exposed states would have followed a similar trajectory. To gauge the validity of the parallel-trend assumption, I investigate whether the estimates of $\{\gamma_q\}$ for the pre-reform quarters (i.e., for $q < 0$) are statistically and economically indistinguishable from zero. The standard errors in (5) and in all subsequent regressions are clustered at the state level.

Employment Effects. The estimates of $\{\gamma_q\}$ are depicted in Figure 13 when employment status is the outcome variable. Panel (a) of Figure 13 shows that the effect on the probability of employment is statistically indistinguishable from zero. Post-reform, the probability of being employed exhibits a consistent trend similar to that observed in the pre-reform period, with a point estimate that is negative but statistically insignificant. This relatively flat trend is consistent across genders, as evidenced by Panels (b) and (c) of Figure 13. These findings reject any short-term changes in employment. Consequently, the observed increase in job creation and the share of part-time jobs resulting from the reform do not seem to translate into a significant increase in the probability of employment.

Informality Effects. The reform notably increased the generation of formal employment, particularly part-time jobs, as evidenced in Section 4. However, the dynamics of employment, whether formal or informal, appear largely unaffected post-reform. Who is filling these newly created positions? I find evidence supporting the hypothesis that formal firms are filling those vacancies with informal workers. Figure 14, Panel (a), depicts the event study coefficients $\{\gamma_q\}$ from equation (5) when the dependent variable denotes worker's informality status. In the pre-reform period, the estimates are small and statistically indistinguishable from zero. Following the reform's enactment, the estimates reveal a notable decline in the likelihood of informal employment within the private sector. I find that 1 percentage point increase in the fraction of jobs offered by

³³If y_{it} denotes employment status, it takes the value of 1 if the individual is employed (either formally or informally) and 0 otherwise. When y_{it} denotes informality status, it takes the value of 1 if the worker-employed in the private sector—is informal and 0 if the worker is formal.

firms with strict hours constraints leads to a decrease in the probability of informal employment by 0.06 percentage points in the one-to-three quarters following the reform. Moreover, consistent with the empirical findings detailed in Section 4.8, Panels (b) and (c) of Figure 14 show that the reform led to a decrease in the likelihood of informal employment among employed women, whereas no statistically significant reduction is observed among male workers (although the estimates are negative). These findings confirm that the reform primarily impacted women, particularly those engaged in informal employment, facilitating their transition to a flexible, part-time job in the formal labor market.

Existing empirical research suggests that women, particularly mothers, exhibit a greater inclination towards seeking labor market flexibility or engaging in informal work to navigate inflexible work schedules (see Berniell et al., 2021). I empirically evaluate this hypothesis and examine potential heterogeneity in the informality effect based on parenthood status. Consistent with prior findings, Appendix F, Panels (a) and (c) of Figure F.10 show that the estimates of $\{\gamma_q\}$ in equation (5) exhibit greater magnitude in the short term for women with at least one child compared to their childless counterparts. This suggests that the reduction in the likelihood of informal employment among women is primarily driven by those with children. Conversely, I find no discernible effect on the probability of working informally among men.

5.2.2. State-level Panel

I now relate the continuous measure for the exposure of state j to the 2017 reform given by equation (4) to state-level outcomes. Specifically, I estimate event-study regressions of the following type:

$$(6) \quad Y_{jt} = A_j + \Phi' Z_{jt} + \Lambda_t + \sum_{\substack{q=-7 \\ k \neq -1}}^{q=9} \Gamma_q \times \overline{\text{Share}}_j \times \mathbb{1}\{t = 2017Q3 + q\} + V_{jt}$$

where Y_{jt} denotes the outcome of interest—either unemployment rate or the share of informal workers—in state j during quarter t . The term A_j denotes state fixed effects, and Λ_t accounts for quarter fixed effects. The vector Z_{jt} contains time-varying controls such as the fraction of individuals living in urban areas, the fraction of female, the fraction of white individuals, the fraction of illiterate, the fraction of individuals below 25, and the fraction of individuals with high school education in state j . Similar to the previous analysis, the event study coefficients $\{\Gamma_q\}$ in equation (6) over the pre-reform period allows me to test the parallel trend assumption. For the post-reform period, estimates yield the causal effect of the reform on state-level outcomes. Following the approach of the empirical analysis in Section 4.3, I detrend the outcomes of interest to address the possibility that highly and barely exposed states evolved differentially prior to the reform's implementation. I regress outcome Y_{jt} on a state specific linear time trend $\theta_j \cdot t$ using pre-reform data from 2016Q1 to 2017Q3. Then, I run specification (6) on the full-period using the residualized outcome $Y_{jt} - \hat{\theta}_j \cdot t$. This allows me to detect any trend breaks in outcomes at the time of the reform's implementation.

Unemployment Effects. Considering the observed increase in employment within firms with strict hours constraints, in comparison to those with looser constraints, after the reform's implementation, one may wonder what the overall effect on aggregate unemployment is. Figure 15 shows estimates from equation (6) using as outcome the detrended unemployment rate (both total and disaggregated by gender) at the state level. The event study coefficients reveal that the reform's effect on unemployment is statistically indistinguishable from zero across for both total and female unemployment rates. In the quarters following the reform's implementation, Figure 15, Panel (a), shows that the total unemployment rate maintains a relatively flat profile, with point estimates that are negative yet statistically insignificant. This trend is mirrored in the unemployment rate among women (see Panel (b) of Figure 15). Conversely, it appears that the reform-induced flexibility shock contributes to a decrease in male unemployment in the medium-run (see Panel (c) of Figure 15). While the relative increase in employment in more (relative to less) constrained firms fails to yield a significant decrease in total unemployment, it does manifest as a reduction in unemployment among men. One plausible interpretation for the former pattern is that firms are predominantly hiring informal workers rather than individuals previously unemployed at the time of the reform's implementation, thereby resulting in no discernible effect on aggregate unemployment.

Informality Effects. Figure 16 provides the corresponding analysis for the informality effects of the reform, where I plot the coefficient estimates of $\{\Gamma_q\}$ from regression equation (6), using the detrended share of informal workers in the state as the dependent variable. Consistent with the individual-panel results, Figure 16, Panel (a), illustrates that the pre-reform conditional informality rates in both heavily exposed and less exposed states were similar. The estimates of $\{\Gamma_q\}$ when $q < 0$ exhibit a relatively flat profile centered around zero. After the reform's implementation, the informality rate fell at a nearly linear rate throughout the entire 2017Q4-2019Q4 period. By the end of 2019, a 1 percentage point increase in the fraction of jobs offered by firms with strict hours constraints led to a decrease in informality by 0.39 percentage points.

To better understand the decline in the aggregate informality rate, I investigate gender disparities. I estimate equation (6) separately for women's and men's informality rates. Panels (b) and (c) of Figure 16 illustrate that the reform led to a significant decline in informality rates for both genders, albeit with a delayed effect observed in the informality rate among women compared to men. This delay may be attributed to women in the formal sector gradually transitioning to part-time positions or flexible schedules. Given that women tend to exhibit a stronger inclination towards flexible work arrangements, such as part-time employment, it is plausible that stringent labor regulations preceding the reform may have led to women being matched to jobs that did not align with their preferences. Consequently, women may have been compelled to take full-time roles or resort to informal employment. This scenario characterized by female labor misallocation, however, changed with the reform easing certain hiring constraints. In response to the reform, women may have reallocated within the formal sector towards part-time jobs, as depicted in Figure 7. This shift within the female labor force potentially resulted in a decrease in job vacancies within the formal sector, displacing opportunities for their informal sector counterparts in the short term, despite

the lower probability of being informal within the private sector for women following the reform (see Panel (b) of Figure 14).

6. Conclusions

This paper provides an assessment of how firms and workers respond to the introduction of flexible labor practices within the context of a concomitant flexible market, namely the informal sector. I analyze the Brazilian labor reform implemented in 2017 that relaxed several legal constraints on hiring workers with flexible work schedules, such as part-time jobs and freelance work. The empirical analysis highlights several key findings on the effects of this reform on formal employment, job creation, and the allocation of labor across genders and sectors.

First, the 2017 reform achieved greater flexibility and facilitated adjustments in the labor market. Following the reform's implementation, firms are more likely to change their worker composition in response to eased constraints on working hours. I find that the reform significantly increased the share of part-time jobs. Although it led to a decline in the number of full-time jobs, I observe a significant increase in employment after the reform. These findings are consistent with the idea of substitution of full-time workers with part-time workers, particularly, within firms with strict hours constraints that are more sensitive to a flexibility shock relative to the already flexible firms. This shift towards more flexible work arrangements aligns with the reform's goal and demonstrates that firms are responsive to changes in labor market flexibility.

Second, the gender-based analysis reveals a nuanced story of how labor market flexibility interacts with gender. While both men and women benefited from the reform in terms of jobs and composition effects, the positive impact on female employment was more prominent. Specifically, I find that small firms in service industries, characterized by a high share of female retail employees, experienced the most pronounced effects in terms of increased employment and shifts towards part-time work. These findings highlight the importance of flexible work arrangements in fostering female labor force participation and suggests that addressing gender-specific preferences for work hours can contribute to reduce labor misallocation.

Third, the state-level analysis conducted in this paper reveals that although the reform did not lead to a reduction in unemployment rates, it did precipitate a decline in informality. After the reform's implementation, I observe a decrease in the likelihood of working informally in the private sector and a decline in state-level informality rates. Further analysis focusing on gender reveals that the reduction in informality is predominantly driven by women, especially mothers, transitioning from informal to formal employment within the private sector in the short term. This trend is followed by both women and men in the state-level economy in the medium term. These findings provide evidence regarding the causal effect of labor market flexibility, especially in developing countries, where it may foster a higher transition from informal to formal employment, as opposed to reducing unemployment.

In summary, the evidence presented in this paper calls for renewed attention to the role of flexibility-enhancing labor market reforms in shaping employment allocation and reducing informality. The findings highlight labor market rigidities as a key source of misallocation, with particularly pronounced consequences for women's la-

bor market outcomes. At the same time, whether greater flexibility in the formal sector translates into broader welfare gains remains an open question, calling for further analysis of earnings dynamics and inequality.

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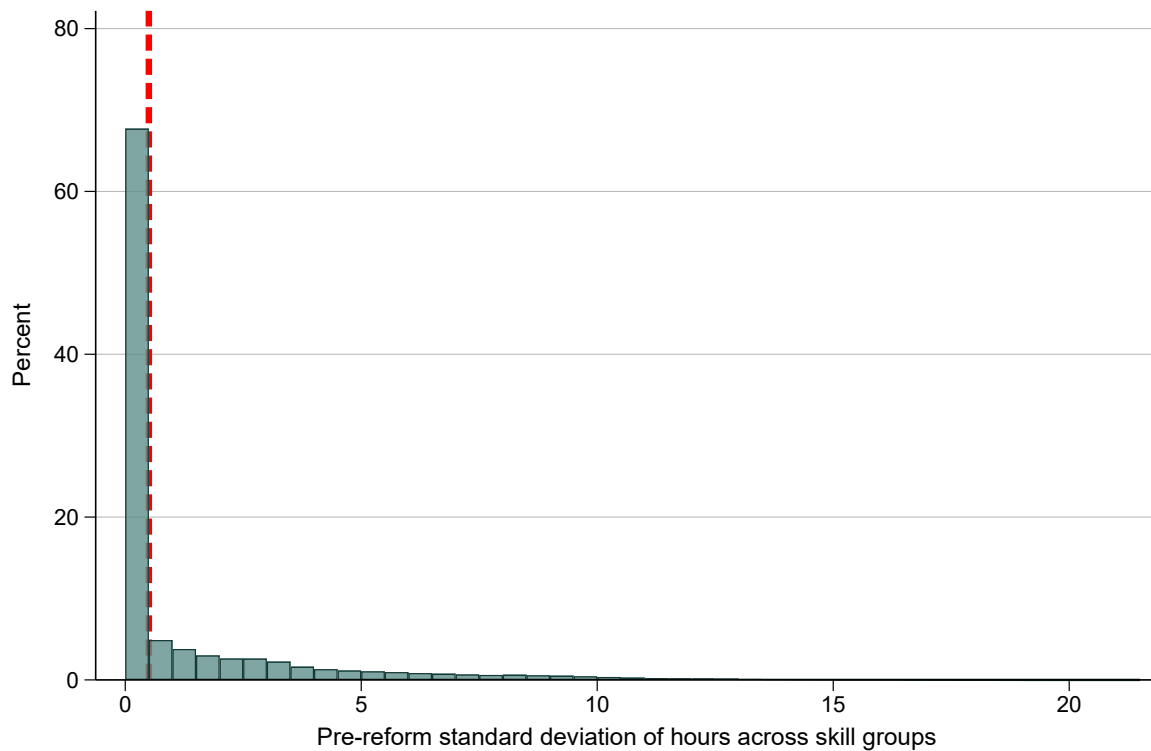
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Figures

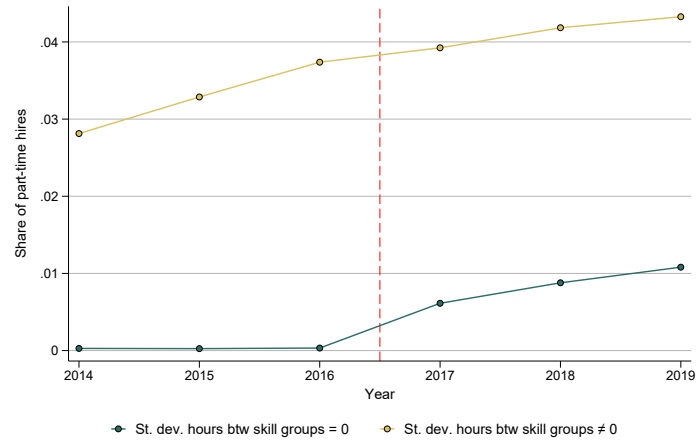
Figure 1. Distribution of Pre-Reform Standard Deviation of Hours across Skill Groups



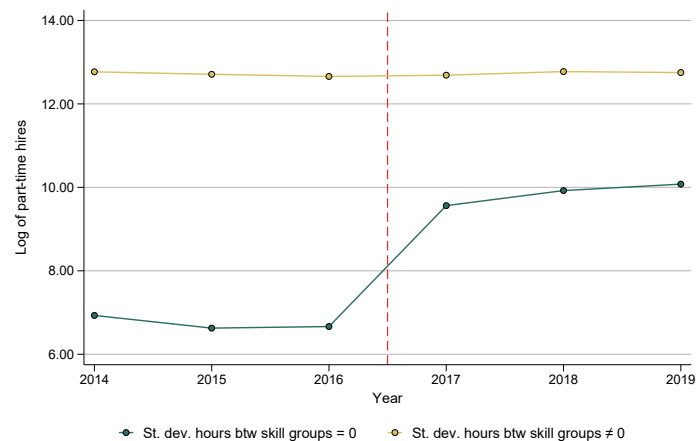
Notes: This figure plots the distribution of hours constraints, measured as the average of the standard deviation of hours across skill groups in a firm over the three pre-reform years (2014-2016). I construct a measure of skill given by the intersection of the education and occupation categories. The education categories are: (1) less than primary education, (2) complete primary education, (3) complete secondary education, and (4) post-secondary education. The occupation categories are: (1) management occupation, (2) professionals, (3) technicians and associate professionals, (4) clerical support workers, (5) service and sales workers, (6) skilled agricultural, forestry and fishery workers, (7) craft related trade workers, (8) plant and machine operators, and assemblers, and (9) elementary occupations. This figure shows the distribution of hours constraints among jobs held by individuals aged 18 to 64 in the private sector, using a balanced panel of firms spanning from 2014 to 2019. Firms with pre-reform size smaller than 5 are excluded.

Figure 2. Evolution of the Share of New Part-Time Jobs and the Number of New Part-Time and Full-Time Jobs

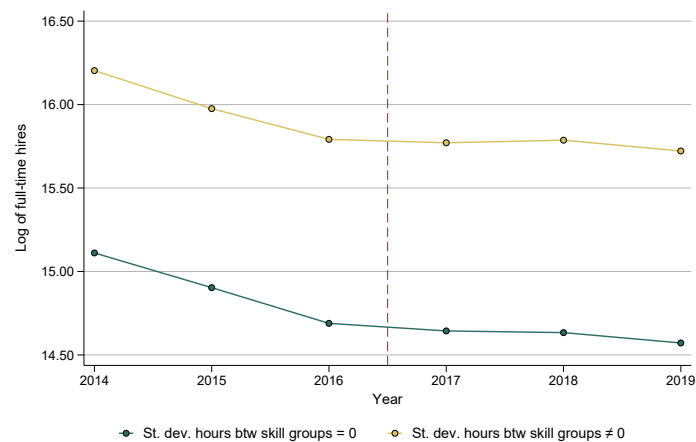
(a) Share of Part-Time Hires ($h \leq 25$)



(b) Log Number of Part-Time Hires ($h \leq 25$)

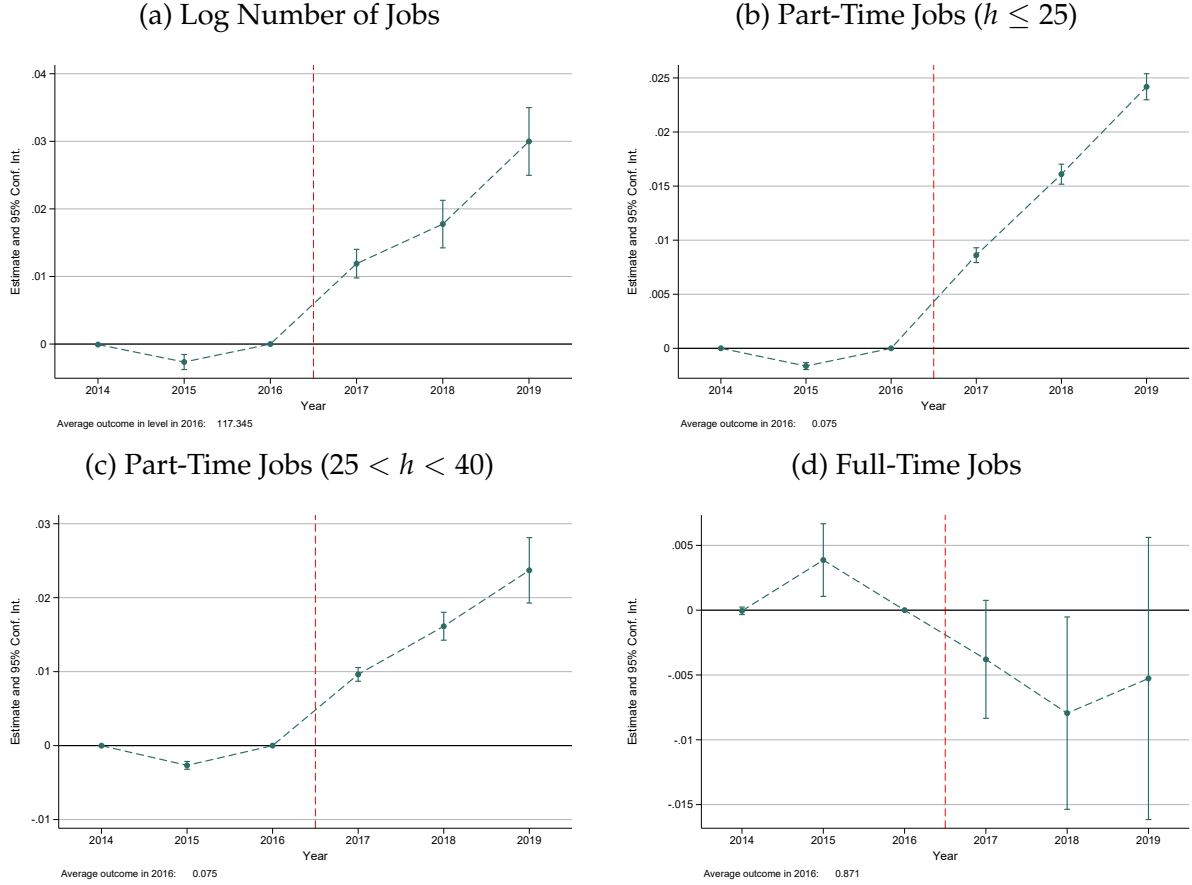


(c) Log Number of Full-Time Hires ($h \geq 40$)



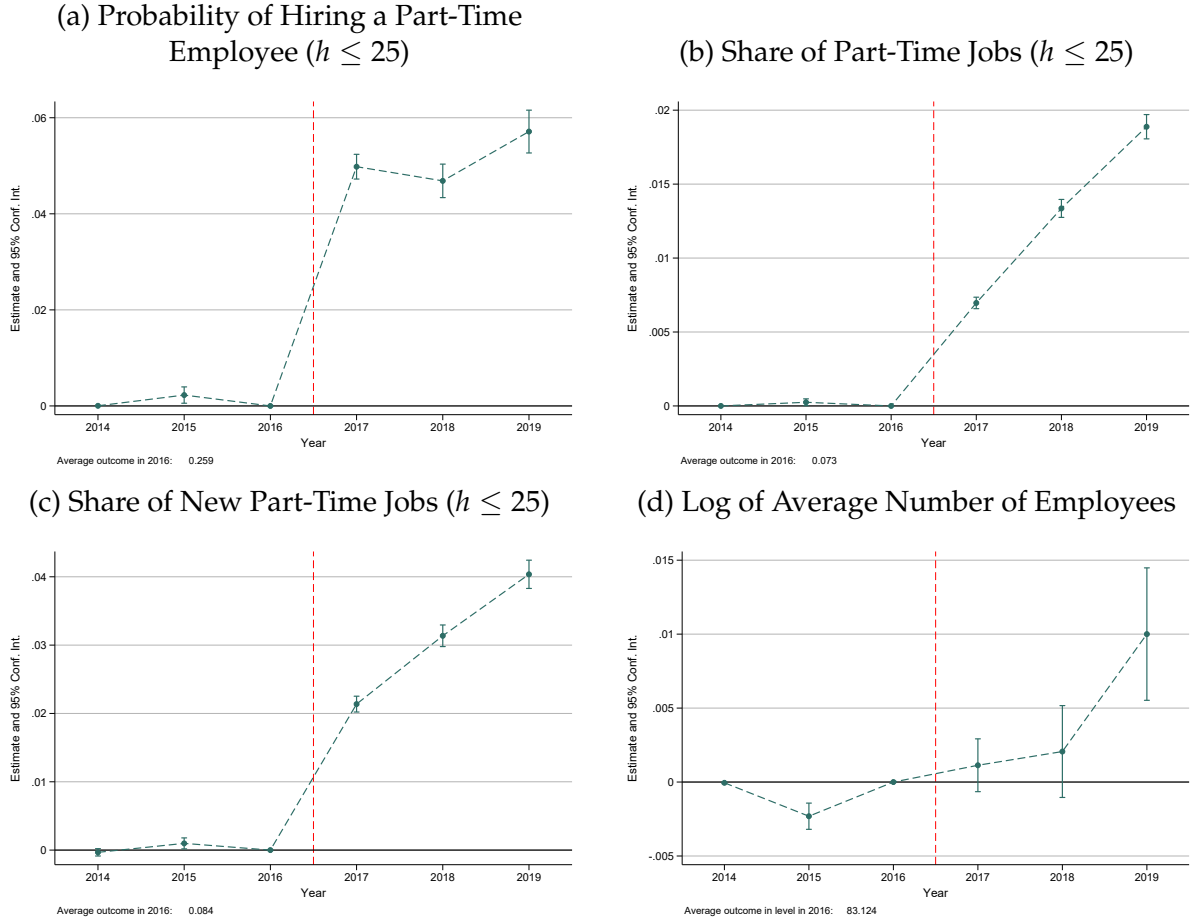
Notes: This figure plots the share of new hires for jobs with an average workweek less than or equal to 25h/wk, log number of new part-time hires (jobs with workweek less than or equal to 25h/wk), and the log number of new full-time hires (jobs with at least 40h/wk), for each year and type of firms. In green, the share or total number of hires for firms with pre-reform standard deviation of hours across skill groups equal to zero. In yellow, the share or total number of hires for firms with pre-reform standard deviation of hours across skill groups greater than zero. Figure shows workers aged 18 to 64 in the private sector, using a balanced panel of firms spanning from 2014 to 2019. Firms with pre-reform size smaller than 5 are excluded.

Figure 3. Effects on Number of Jobs



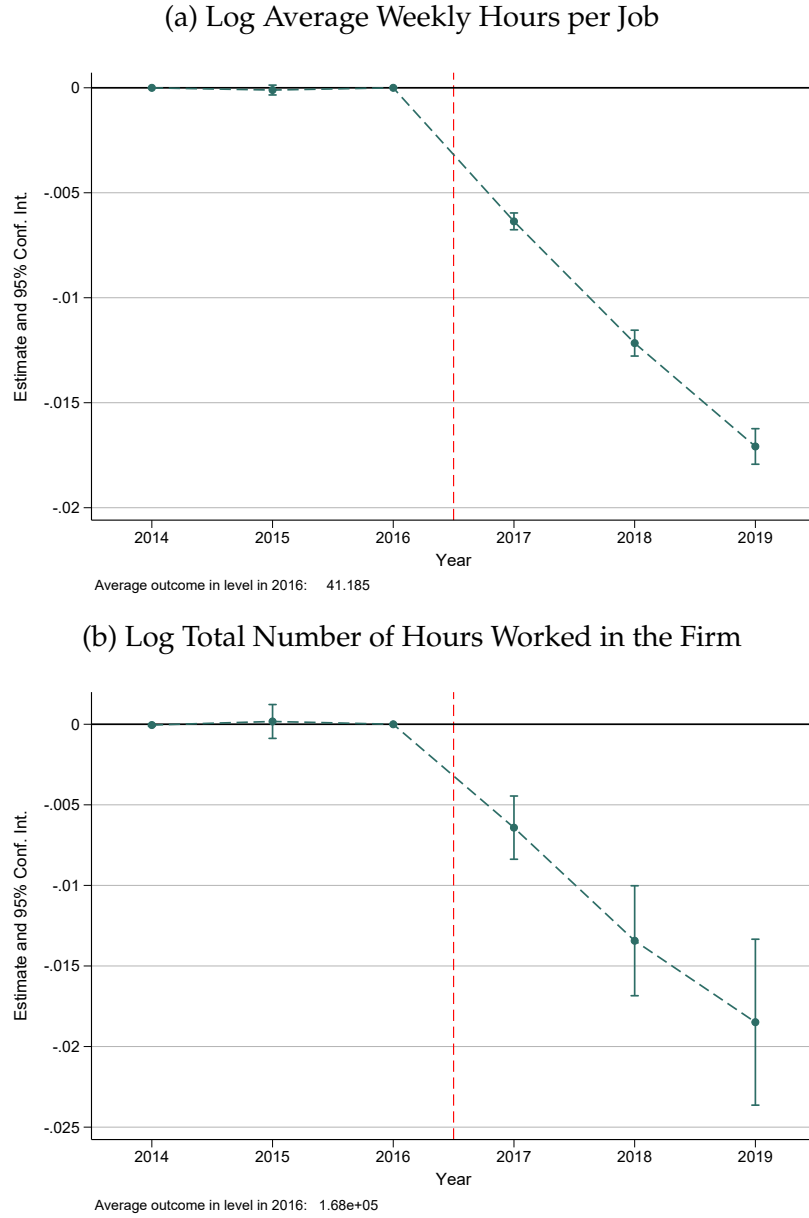
Notes: This figure reports estimates from equation (2) after detrending the outcome. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended log of the annual number of jobs. In Panel (b), the outcome variable is the detrended annual number of part-time jobs with a workweek below or equal to 25 hours divided by the annual number of jobs in 2014. Panel (c) reports the effects on the detrended annual number of part-time jobs with a workweek between 26 and 39 hours divided by the annual number of jobs in 2014. In Panel (d), the outcome variable is the detrended annual number of full-time jobs with a workweek above or equal to 40 hours divided by the annual number of jobs in 2014. Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

Figure 4. Effects on Share and Creation of Part-Time Jobs and Employment



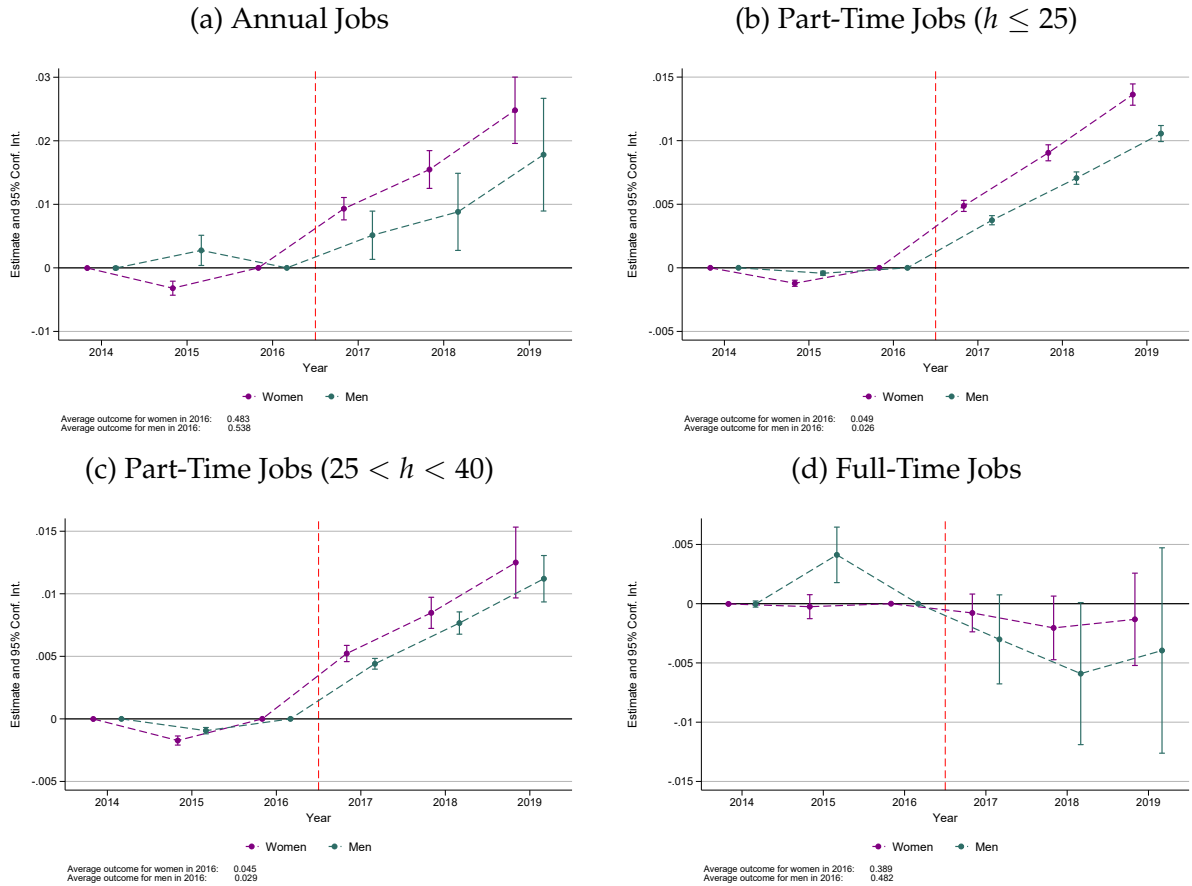
Notes: This figure reports estimates from equation (2) after detrending the outcome. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended indicator of at least one part-time hire with workweek below or equal to 25 hours. In Panel (b), the outcome variable is the detrended share of part-time jobs defined as the annual number of part-time jobs with a workweek below or equal to 25 hours divided by the annual number of observed jobs. Panel (c) reports the effects on the detrended share of new part-time jobs defined as the annual number of part-time hires with a workweek below or equal to 25 hours divided by the annual number of observed hires. In Panel (d), the outcome variable is the detrended log of employment (average number of workers in a month). Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

Figure 5. Effects on the Average Weekly Hours per Job and Total Number of Hours Worked



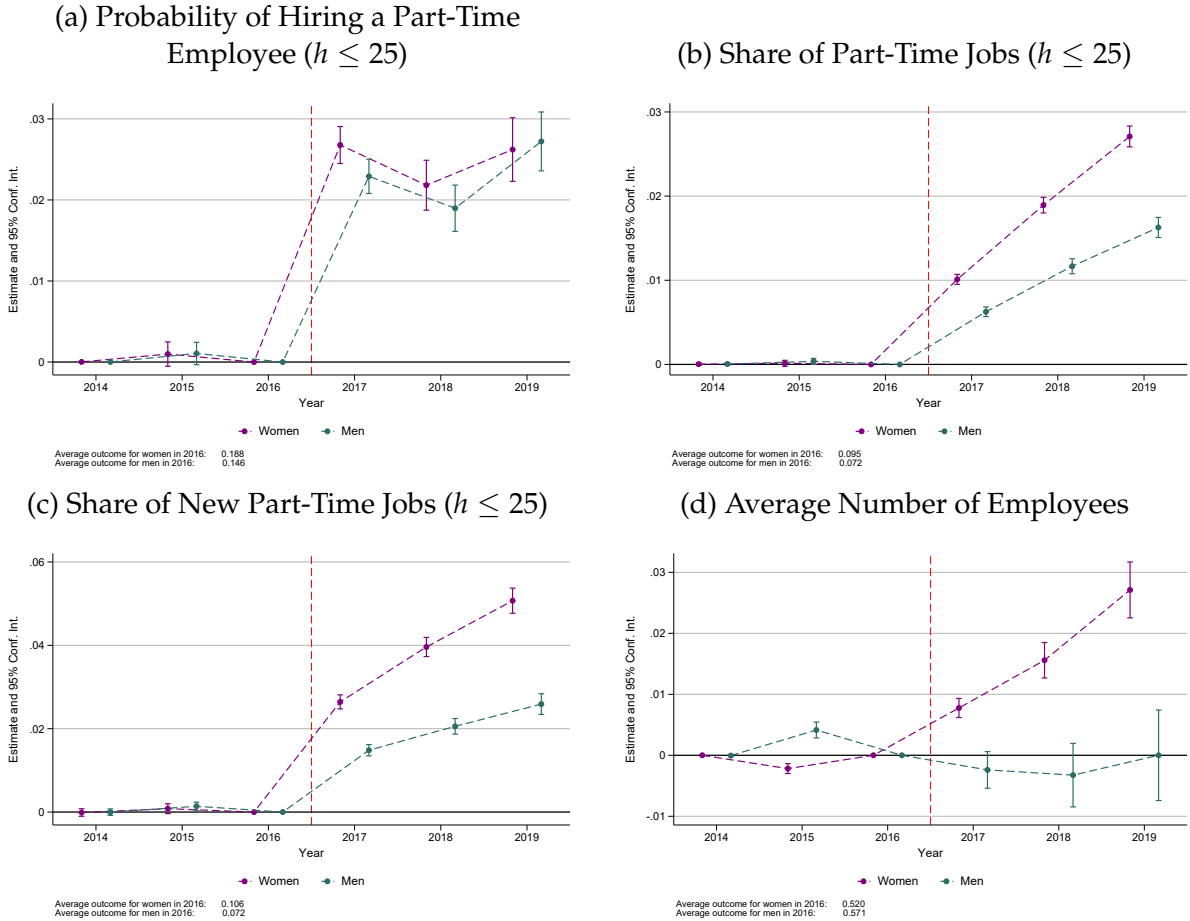
Notes: This figure reports estimates from equation (2) after detrending the outcome. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended log of average weekly hours per job. In Panel (b), the outcome variable is the detrended log of the annual number of hours worked in the firm across all jobs (spells). Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

Figure 6. Effects on Number of Jobs for Men and Women



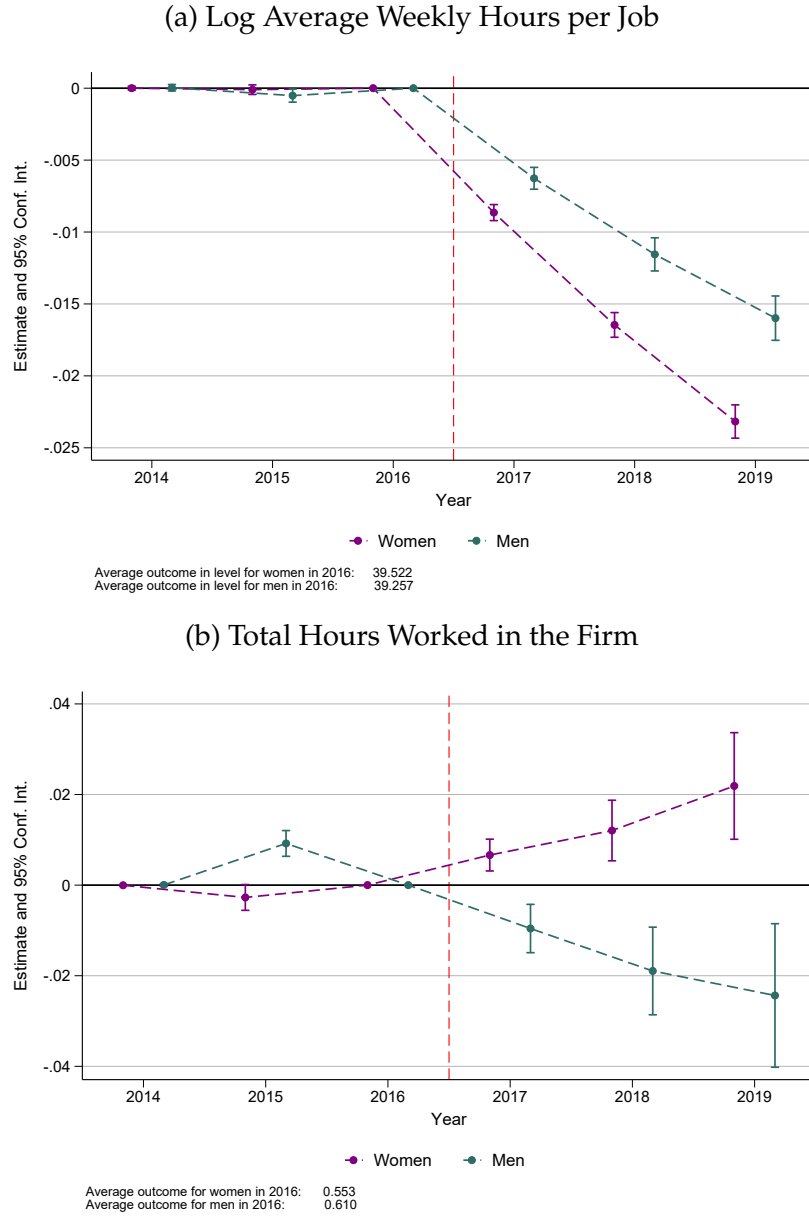
Notes: This figure reports estimates from equation (2) after detrending the outcome that is gender specific. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended annual number of jobs for women (men) divided by the annual number of jobs in 2014. In Panel (b), the outcome variable is the detrended annual number of part-time jobs for women (men) with a workweek below or equal to 25 hours divided by the annual number of jobs in 2014. Panel (c) reports the effects on the detrended annual number of part-time jobs for women (men) with a workweek between 26 and 39 hours divided by the annual number of jobs in 2014. In Panel (d), the outcome variable is the detrended annual number of full-time jobs for women (men) with a workweek above or equal to 40 hours divided by the annual number of jobs in 2014. Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

Figure 7. Effects on Share and Creation of Part-Time Jobs and Employment for Men and Women



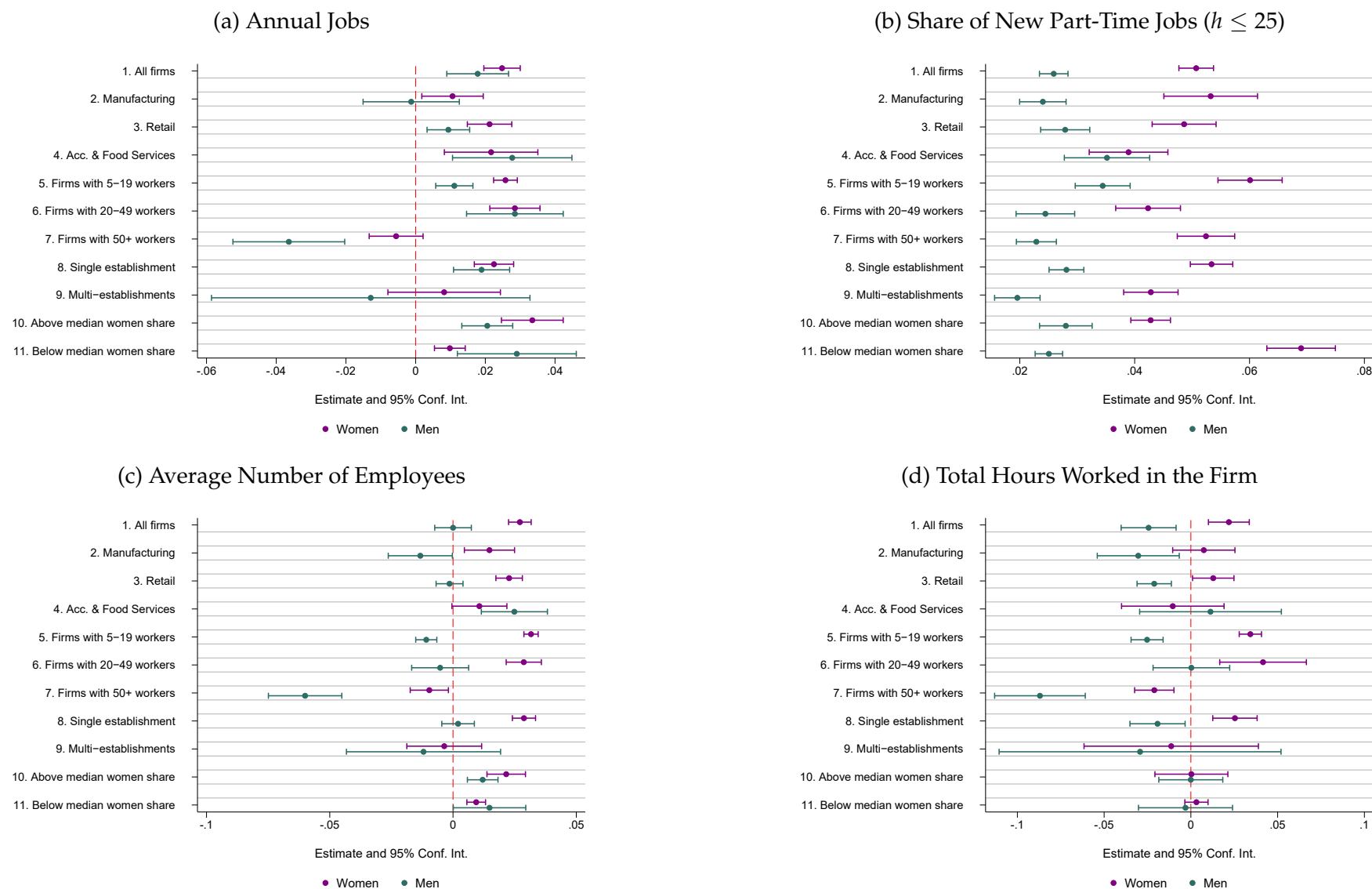
Notes: This figure reports estimates from equation (2) after detrending the outcome that is gender specific. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended indicator of at least one part-time female (male) hire with workweek below or equal to 25 hours. In Panel (b), the outcome variable is the detrended share of part-time jobs for women (men) defined as the annual number of part-time jobs for women (men) with a workweek below or equal to 25 hours divided by the annual number of observed jobs for women (men). Panel (c) reports the effects on the detrended share of new part-time jobs for women (men) defined as the annual number of part-time female (male) hires with a workweek below or equal to 25 hours divided by the annual number of observed female (male) hires. In Panel (d), the outcome variable is the detrended monthly average number of female (male) workers divided by the monthly average number of workers in 2014. Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

Figure 8. Effects on the Average Weekly Hours per Job and Total Number of Hours Worked



Notes: This figure reports estimates from equation (2) after detrending the outcome that is gender specific. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended log of average weekly hours per job for men and women. In Panel (b), the outcome variable is the detrended annual number of hours worked by women (men) in the firm across all female (male) jobs divided by the annual number of hours worked in the firm across all jobs (spells) in 2014. Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

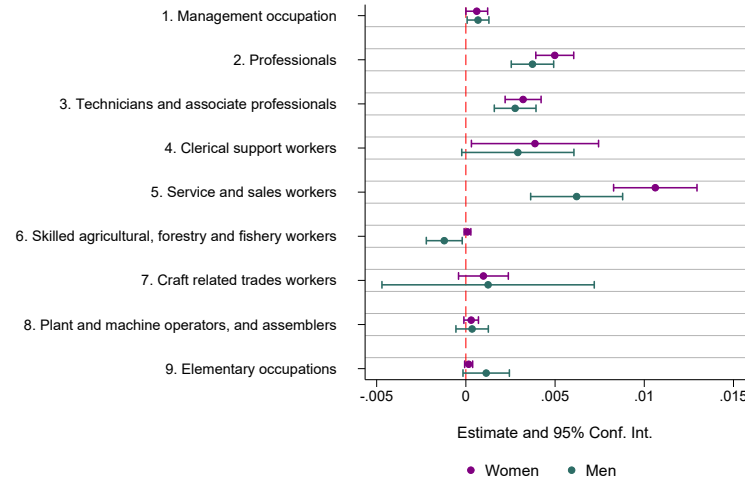
Figure 9. Firm Heterogeneity, by Gender



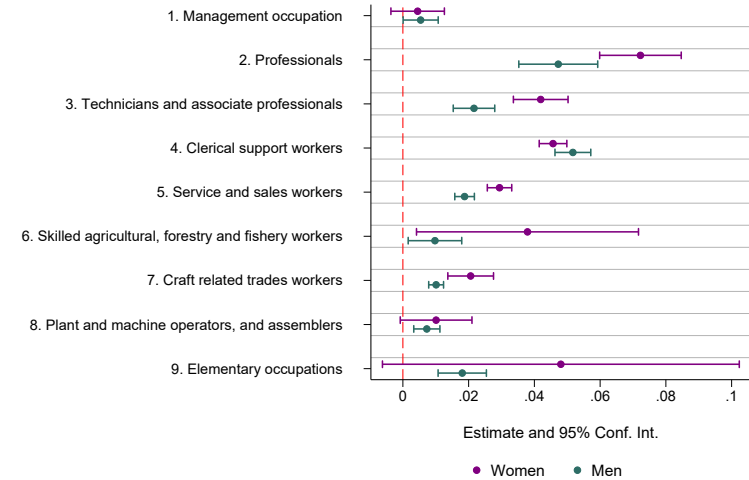
Notes: This figure reports estimates of β_2 from equation (2) for different subgroups of firms after detrending the outcome. Each regression has been estimated separately in the corresponding subsample of firms. In Panel (a), the outcome variable is the detrended annual number of jobs for women (men) divided by the annual number of jobs in 2014. In Panel (b), the outcome variable is the detrended share of new part-time jobs for women (men) defined as the annual number of part-time female (male) hires with a workweek below or equal to 25 hours divided by the annual number of observed female (male) hires. Panel (c) reports the effects on the detrended monthly average number of female (male) workers divided by the monthly average number of workers in 2014. In Panel (d), the outcome variable is the detrended log of the annual number of hours worked by women (men) in the firm across all female (male) jobs divided by the annual number of hours worked in the firm across all jobs (spells) in 2014. Effects are estimated on a balanced panel of firms (2014–2019) with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

Figure 10. Within-Occupation Heterogeneity, by Gender

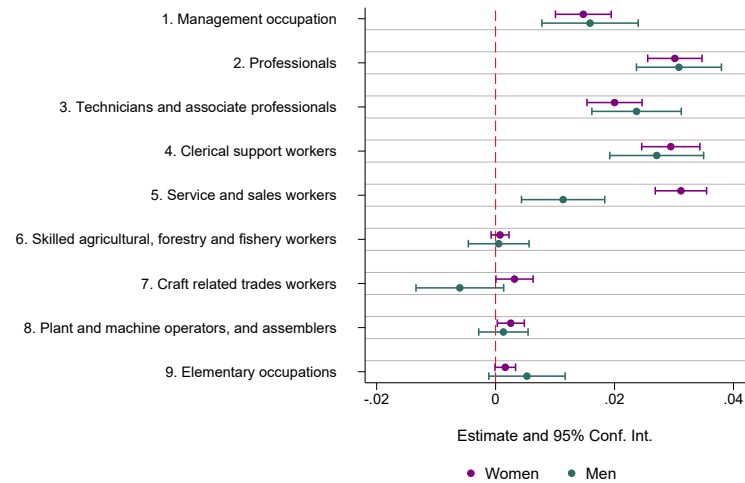
(a) Annual Jobs



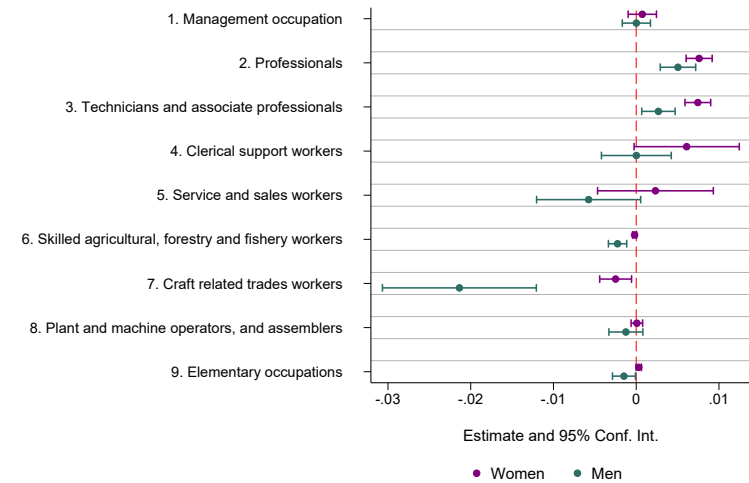
(b) Share of New Part-Time Jobs ($h \leq 25$)



(c) Average Number of Employees



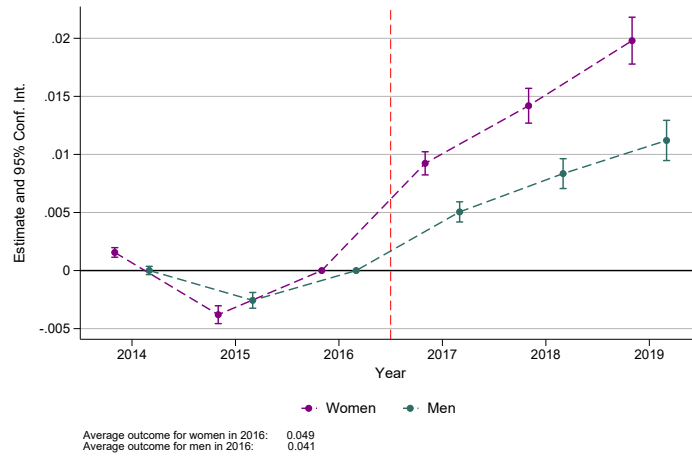
(d) Total Hours Worked in the Firm



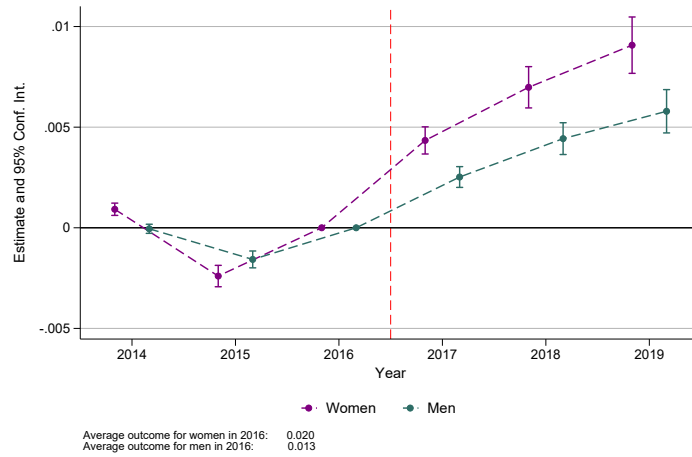
Notes: This figure reports estimates from equation (2) after detrending the outcome that is gender and occupation specific. In Panel (a), the outcome variable is the detrended annual number of jobs for women (men) for each occupation category divided by the annual number of jobs in 2014. In Panel (b), the outcome variable is the detrended share of new part-time jobs for women (men) defined as the annual number of part-time female (male) hires with a workweek below or equal to 25 hours divided by the annual number of observed female (male) hires for each occupation category. Panel (c) reports the effects on the detrended monthly average number of female (male) workers for each occupation category divided by the monthly average number of workers in 2014. In Panel (d), the outcome variable is the detrended log of the annual number of hours worked by women (men) in the firm across all female (male) jobs for each occupation category divided by the annual number of hours worked in the firm across all jobs (spells) in 2014. Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

Figure 11. Effects on the Share of Employees Entering the Formal Labor Market for the First Time since 2013

(a) All Employees

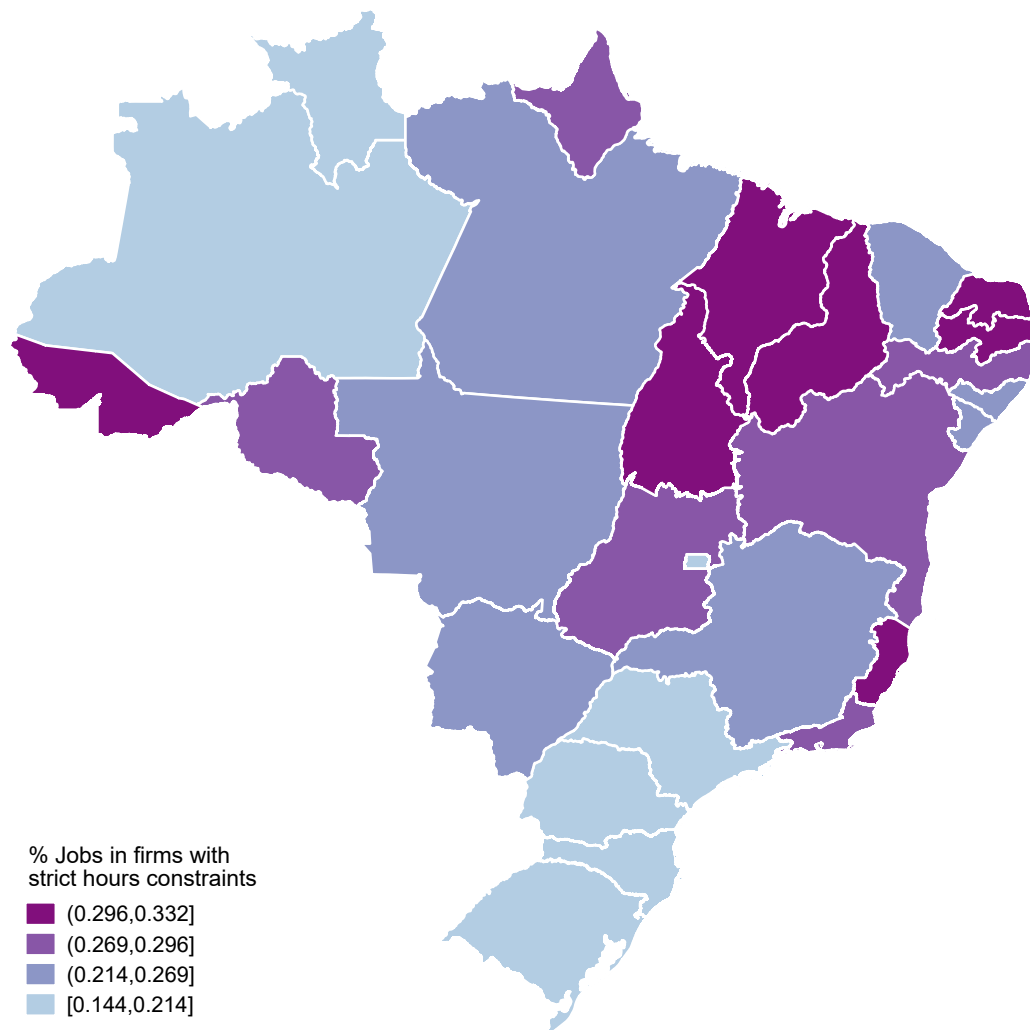


(b) Employees above 35



Notes: This figure reports estimates from equation (2) after detrending the outcome. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended fraction of employees newly entering the formal labor market—represented by their appearance in RAIS—since 2013. In Panel (b), the outcome variable is the detrended fraction of employees above 35 years old newly entering the formal labor market since 2013. Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

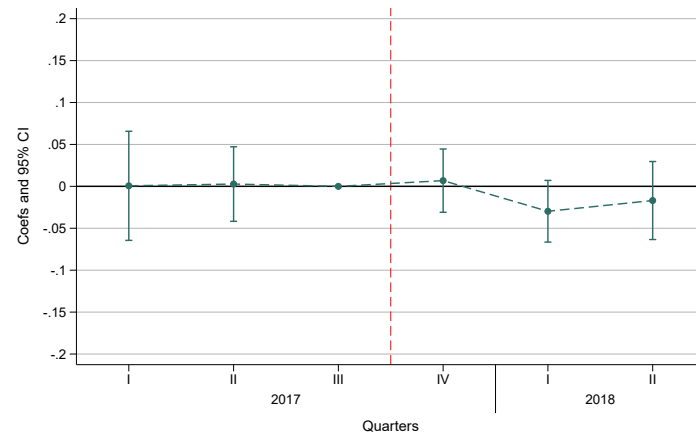
Figure 12. Exposure to the 2017 Labor Reform across Brazilian States



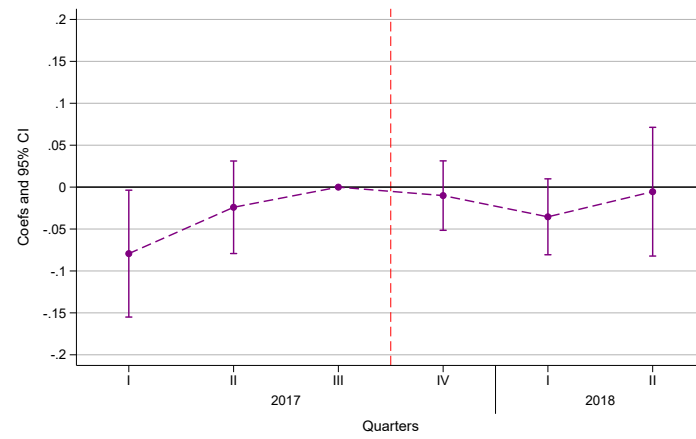
Notes: This figure shows the exposure to the 2017 labor reform across 27 Brazilian states. State-level exposure to the reform is measured using equation (4).

Figure 13. Effects on the Probability of Being Employed

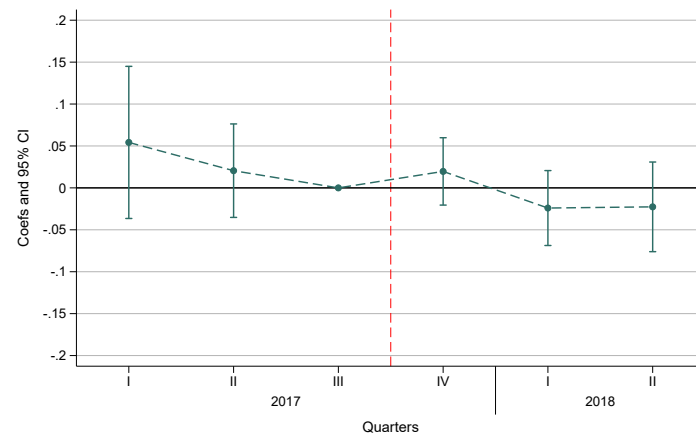
(a) All Workers



(b) Women



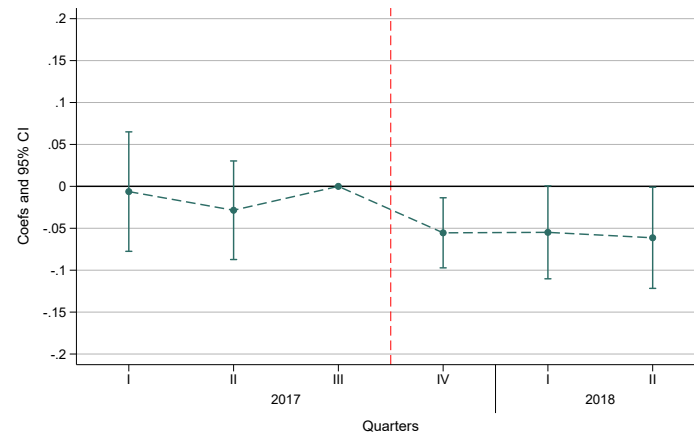
(c) Men



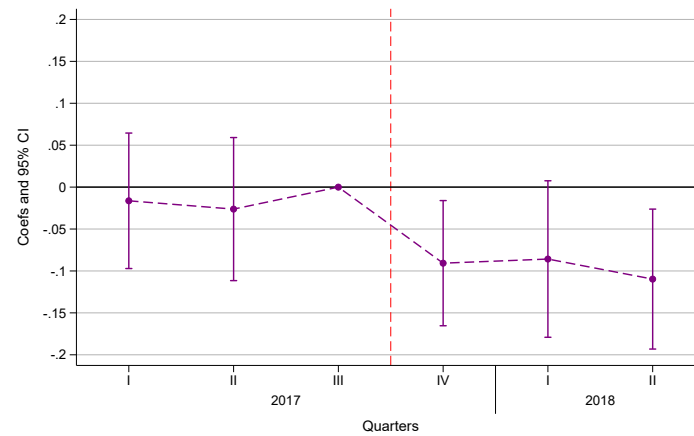
Notes: This figure reports estimates from equation (5). In Panel (a), the outcome variable is a dummy variable that indicates whether the individual was employed (employment status), either formally or informally. Panel (b) reports the effects on employment status among women. In Panel (c), the outcome variable is the employment status among men. Effects are estimated on a quarterly panel of individuals aged between 18 and 64 years old. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the state level.

Figure 14. Effects on the Probability of Working Informally

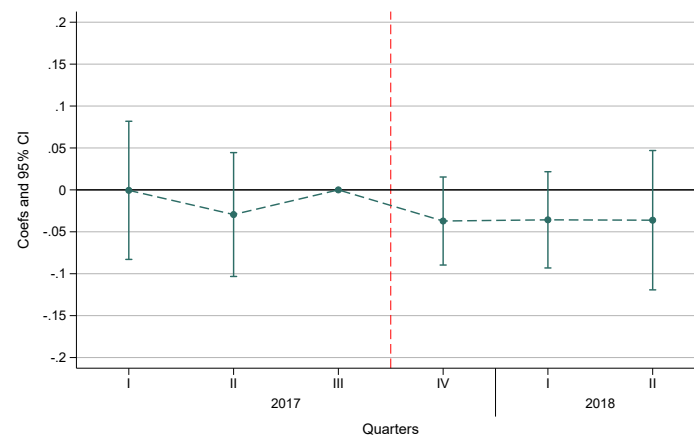
(a) All Workers in the Private Sector



(b) Women in the Private Sector



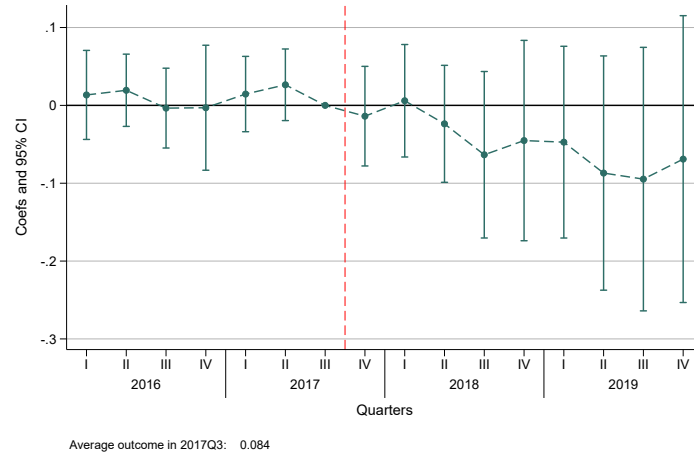
(c) Men in the Private Sector



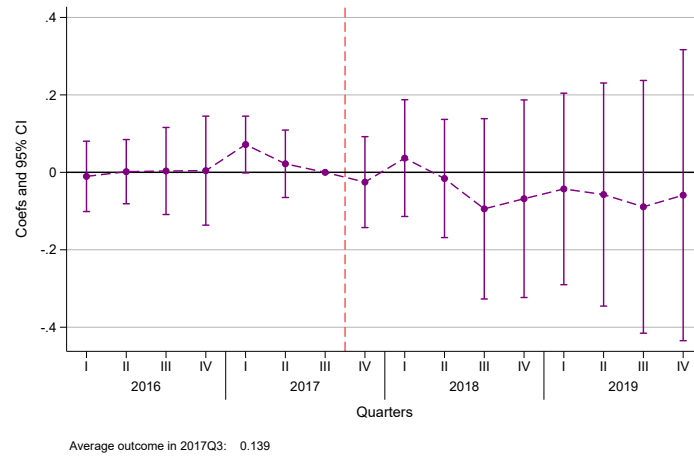
Notes: This figure reports estimates from equation (5). In Panel (a), the outcome variable is a dummy variable that takes the value of one if the individual was informally employed—without a signed booklet—and zero if the individual was formally employed. Panel (b) reports the effects on informality status among women. In Panel (c), the outcome variable is the informality status among men. Effects are estimated on a quarterly panel of individuals aged between 18 and 64 years old. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the state level.

Figure 15. Employment Effects of the Labor Reform at the State Level

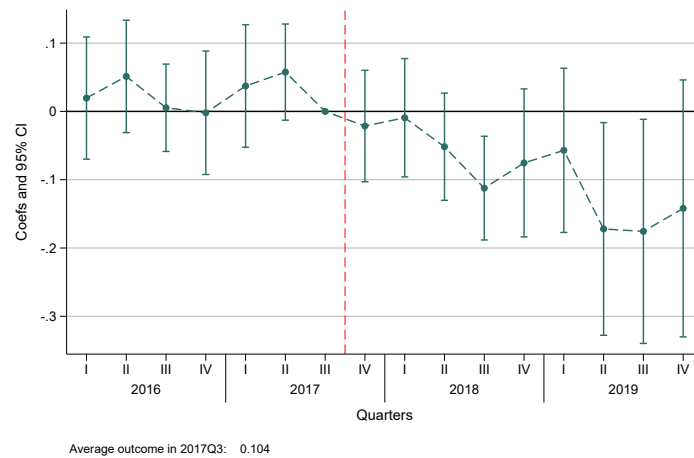
(a) Unemployment Rate (Total)



(b) Unemployment Rate for Women



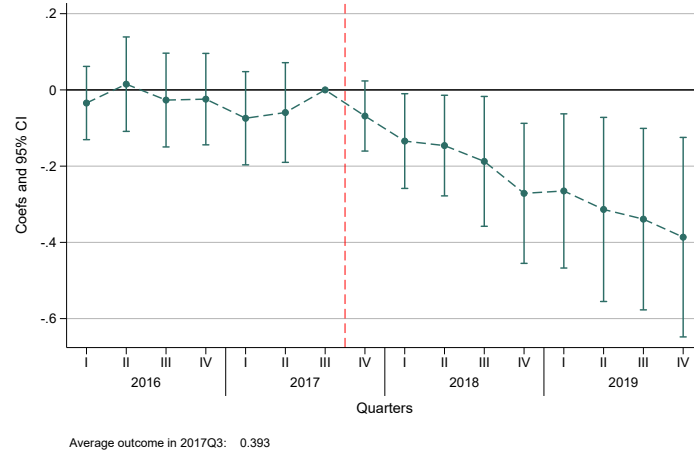
(c) Unemployment Rate for Men



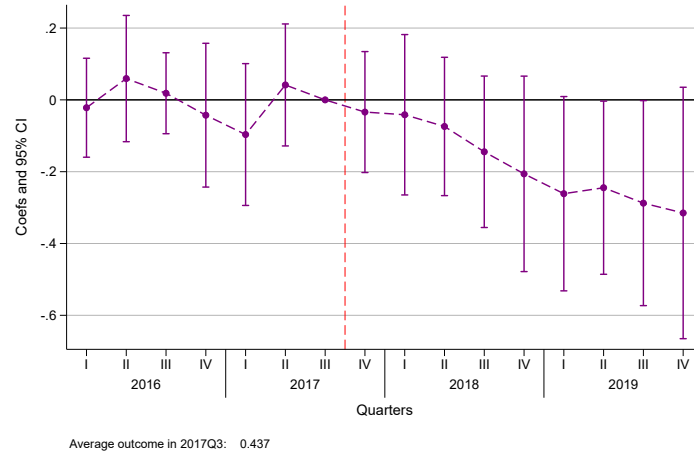
Notes: This figure reports estimates from equation (6) after detrending the outcome. Specifically, I regress outcome Y_{jt} on a state specific linear time trend $\theta_j \cdot t$ using pre-reform data from 2016Q1 to 2017Q3. Then, I run specification (6) on the full-period using the residualized outcome $Y_{jt} - \hat{\theta}_j \cdot t$. In Panel (a), the outcome variable is the detrended unemployment rate. Panel (b) reports the effects on the detrended unemployment rate for women. In Panel (c), the outcome variable is the detrended unemployment rate for men. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the state level.

Figure 16. Informality Effects of the Labor Reform at the State level

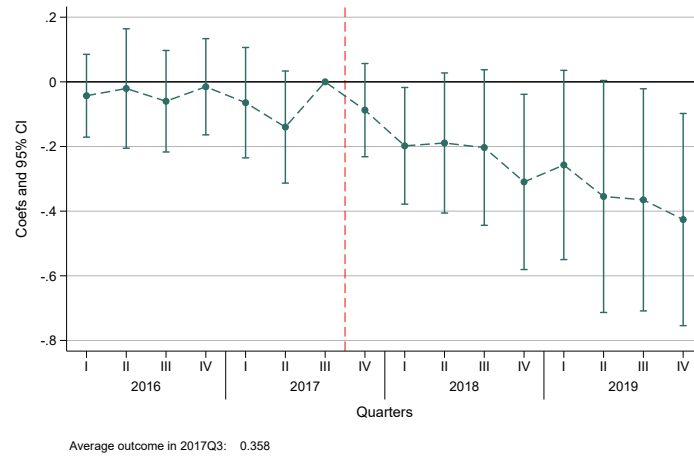
(a) Informality Rate (Total)



(b) Informality Rate for Women



(c) Informality Rate for Men



Notes: This figure reports estimates from equation (6) after detrending the outcome. Specifically, I regress outcome Y_{jt} on a state specific linear time trend $\theta_j \cdot t$ using pre-reform data from 2016Q1 to 2017Q3. Then, I run specification (6) on the full-period using the residualized outcome $Y_{jt} - \hat{\theta}_j \cdot t$. In Panel (a), the outcome variable is the detrended informality rate. Panel (b) reports the effects on the detrended informality rate for women. In Panel (c), the outcome variable is the detrended informality rate for men. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the state level.

Tables

Table 1. Average Work Hours of Formal Workers by Skill Group

	Obs.	Mean weekly work hours	
<u>1. Management occupation</u>			
No Secondary Education	9,742	47.084	47.086
Secondary Education	9,541	43.255	43.255
Post-Secondary Education	3,114	43.117	43.129
<u>2. Professionals</u>			
No Secondary Education, Professionals	3,646	37.178	37.173
Secondary Education	32,620	37.373	37.372
Post-Secondary Education	14,054	37.006	37.001
<u>3. Technicians and associate professionals</u>			
No Secondary Education	35,512	42.550	42.550
Secondary Education	13,867	40.917	40.916
Post-Secondary Education	3,120	40.083	40.063
<u>4. Clerical support workers</u>			
No Secondary Education	56,030	41.564	41.564
Secondary Education	21,440	40.608	40.608
Post-Secondary Education	4,133	40.902	40.903
<u>5. Service and sales workers</u>			
No Secondary Education	166,177	44.949	44.949
Secondary Education	41,078	43.851	43.851
Post-Secondary Education	4,513	43.442	43.441
<u>6. Skilled agricultural, forestry and fishery workers</u>			
No Secondary Education	14,626	46.620	46.621
Secondary Education	728	45.819	45.820
Post-Secondary Education	32	45.377	46.026
<u>7. Craft related trades workers</u>			
No Secondary Education	118,884	43.834	43.834
Secondary Education	11,697	43.401	43.402
Post-Secondary Education	544	43.129	43.151
<u>8. Plant and machine operators, and assemblers</u>			
No Secondary Education	119,080	45.729	45.729
Secondary Education	13,753	44.324	44.326
Post-Secondary Education	666	44.257	44.275
<u>9. Elementary occupations</u>			
No Secondary Education	141,440	43.830	43.830
Secondary Education	13,473	43.384	43.385
Post-Secondary Education	548	43.332	43.319

Notes: This table shows the average weekly work hours by skill groups for formal workers. Data on workers' hours are drawn from the National Household Survey (PNAD-continua) from rounds 2012Q1 to 2017Q3. Skill groups are defined at the intersection of the education and occupation categories. The table shows statistics for workers in the private formal sector. Using the most detailed industry classification (5-digit) and the most detailed occupational classification (4-digit) in the PNAD-continua, I focus on occupation-industry pairs that have both formal and informal workers in each education category. Indeed, only for these workers I can meaningfully compare the variability of work hours between informal and formal workers. I restrict the sample to workers in private sector firms. The sample includes working-age population aged between 18 and 64. Workers in firms with size smaller than 5 are excluded.

Table 2. Summary Statistics at the Firm Level in 2016

	St. dev. hours = 0 (1)	St. dev. hours > 0 (2)
Number of jobs (spells)	22.681 (43.040)	117.345 (1195.548)
Number of part-time jobs (h<=25)	0.008 (0.488)	4.041 (86.186)
Number of full-time jobs (h>=40)	22.649 (42.998)	101.924 (937.221)
Percentage of part-time employees (h<=25)	0.001 (0.023)	0.073 (0.151)
Number of employees	16.345 (29.973)	83.124 (865.072)
Number of establishments	1.140 (0.658)	2.061 (124.222)
Percentage of female employees	0.393 (0.305)	0.475 (0.297)
Percentage of black employees	0.043 (0.083)	0.040 (0.070)
Percentage of white employees	0.544 (0.367)	0.575 (0.353)
Percentage of mixed-race employees	0.299 (0.336)	0.260 (0.313)
Percentage of employees with less than 25 years old	0.194 (0.163)	0.199 (0.152)
Percentage of employees between 25 and 34 years old	0.346 (0.163)	0.354 (0.144)
Percentage of employees between 35 and 45 years old	0.268 (0.147)	0.266 (0.127)
Percentage of employees above 45 years old	0.192 (0.166)	0.182 (0.143)
Percentage of employees with less than primary education	0.093 (0.166)	0.084 (0.144)
Percentage of employees with complete primary education	0.219 (0.231)	0.176 (0.186)
Percentage of employees with complete secondary education	0.605 (0.294)	0.544 (0.271)
Percentage of employees with post-secondary education	0.083 (0.156)	0.196 (0.245)
Average hours per job	43.950 (0.796)	41.185 (4.671)
Average tenure (in months)	41.302 (28.166)	39.992 (26.445)
Average Monthly Earnings (R\$ 2016)	1,569.995 (876.865)	1,792.501 (1302.084)
Percentage of workers whose hourly wage < 1.2 x Min wage	0.248 (0.297)	0.187 (0.231)
Percentage of Workers above 30: First time in RAIS since 2013	0.015 (0.035)	0.016 (0.032)
Observations	353,871	233,181

Notes: This table reports descriptive statistics for firms using information from the 2016 RAIS data. Column (1) shows summary statistics for firms with a pre-reform standard deviation of hours across skill groups equal to zero. Column (2) reports summary statistics for firms with a pre-reform standard deviation greater than zero. Values are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Standard deviations are in parentheses.

Table 3. The Effects of the Reform on Firms' Outcomes

	Log of Number of Jobs (1)	Part-time Jobs ($h \leq 25$) in terms of 2014 Jobs (2)	Part-time Jobs (25 < $h < 40$) in terms of 2014 Jobs (3)	Full-time Jobs ($h \geq 40$) in terms of 2014 Jobs (4)	Share of Part-Time Jobs ($h \leq 25$) (5)	Share of New Part-Time Jobs ($h \leq 25$) (6)	Log of Average Number of Employees (7)	Log Average Weekly Hours Worked per Job (8)	Log Total Hours Worked (9)
Short Run Effect of the Reform	0.012*** (0.001)	0.009*** (0.000)	0.010*** (0.000)	-0.004 (0.002)	0.007*** (0.000)	0.021*** (0.001)	0.001 (0.001)	-0.006*** (0.000)	-0.006*** (0.001)
Medium Run Effect of the Reform	0.030*** (0.003)	0.024*** (0.001)	0.024*** (0.002)	-0.005 (0.006)	0.019*** (0.000)	0.040*** (0.001)	0.010*** (0.002)	-0.017*** (0.000)	-0.018*** (0.003)
Pre-reform Mean in Firms with $\bar{\sigma}_f > 0$	3.431	0.075	0.075	0.871	0.073	0.084	3.122	3.709	10.706
Observations	3,522,293	3,522,318	3,522,318	3,522,318	3,522,293	2,940,375	3,522,293	3,522,293	3,522,293

Notes: This table reports the coefficient β_0 —defined as “Short Run Effect of the Reform”—and β_2 —defined as the “Medium Run Effect of the Reform”—from equation (2) after detrending the outcome variable. The former refers to the effect of the reform at enactment, while the latter refers to the effect two years after the reform was implemented. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. All displayed results control for firm fixed effects and state-by-year fixed effects. Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Standard errors are displayed in parentheses and are clustered at the firm level.

Table 4. The Effects of the Reform on Firms' Outcomes, by Employees' Gender

	Number of Jobs ($h \leq 25$) in terms of 2014 Jobs	Part-time Jobs ($h \leq 25$) in terms of 2014 Jobs	Part-time Jobs (25 < $h < 40$) in terms of 2014 Jobs	Full-time Jobs ($h \geq 40$) in terms of 2014 Jobs	Share of Part-Time Jobs ($h \leq 25$)	Share of New Part-Time Jobs ($h \leq 25$)	Employ- ment in terms of 2014 Average Number of Employees	Log Average Weekly Hours Worked per Jobs	Total Hours Worked in terms of 2014 Total Hours Worked
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A. Women									
Short Run Effect of the Reform	0.009*** (0.001)	0.005*** (0.000)	0.005*** (0.000)	-0.001 (0.001)	0.010*** (0.000)	0.026*** (0.001)	0.008*** (0.001)	-0.009*** (0.000)	0.007*** (0.002)
Medium Run Effect of the Reform	0.025*** (0.003)	0.014*** (0.000)	0.012*** (0.001)	-0.001 (0.002)	0.027*** (0.001)	0.051*** (0.002)	0.027*** (0.002)	-0.023*** (0.001)	0.022*** (0.006)
Pre-reform Mean in Firms with $\bar{\sigma}_f > 0$ Observations	0.483 3,522,318	0.049 3,522,318	0.045 3,522,318	0.389 3,522,318	0.095 3,196,099	0.106 2,005,452	0.520 3,522,318	3.692 3,196,099	0.553 3,522,318
Panel B. Men									
Short Run Effect of the Reform	0.005*** (0.002)	0.004*** (0.000)	0.004*** (0.000)	-0.003 (0.002)	0.006*** (0.000)	0.015*** (0.001)	-0.002 (0.002)	-0.006*** (0.000)	-0.010*** (0.003)
Medium Run Effect of the Reform	0.018*** (0.005)	0.011*** (0.000)	0.011*** (0.001)	-0.004 (0.004)	0.016*** (0.001)	0.026*** (0.001)	0.000 (0.004)	-0.016*** (0.001)	-0.024*** (0.008)
Pre-reform Mean in Firms with $\bar{\sigma}_f > 0$ Observations	0.538 3,522,318	0.026 3,522,318	0.029 3,522,318	0.482 3,522,318	0.072 3,315,714	0.072 2,335,237	0.571 3,522,318	3.699 3,315,714	0.610 3,522,318

Notes: This table reports the coefficient β_0 —defined as “Short Run Effect of the Reform”—and β_2 —defined as the “Medium Run Effect of the Reform”—from equation (2) after detrending the outcome variable that is gender specific. The former refers to the effect of the reform at enactment, while the latter refers to the effect two years after the reform was implemented. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. All displayed results control for firm fixed effects and state-by-year fixed effects. Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Standard errors are displayed in parentheses and are clustered at the firm level.

Appendices

A. Work Hours in the Informal Sector

Table A.1. Average Work Hours of Informal Workers by Skill Group

	Obs.	Mean weekly work hours	
<u>1. Management occupation</u>			
No Secondary Education	1,921	45.372	45.379
Secondary Education	1,711	42.723	42.720
Post-Secondary Education	605	43.206	43.156
<u>2. Professionals</u>			
No Secondary Education, Professionals	2,832	32.221	32.203
Secondary Education	7,269	36.037	36.039
Post-Secondary Education	2,920	36.704	36.705
<u>3. Technicians and associate professionals</u>			
No Secondary Education	6,836	39.814	39.814
Secondary Education	3,507	38.189	38.188
Post-Secondary Education	852	38.346	38.293
<u>4. Clerical support workers</u>			
No Secondary Education	6,808	39.320	39.320
Secondary Education	2,617	37.709	37.704
Post-Secondary Education	510	36.145	35.977
<u>5. Service and sales workers</u>			
No Secondary Education	25,459	41.200	41.199
Secondary Education	5,767	39.518	39.524
Post-Secondary Education	757	38.676	38.634
<u>6. Skilled agricultural, forestry and fishery workers</u>			
No Secondary Education	6,568	44.124	44.124
Secondary Education	374	41.131	41.065
Post-Secondary Education	17	46.194	51.839
<u>7. Craft related trades workers</u>			
No Secondary Education	21,755	42.885	42.886
Secondary Education	2,160	41.867	41.885
Post-Secondary Education	162	41.046	40.878
<u>8. Plant and machine operators, and assemblers</u>			
No Secondary Education	20,272	44.244	44.244
Secondary Education	2,251	43.147	43.176
Post-Secondary Education	166	42.460	42.823
<u>9. Elementary occupations</u>			
No Secondary Education	44,317	39.741	39.741
Secondary Education	3,232	38.526	38.535
Post-Secondary Education	158	35.953	35.556

Notes: This table shows the average weekly work hours by skill groups for informal workers. Data on workers' hours are drawn from the National Household Survey (PNAD-continua) from rounds 2012Q1 to 2017Q3. Skill groups are defined at the intersection of the education and occupation categories. The table shows statistics for workers in the private formal sector. Using the most detailed industry classification (5-digit) and the most detailed occupational classification (4-digit) in the PNAD-continua, I focus on occupation-industry pairs that have both formal and informal workers in each education category. Indeed, only for these workers I can meaningfully compare the variability of work hours between informal and formal workers. I restrict the sample to workers in private sector firms. The sample includes working-age population aged between 18 and 64. Workers in firms with size smaller than 5 are excluded.

B. Firm Heterogeneity

Table B.2. Heterogeneity in the Effect of the Reform on Firms' Outcomes, Based on Industry and Pre-Reform Fraction of Female Jobs

	Log of Number of Jobs (1)	Part-time Jobs ($h \leq 25$) in terms of 2014 Jobs (2)	Part-time Jobs ($25 < h < 40$) in terms of 2014 Jobs (3)	Full-time Jobs ($h \geq 40$) in terms of 2014 Jobs (4)	Share of Part-Time Jobs ($h \leq 25$) (5)	Share of New Part-Time Jobs ($h \leq 25$) (6)	Log of Average Number of Employees (7)	Log Average Hours Worked per Job (8)	Log Total Hours Worked (9)
Panel A. Manufacturing Industry									
Medium Run Effect of the Reform	-0.015** (0.006)	0.014*** (0.001)	0.006*** (0.001)	-0.011 (0.009)	0.015*** (0.001)	0.035*** (0.002)	-0.030*** (0.006)	-0.011*** (0.001)	-0.053*** (0.006)
Pre-reform Mean in Firms with $\bar{\sigma}_f > 0$	3.742	0.028	0.026	0.900	0.030	0.044	3.470	3.758	11.100
Observations	592,856	592,858	592,858	592,858	592,856	483,119	592,856	592,856	592,856
Panel B. Retail Industry									
Medium Run Effect of the Reform	0.035*** (0.004)	0.021*** (0.001)	0.030*** (0.002)	-0.020*** (0.005)	0.020*** (0.001)	0.042*** (0.002)	0.021*** (0.004)	-0.018*** (0.001)	-0.002 (0.004)
Pre-reform Mean in Firms with $\bar{\sigma}_f > 0$	3.125	0.043	0.074	0.873	0.044	0.056	2.790	3.740	10.404
Observations	1,068,323	1,068,324	1,068,324	1,068,324	1,068,323	896,872	1,068,323	1,068,323	1,068,323
Panel C. Accommodation and Food Industry									
Medium Run Effect of the Reform	0.035*** (0.007)	0.017*** (0.001)	0.041*** (0.006)	-0.009 (0.013)	0.017*** (0.001)	0.039*** (0.003)	0.022*** (0.006)	-0.018*** (0.001)	-0.010 (0.008)
Pre-reform Mean in Firms with $\bar{\sigma}_f > 0$	3.252	0.048	0.114	0.825	0.049	0.062	2.888	3.726	10.487
Observations	379,455	379,460	379,460	379,460	379,455	333,676	379,455	379,455	379,455
Panel D. High Fraction of Women									
Medium Run Effect of the Reform	0.048*** (0.003)	0.028*** (0.001)	0.029*** (0.004)	-0.003 (0.006)	0.021*** (0.001)	0.042*** (0.002)	0.025*** (0.003)	-0.020*** (0.001)	-0.007** (0.003)
Pre-reform Mean in Firms with $\bar{\sigma}_f > 0$	3.266	0.103	0.095	0.824	0.099	0.108	2.958	3.682	10.515
Observations	1,761,155	1,761,162	1,761,162	1,761,162	1,761,155	1,500,955	1,761,155	1,761,155	1,761,155
Panel E. Low Fraction of Women									
Medium Run Effect of the Reform	0.019*** (0.004)	0.020*** (0.001)	0.018*** (0.001)	0.001 (0.010)	0.017*** (0.000)	0.038*** (0.001)	-0.002 (0.004)	-0.014*** (0.000)	-0.030*** (0.004)
Pre-reform Mean in Firms with $\bar{\sigma}_f > 0$	3.654	0.038	0.047	0.934	0.038	0.051	3.343	3.745	10.962
Observations	1,761,138	1,761,156	1,761,156	1,761,156	1,761,138	1,439,420	1,761,138	1,761,138	1,761,138

Notes: This table reports the coefficient β_2 —defined as the “Medium Run Effect of the Reform”—from equation (2) after detrending the outcome variable. The coefficient refers to the effect of the reform two years after the reform was implemented. Panel A, Panel B, and Panel C show the event-study coefficient among the sample of firms belonging to the manufacturing industry, retail industry, and accommodation and food industry, respectively. In Panel D and Panel E, I estimate the firm-level event study specification among firms where the pre-reform fraction of female workers is above and below the median of the pre-reform distribution of women, respectively. All underlying regressions control for firm fixed effects and state-by-year fixed effects. Effects are estimated on a balanced panel of firms (2014–2019) with at least 5 workers before the implementation of the reform. Standard errors are displayed in parentheses and are clustered at the firm level.

Table B.3. Heterogeneity in the Effect of the Reform on Firms' Outcomes, Based on Pre-Reform Firm Size

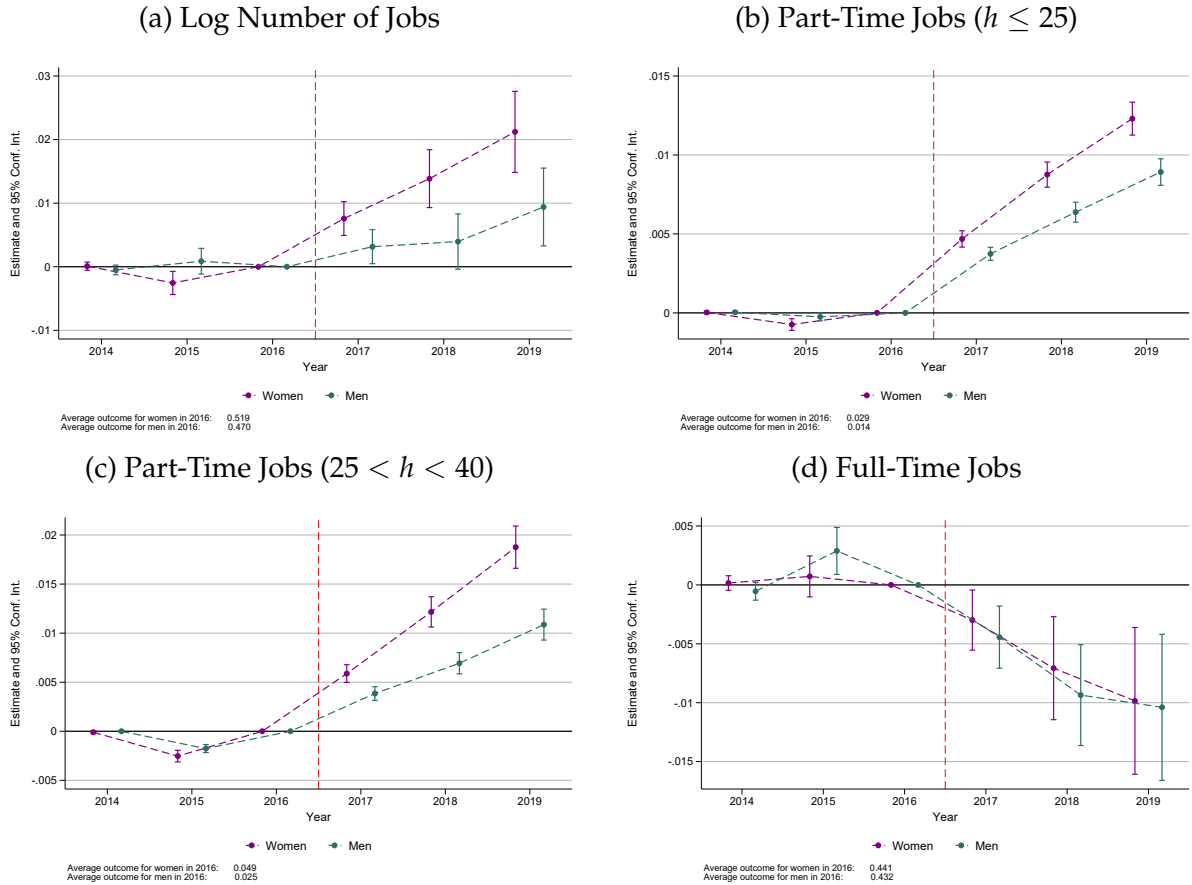
	Log of Number of Jobs (1)	Part-time Jobs ($h \leq 25$) in terms of 2014 Jobs (2)	Part-time Jobs (25 < $h < 40$) in terms of 2014 Jobs (3)	Full-time Jobs ($h \geq 40$) in terms of 2014 Jobs (4)	Share of Part-Time Jobs ($h \leq 25$) (5)	Share of New Part-Time Jobs ($h \leq 25$) (6)	Log of Average Number of Employees (7)	Log Average Hours Worked per Job (8)	Log Total Hours Worked (9)
Panel A. Firms with 5–19 workers									
Medium Run Effect of the Reform	0.029*** (0.003)	0.030*** (0.001)	0.026*** (0.001)	-0.019*** (0.003)	0.025*** (0.001)	0.053*** (0.002)	0.004 (0.003)	-0.022*** (0.001)	-0.016*** (0.003)
Pre-reform Mean in Firms with $\bar{\sigma}_f > 0$	2.593	0.082	0.072	0.799	0.083	0.093	2.295	3.703	9.872
Observations	2,431,273	2,431,273	2,431,273	2,431,273	2,431,273	1,877,518	2,431,273	2,431,273	2,431,273
Panel B. Firms with 20–49 workers									
Medium Run Effect of the Reform	0.009 (0.006)	0.016*** (0.001)	0.018*** (0.002)	0.023** (0.009)	0.012*** (0.001)	0.032*** (0.002)	-0.023*** (0.006)	-0.012*** (0.001)	-0.037*** (0.006)
Pre-reform Mean in Firms with $\bar{\sigma}_f > 0$	3.832	0.076	0.063	0.846	0.075	0.088	3.522	3.704	11.101
Observations	566,973	566,976	566,976	566,976	566,973	530,332	566,973	566,973	566,973
Panel C. Firms with 50+ workers									
Medium Run Effect of the Reform	-0.080*** (0.012)	0.009*** (0.001)	0.011*** (0.002)	-0.063*** (0.010)	0.009*** (0.001)	0.029*** (0.001)	-0.114*** (0.011)	-0.011*** (0.001)	-0.137*** (0.012)
Pre-reform Mean in Firms with $\bar{\sigma}_f > 0$	5.382	0.039	0.066	0.830	0.041	0.057	5.063	3.733	12.672
Observations	314,388	314,388	314,388	314,388	314,388	306,879	314,388	314,388	314,388
Panel D. Single establishment									
Medium Run Effect of the Reform	0.032*** (0.003)	0.025*** (0.001)	0.022*** (0.003)	-0.006 (0.005)	0.020*** (0.000)	0.043*** (0.001)	0.014*** (0.002)	-0.018*** (0.001)	-0.012*** (0.003)
Pre-reform Mean in Firms with $\bar{\sigma}_f > 0$	3.217	0.082	0.074	0.855	0.080	0.091	2.917	3.703	10.494
Observations	3,036,095	3,036,120	3,036,120	3,036,120	3,036,095	2,496,813	3,036,095	3,036,095	3,036,095
Panel E. Multi-establishments									
Medium Run Effect of the Reform	-0.024*** (0.007)	0.020*** (0.001)	0.030*** (0.004)	-0.055** (0.028)	0.014*** (0.001)	0.031*** (0.002)	-0.039*** (0.006)	-0.013*** (0.001)	-0.070*** (0.007)
Pre-reform Mean in Firms with $\bar{\sigma}_f > 0$	4.303	0.047	0.076	0.935	0.045	0.057	3.958	3.733	11.569
Observations	486,198	486,198	486,198	486,198	486,198	443,562	486,198	486,198	486,198

Notes: This table reports the coefficient β_2 —defined as the “Medium Run Effect of the Reform”—from equation (2) after detrending the outcome variable. The coefficient refers to the effect of the reform two years after the reform was implemented. Panel A, Panel B, and Panel C show the event-study coefficient among the sample of firms with pre-reform size of 5–19 workers, 20–49 workers, and above 49 workers, respectively. In Panel D and Panel E, I estimate the firm-level event study specification among firms with single establishments before the reform and firms with multiple establishments before the reform, respectively. All underlying regressions control for firm fixed effects and state-by-year fixed effects. Effects are estimated on a balanced panel of firms (2014–2019) with at least 5 workers before the implementation of the reform. Standard errors are displayed in parentheses and are clustered at the firm level.

C. 2017 Reform Effect across Industries

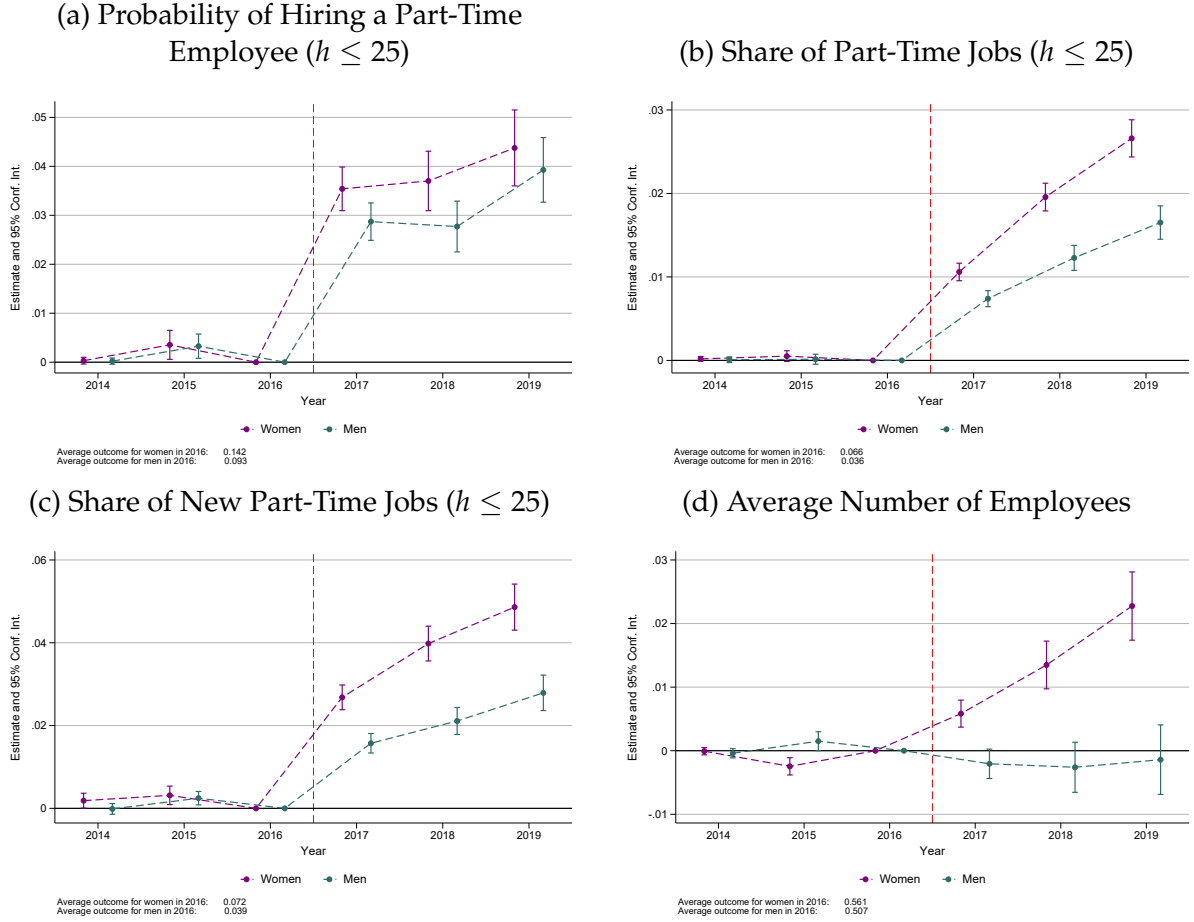
C.1. Retail Trade

Figure C.1. Effects on Number of Jobs in Retail Trade



Notes: This figure reports estimates from equation (2) after detrending the outcome that is gender specific. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended annual number of jobs for women (men) divided by the annual number of jobs in 2014. In Panel (b), the outcome variable is the detrended annual number of part-time jobs for women (men) with a workweek below or equal to 25 hours divided by the annual number of jobs in 2014. Panel (c) reports the effects on the detrended annual number of part-time jobs for women (men) with a workweek between 26 and 39 hours divided by the annual number of jobs in 2014. In Panel (d), the outcome variable is the detrended annual number of full-time jobs for women (men) with a workweek above or equal to 40 hours divided by the annual number of jobs in 2014. Effects are estimated on a balanced panel of firms (2014-2019) belonging to the retail industry with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

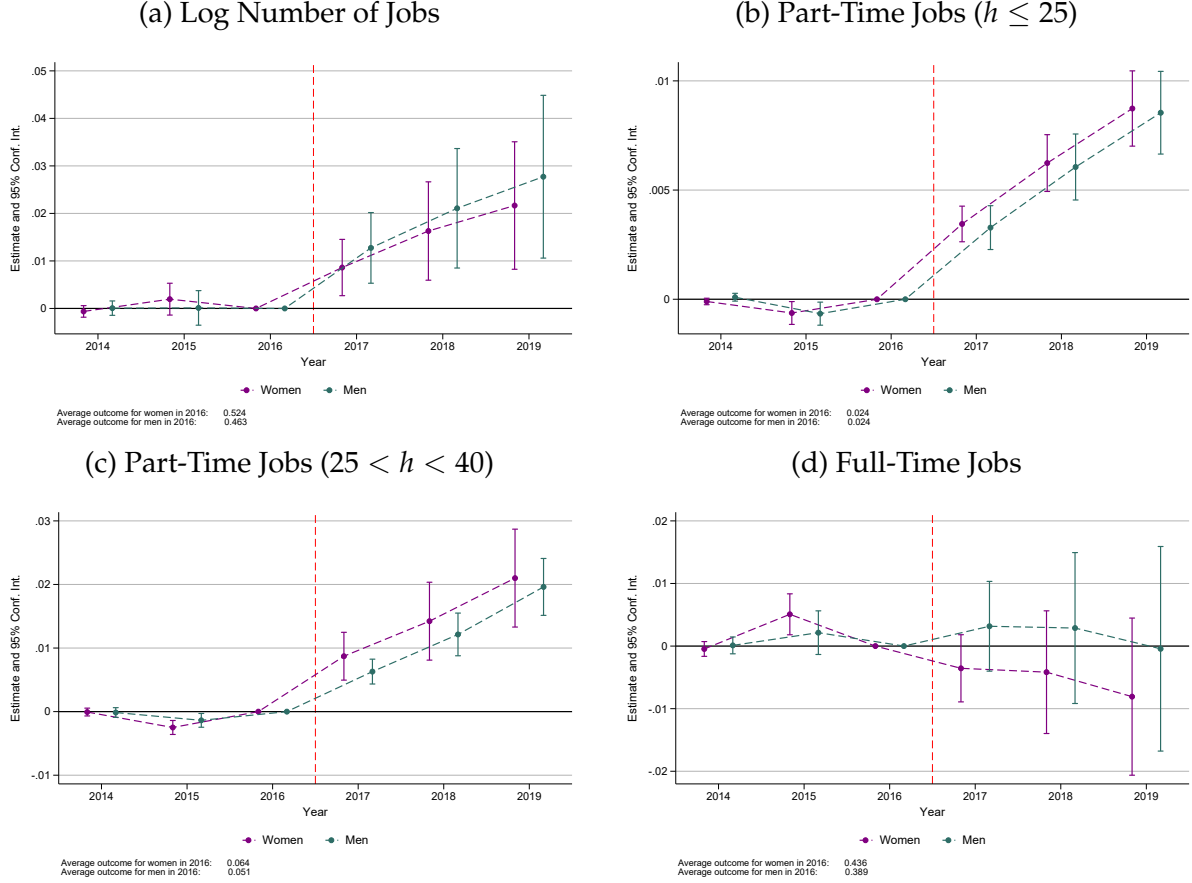
Figure C.2. Effects on Share and Creation of Part-Time Jobs and Employment in Retail Trade



Notes: This figure reports estimates from equation (2) after detrending the outcome that is gender specific. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended indicator of at least one part-time female (male) hire with workweek below or equal to 25 hours. In Panel (b), the outcome variable is the detrended share of part-time jobs for women (men) defined as the annual number of part-time jobs for women (men) with a workweek below or equal to 25 hours divided by the annual number of observed jobs for women (men). Panel (c) reports the effects on the detrended share of new part-time jobs for women (men) defined as the annual number of part-time female (male) hires with a workweek below or equal to 25 hours divided by the annual number of observed female (male) hires. In Panel (d), the outcome variable is the detrended monthly average number of female (male) workers divided by the monthly average number of workers in 2014. Effects are estimated on a balanced panel of firms (2014-2019) belonging to the retail industry with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

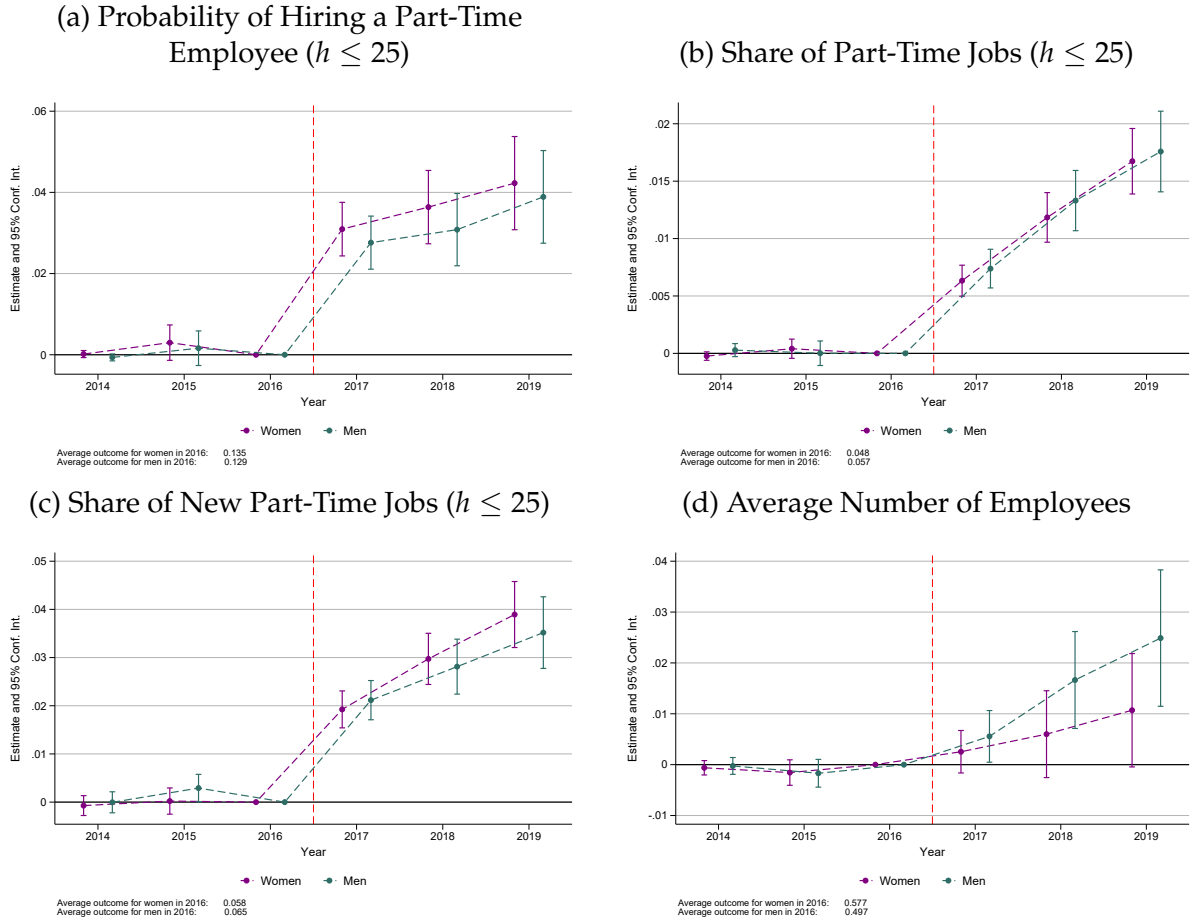
C.2. Accommodation and Food Services

Figure C.3. Effects on Number of Jobs in Accommodation and Food Services



Notes: This figure reports estimates from equation (2) after detrending the outcome that is gender specific. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended annual number of jobs for women (men) divided by the annual number of jobs in 2014. In Panel (b), the outcome variable is the detrended annual number of part-time jobs for women (men) with a workweek below or equal to 25 hours divided by the annual number of jobs in 2014. Panel (c) reports the effects on the detrended annual number of part-time jobs for women (men) with a workweek between 26 and 39 hours divided by the annual number of jobs in 2014. In Panel (d), the outcome variable is the detrended annual number of full-time jobs for women (men) with a workweek above or equal to 40 hours divided by the annual number of jobs in 2014. Effects are estimated on a balanced panel of firms (2014-2019) belonging to the accommodation and food industry with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

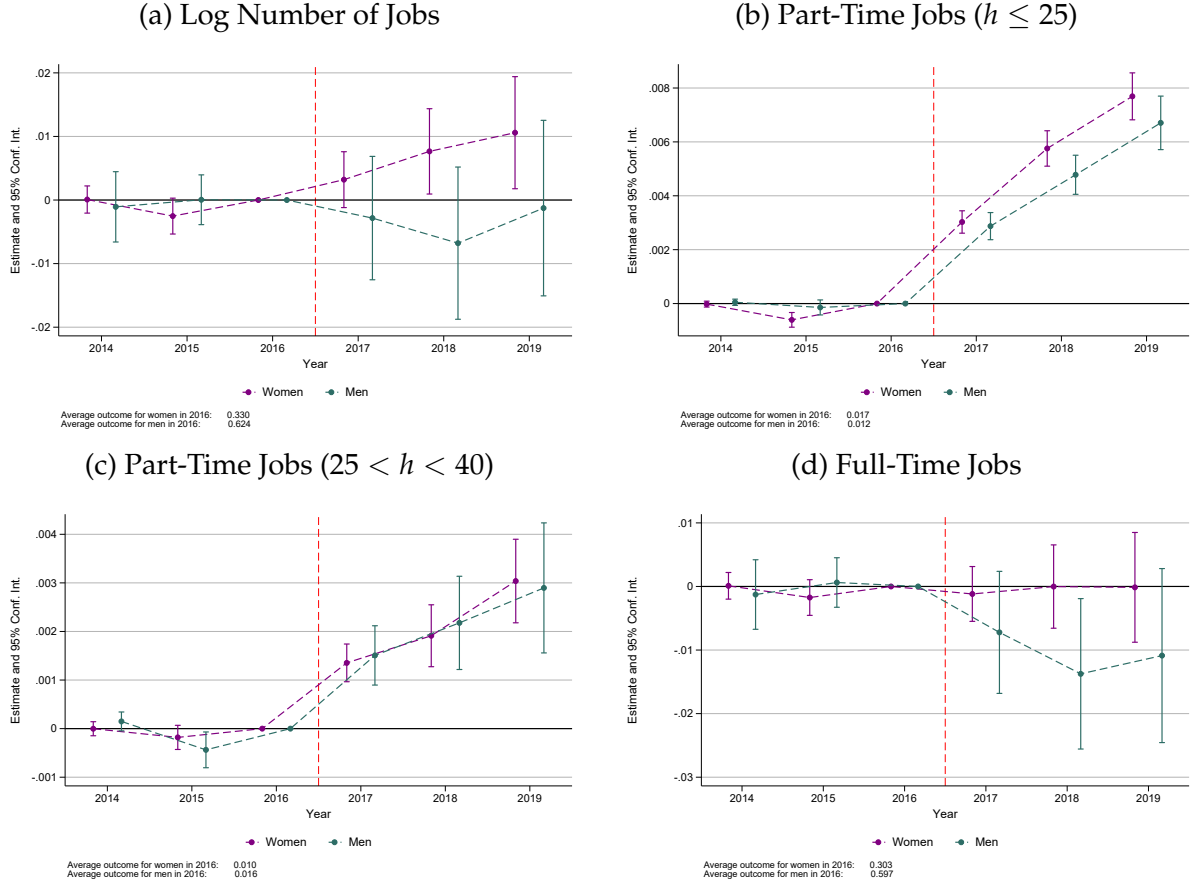
Figure C.4. Effects on Share and Creation of Part-Time Jobs and Employment in Accommodation and Food Services



Notes: This figure reports estimates from equation (2) after detrending the outcome that is gender specific. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended indicator of at least one part-time female (male) hire with workweek below or equal to 25 hours. In Panel (b), the outcome variable is the detrended share of part-time jobs for women (men) defined as the annual number of part-time jobs for women (men) with a workweek below or equal to 25 hours divided by the annual number of observed jobs for women (men). Panel (c) reports the effects on the detrended share of new part-time jobs for women (men) defined as the annual number of part-time female (male) hires with a workweek below or equal to 25 hours divided by the annual number of observed female (male) hires. In Panel (d), the outcome variable is the detrended monthly average number of female (male) workers divided by the monthly average number of workers in 2014. Effects are estimated on a balanced panel of firms (2014-2019) belonging to the accommodation and food industry with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

C.3. Manufacturing

Figure C.5. Effects on Number of Jobs in Manufacturing



Notes: This figure reports estimates from equation (2) after detrending the outcome that is gender specific. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended annual number of jobs for women (men) divided by the annual number of jobs in 2014. In Panel (b), the outcome variable is the detrended annual number of part-time jobs for women (men) with a workweek below or equal to 25 hours divided by the annual number of jobs in 2014. Panel (c) reports the effects on the detrended annual number of part-time jobs for women (men) with a workweek between 26 and 39 hours divided by the annual number of jobs in 2014. In Panel (d), the outcome variable is the detrended annual number of full-time jobs for women (men) with a workweek above or equal to 40 hours divided by the annual number of jobs in 2014. Effects are estimated on a balanced panel of firms (2014-2019) belonging to the manufacturing industry with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

Figure C.6. Effects on Share and Creation of Part-Time Jobs and Employment in Manufacturing



D. Part-Time Jobs across Industries

Table D.1. Percentage of Part-Time Jobs across Industries

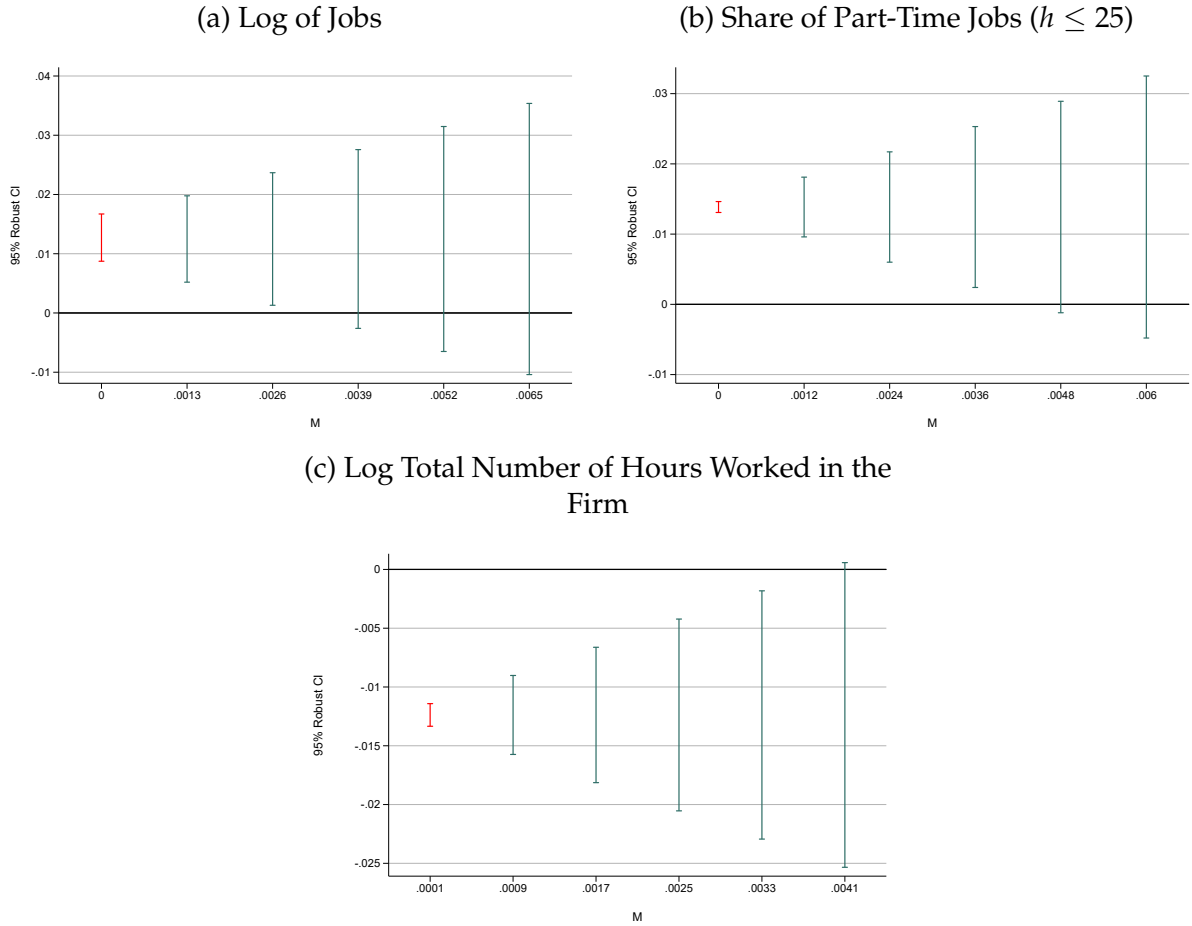
Industry	2016 RAIS
Agriculture, Forestry, Fishing and Hunting	0.008
Mining, Quarrying, and Oil and Gas Extraction	0.016
Construction	0.012
Manufacturing	0.012
Wholesale Trade	0.014
Retail Trade	0.015
Transportation and Warehousing	0.016
Utilities	0.011
Information	0.021
Finance and Insurance	0.009
Real Estate and Rental and Leasing	0.014
Professional, Scientific, and Technical Services	0.017
Administrative and support services	0.020
Educational Services	0.336
Health Care and Social Assistance	0.059
Arts, Entertainment, and Recreation	0.226
Accommodation and Food Services	0.034

Notes: This table shows the average share of part-time jobs with workweek less than or equal to 25 hours per week across 1-digit industries. For the sake of maintaining consistency with the preceding firm sample used in estimation, I use a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform.

E. Robustness

E.1. Credible Approach to Parallel Trends

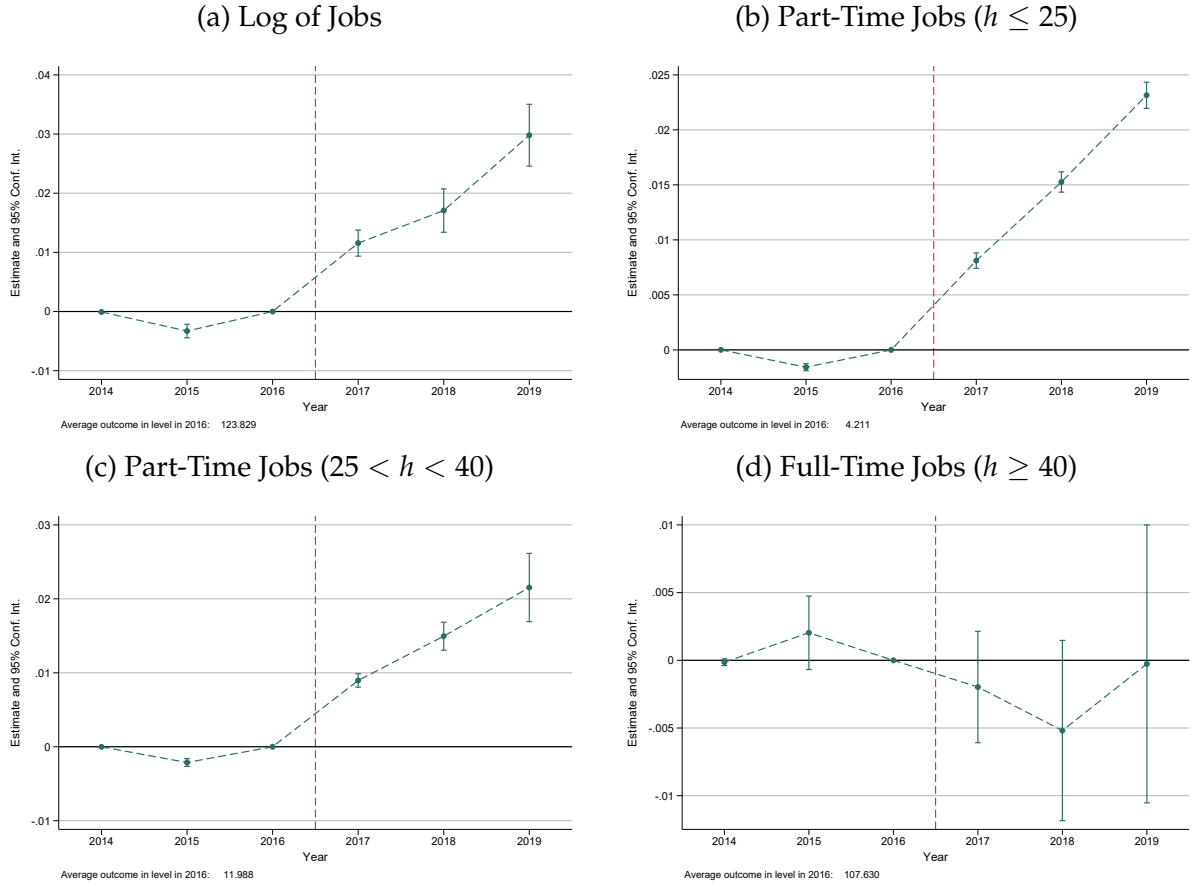
Figure E.1. Honest Pre-Trends on the Effect of the Reform



Notes: These three figures plot the coefficient β_1 from equation (2) for year 2018 (a year after the implementation of the reform). In Panel (a), the outcome variable is the log of the annual number of jobs. In Panel (b), the outcome variable is the share of part-time jobs defined as the annual number of part-time jobs with a workweek below or equal to 25 hours divided by the annual number of observed jobs. In panel (c), the outcome variable is the annual number of hours worked in the firm across all jobs (spells). Following the procedure in Rambachan and Roth (2023), this figure plots alternative confidence intervals that allow for deviations from parallel trends in the pre-reform period. In red, I plot the OLS confidence intervals from equation (2) after detrending the outcome, which allows only for linear violations of parallel trends. In green, larger values of M (>0) allow for larger deviations from linearity. 95% confidence intervals are shown. Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

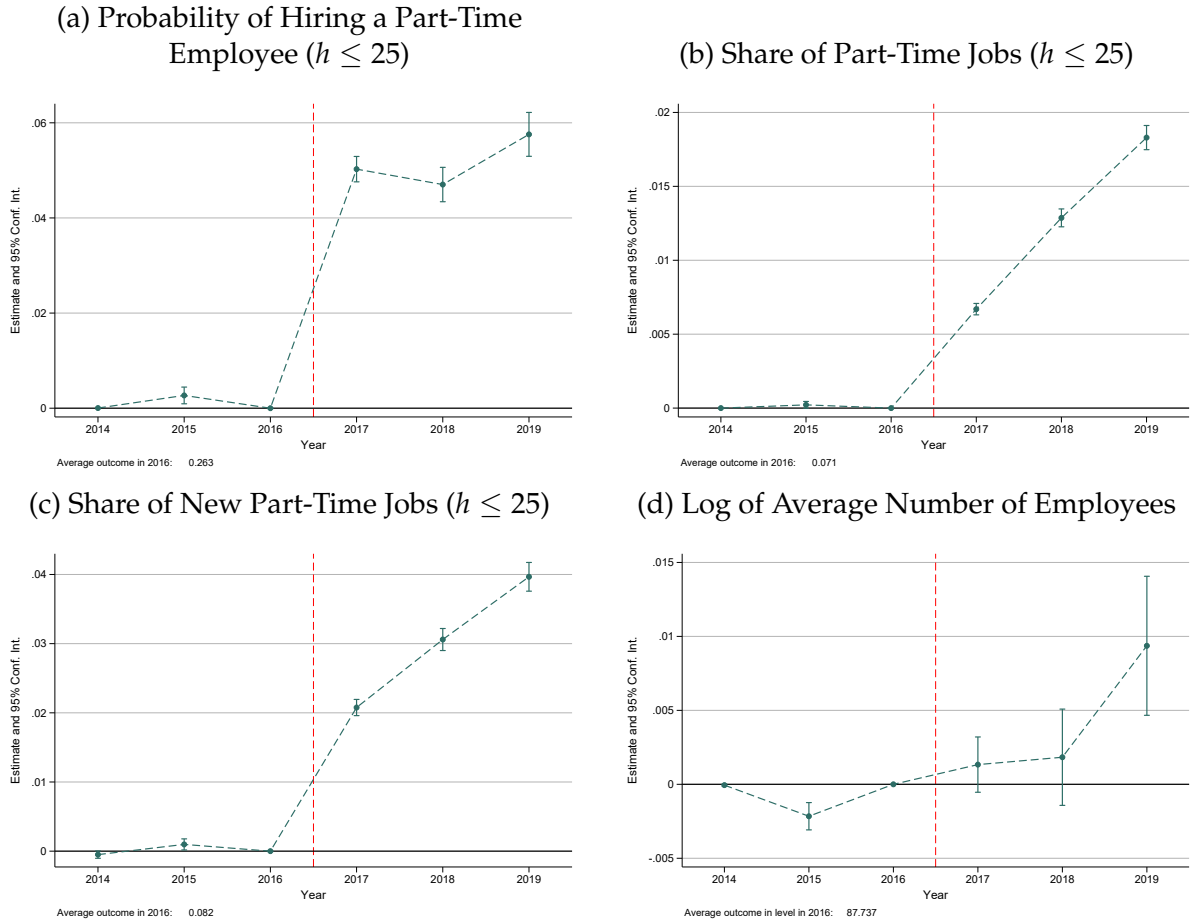
E.2. Many-Skill-Group Firms

Figure E.2. Effects on Number of Jobs in Firms with at least 3 Skill Groups



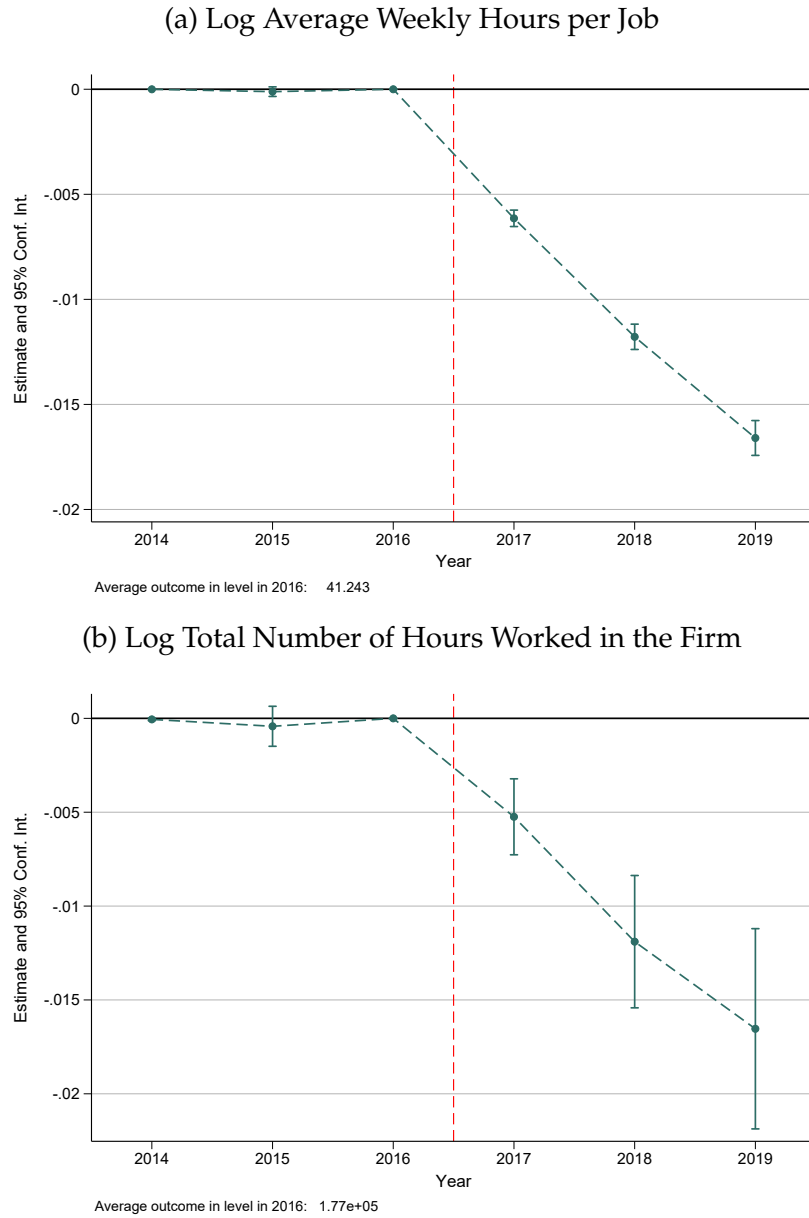
Notes: This figure reports estimates from equation (2) after detrending the outcome. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended log of the annual number of jobs. In Panel (b), the outcome variable is the detrended annual number of part-time jobs with a workweek below or equal to 25 hours divided by the annual number of jobs in 2014. Panel (c) reports the effects on the detrended annual number of part-time jobs with a workweek between 26 and 39 hours divided by the annual number of jobs in 2014. In Panel (d), the outcome variable is the detrended annual number of full-time jobs with a workweek above or equal to 40 hours divided by the annual number of jobs in 2014. Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers and 3 skill groups before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

Figure E.3. Effects on Share and Creation of Part-Time Jobs and Employment in Firms with at least 3 Skill Groups



Notes: This figure reports estimates from equation (2) after detrending the outcome. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended indicator of at least one part-time hire with workweek below or equal to 25 hours. In Panel (b), the outcome variable is the detrended share of part-time jobs defined as the annual number of part-time jobs with a workweek below or equal to 25 hours divided by the annual number of observed jobs. Panel (c) reports the effects on the detrended share of new part-time jobs defined as the annual number of part-time hires with a workweek below or equal to 25 hours divided by the annual number of observed hires. In Panel (d), the outcome variable is the detrended log of employment (average number of workers in a month). Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers and 3 skill groups before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

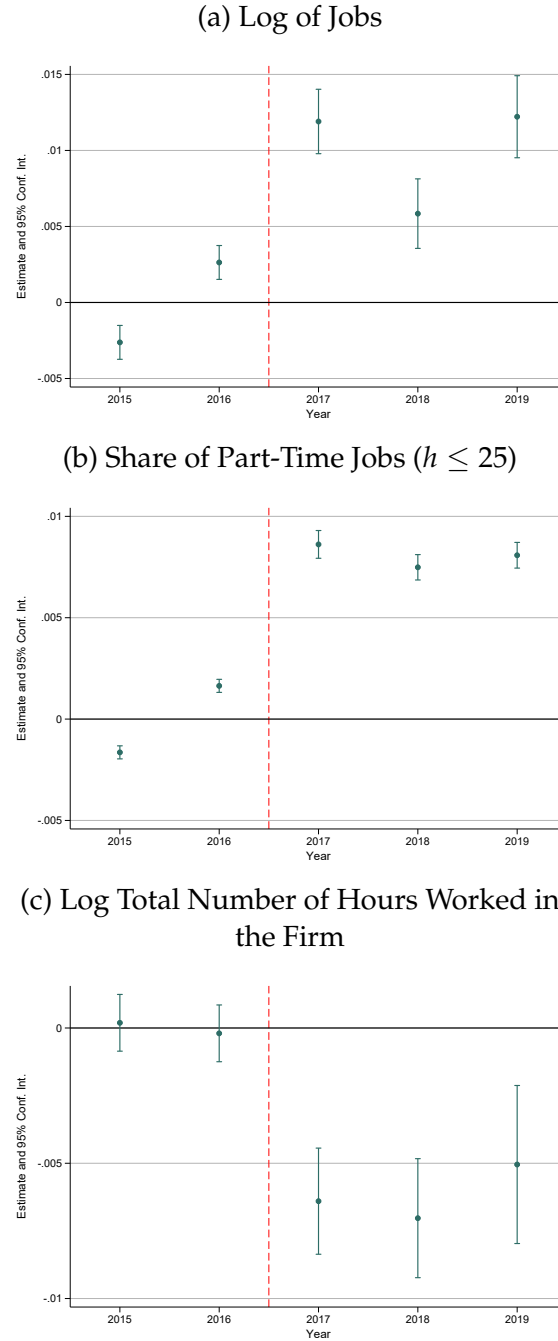
Figure E.4. Effects on the Average Weekly Hours per Job and Total Number of Hours Worked in Firms with at least 3 Skill Groups



Notes: This figure reports estimates from equation (2) after detrending the outcome. Specifically, I regress outcome y_{ft} on a firm specific linear time trend $\theta_f \cdot t$ using pre-reform data from 2014 to 2016. Then, I run specification (2) on the full-period using the residualized outcome $y_{ft} - \hat{\theta}_f \cdot t$. In Panel (a), the outcome variable is the detrended log of average weekly hours per job. In Panel (b), the outcome variable is the detrended log of the annual number of hours worked in the firm across all jobs (spells). Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers and 3 skill groups before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

E.3. Mean Reversion

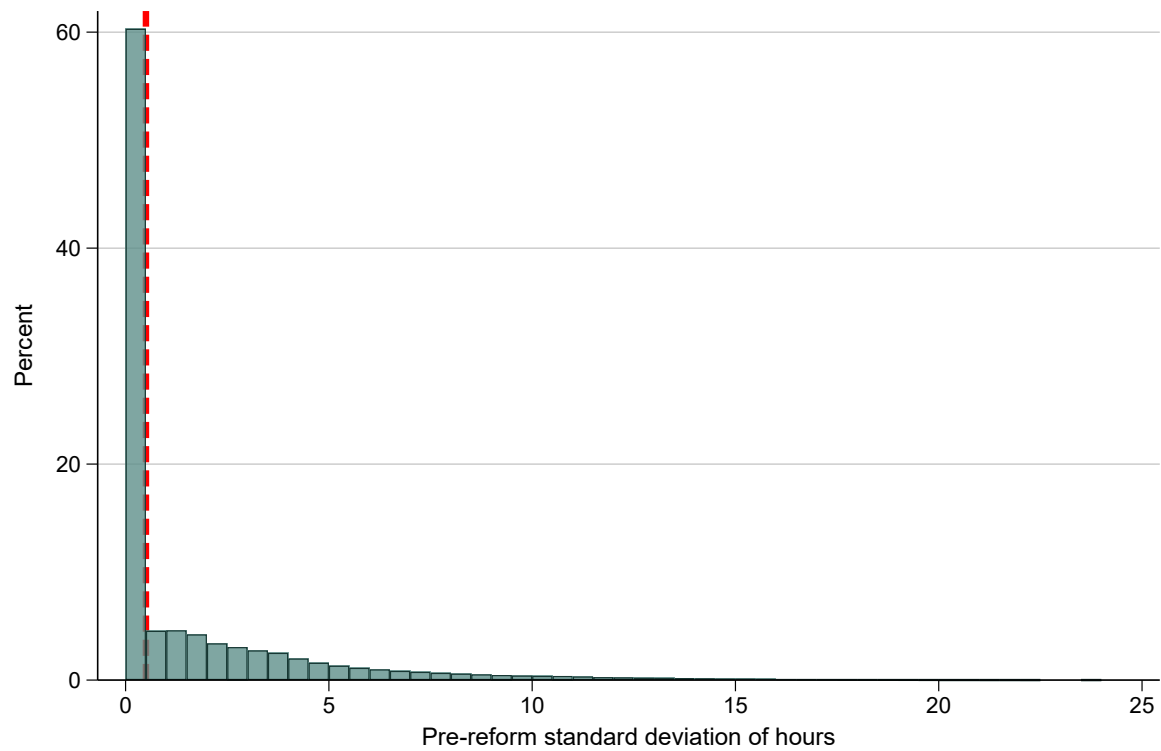
Figure E.5. Difference-in-Difference Estimates Over 2-Year Rolling Periods



Notes: This figure reports estimates from a difference-in-difference regression over two-year rolling periods after detrending the outcome using pre-reform data from 2014 to 2016. Each estimate is computed in a separate regression. For instance, the estimate in 2016 stems from a difference-in-difference regression estimated on years 2015 and 2016, comparing the outcomes of firms subject to strict hours constraints ($\bar{\sigma}_f = 0$) with those facing looser constraints ($\bar{\sigma}_f > 0$). In Panel (a), the outcome variable is the detrended log of the annual number of jobs. In Panel (b), the outcome variable is the detrended share of part-time jobs defined as the annual number of part-time jobs with a workweek below or equal to 25 hours divided by the annual number of observed jobs. In Panel (c), the outcome variable is the detrended log of annual number of hours worked in the firm across all jobs (spells). Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

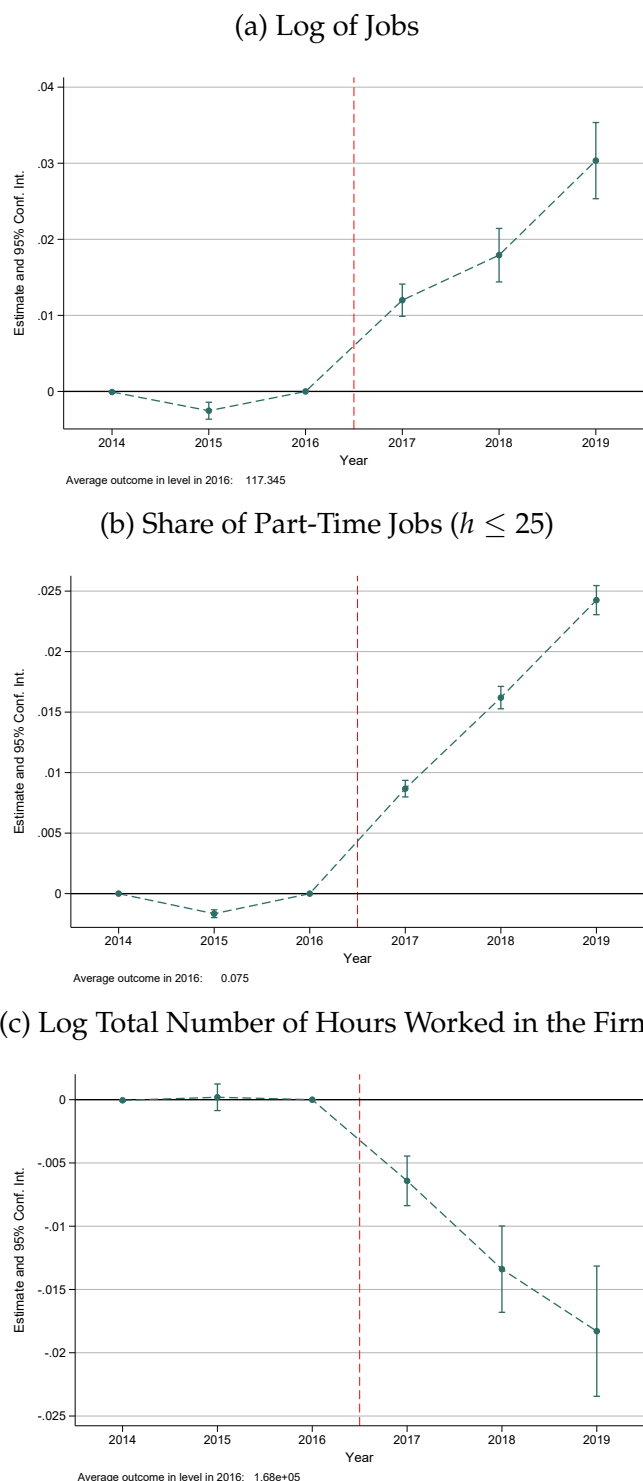
E.4. Alternative Measure of Hour Constraints: Standard Deviation of Hours across All Workers

Figure E.6. Distribution of Pre-Reform Standard Deviation of Hours



Notes: This figure plots the distribution of hours constraints measured as the average of the standard deviation of hours in the firm over the three pre-reform years (2014-2016). This figure shows the distribution of hours constraints among jobs held by individuals aged 18 to 64 in the private sector, using a balanced panel of firms spanning from 2014 to 2019. Firms with pre-reform size smaller than 5 are excluded.

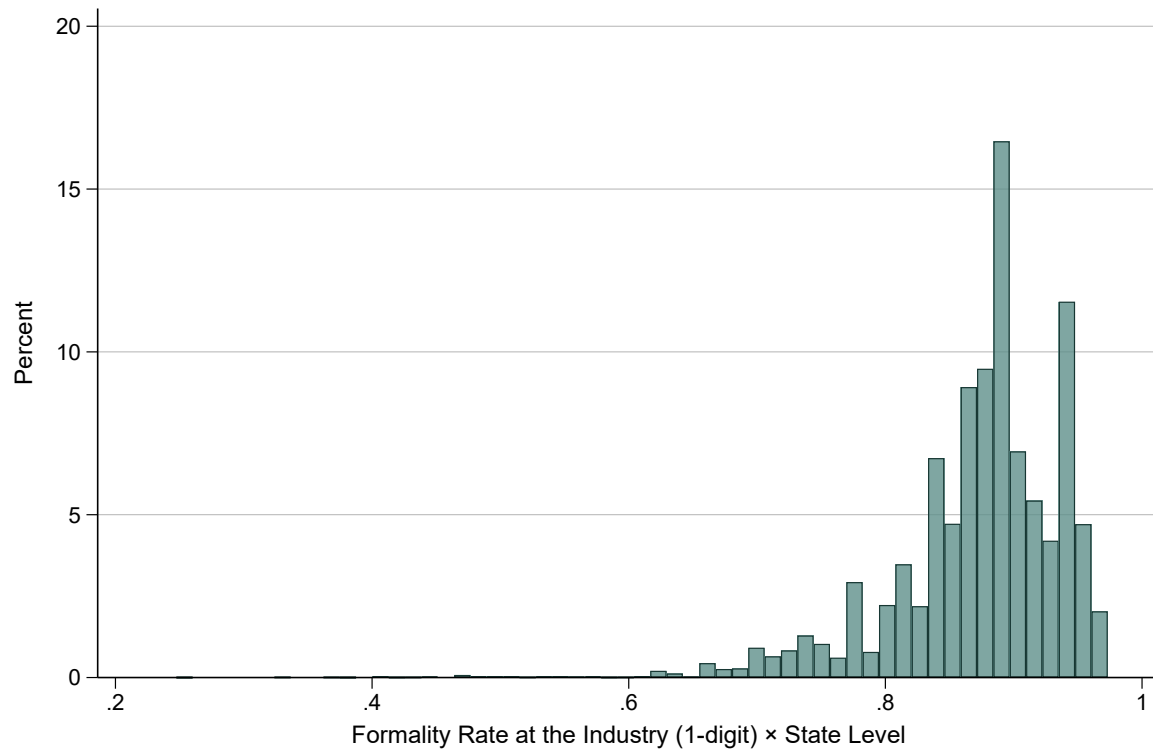
Figure E.7. Effects on Number of Jobs, Creation of Part-Time Jobs and Total Number of Hours Worked using the Alternative Measure of Constraints



Notes: This figure reports estimates from equation (2) after detrending the outcome, where $\bar{\sigma}_f$ is the average of the standard deviation of hours across all workers in the firm over the three pre-reform years (2014-2016). In Panel (a), the outcome variable is the detrended log of the annual number of jobs. In Panel (b), the outcome variable is the detrended share of part-time jobs defined as the annual number of part-time jobs with a workweek below or equal to 25 hours divided by the annual number of observed jobs. In Panel (c), the outcome variable is the detrended log of annual number of hours worked in the firm across all jobs (spells). Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

E.5. Alternative Measure of Exposure: Formality Rate at the Industry-State Level

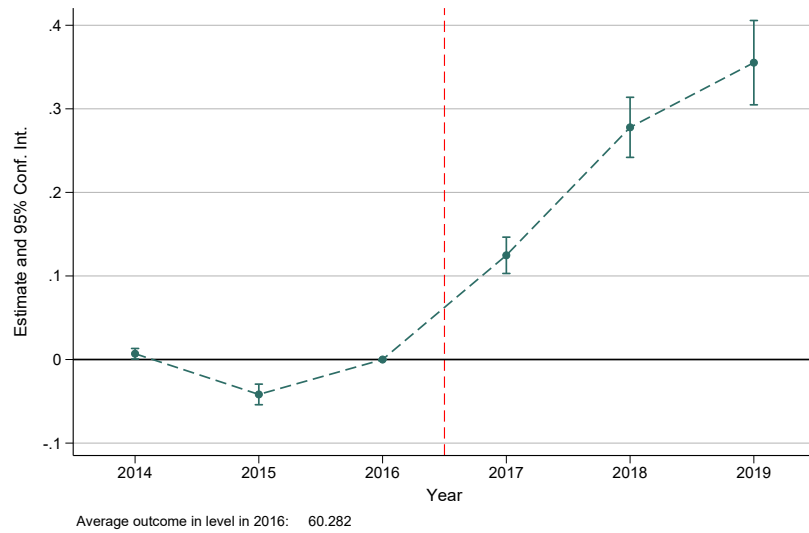
Figure E.8. Distribution of Formality Rate



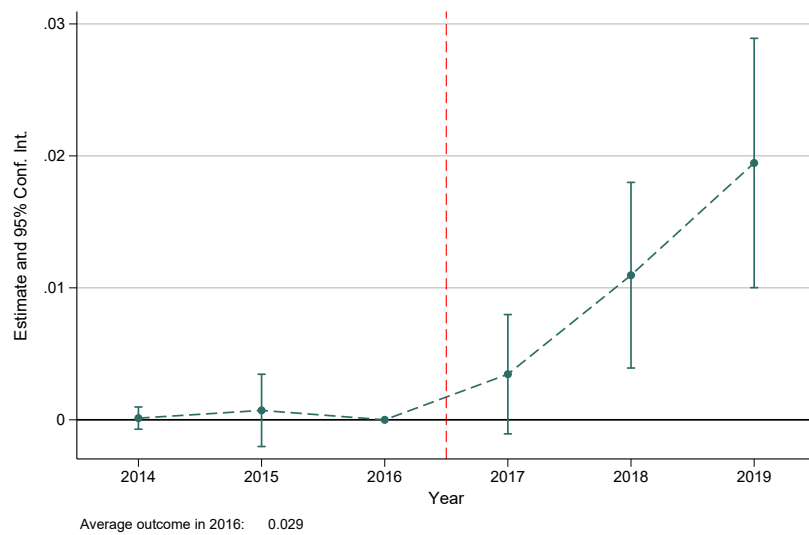
Notes: This figure plots the distribution of the pre-reform formality rate assigned to each firm based on its industry and state location. The formality rate denotes the average fraction of formal workers in the industry and state over the period 2012-2016. The figure uses a balanced panel of firms spanning from 2014 to 2019. Firms with a pre-reform size smaller than 5 are excluded.

Figure E.9. Effects on Number of Jobs and Share of Part-Time Jobs using Industry-State Formality Rates

(a) Log of Jobs



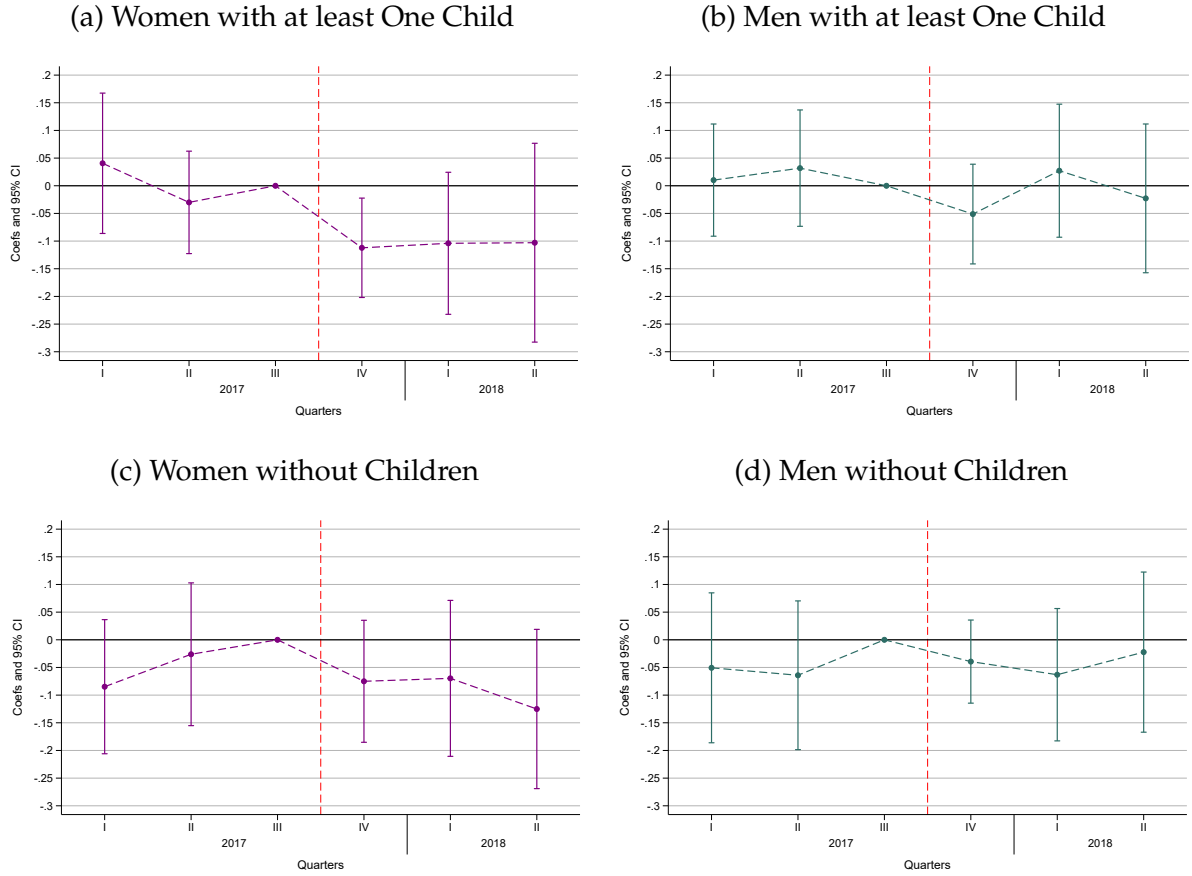
(b) Share of Part-Time Jobs ($h \leq 25$)



Notes: This figure reports estimates from equation (3) after detrending the outcome variable. In Panel (a), the outcome variable is the detrended log of the annual number of jobs. In Panel (b), the outcome variable is the detrended share of part-time jobs defined as the annual number of part-time jobs with a workweek below or equal to 25 hours divided by the annual number of observed jobs. Effects are estimated on a balanced panel of firms (2014-2019) with at least 5 workers before the implementation of the reform. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the firm level.

F. Within-Gender Heterogeneity in Informality

Figure F.10. Within-Gender Heterogeneity in Informality



Notes: This figure reports estimates from equation (5) when the outcome variable is a dummy variable that takes the value of one if the individual was informally employed—without a signed booklet—and zero if the individual was formally employed. Panel (a) and Panel (b) report the effects on informality status among the sample of women and men, respectively, that have at least one child. Panel (c) and Panel (d) report the effects on informality status among the sample of childless women and men, respectively. Effects are estimated on a quarterly panel of individuals aged between 18 and 64 years old. Ninety-five percent confidence intervals are obtained after clustering the standard errors at the state level.